CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY
FACULTY OF SOFTWARE ENGINEERING

GRADUATION THESIS

BACHELOR OF ENGINEERING IN

SOFTWARE ENGINEERING

(HIGH-QUALITY PROGRAM)

DESIGNING AN AUTOMATED PIPELINE FOR ANALYZING BACTERIAL BLIGHT RESISTANCE GENES IN RICE

Student: Thach Thanh Nhi
Student ID: B2203572
Class: 2022 – 2026 (K48)
Advisor: Ho Quang Thai

Can Tho, 08/2026

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CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

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Despite my best efforts, this thesis may still contain shortcomings. I sincerely welcome any valuable feedback from the faculty to help improve this work.

Thank you sincerely.

Can Tho, August 16, 2026

Thach Thanh Nhi

COMMITMENT TO RESULTS

I hereby certify that the graduation thesis titled " Designing an Automated Pipeline for Analyzing Bacterial Blight Resistance Genes in Rice" is the result of my own research and has not been submitted for any other degree or qualification.

All references used in the thesis are clearly cited, including the author's name, the title of the work, and the time and place of publication. I accept full responsibility for any unauthorized copying, violations of academic regulations, or instances of academic dishonesty.

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Thach Thanh Nhi

ABSTRACT

This thesis designs, implements, and evaluates three reproducible genomic prediction pipelines for rice bacterial blight (BB) resistance using genomic resources from the 3,000 Rice Genomes Project (3K-RG) and the 340-accession phenotype panel reported by Lu et al. (2021). Pipeline A provides a multi-representation benchmark that compares genome-wide SNPs, Plant-DNABERT embeddings of homology-anchored resistance-gene loci, and raw one-hot DNA under both random and population-structure-aware cross-validation. Pipeline B uses a controlled negative-control design that compares DNABERT-2 embeddings of candidate resistance/QTL regions with matched random control regions, regional SNPs, genome-wide SNPs, and population/sequence-quality baselines. Pipeline C proposes an Enhanced Genomic Transformer (EGT) that fuses 2,048 training-only ANOVA-selected SNPs, seeded by 20 resistance-gene/QTL regions, with eleven frozen Nucleotide Transformer 500M resistance-gene priors through self-attention and cross-attention. The experiments show that evaluation design materially changes the apparent predictive performance. In Pipeline B, candidate-region embeddings exceeded matched control-region embeddings in GroupKFold MCC for all four Xanthomonas oryzae pv. oryzae strains, supporting resistance-associated sequence signal beyond the matched genomic background. Under the EGT 80/20 teacher-style validation protocol, the selected Ver4.1 configuration achieved a validation mean MCC of approximately 0.548 across C4, C5, V, and P9a. These EGT values are reported as validation performance because the same validation set is used for checkpoint, threshold, and task-weight selection. To make the research executable outside the training notebooks, the study also delivers a command-line interface, a bilingual Streamlit dashboard, and a 22-file final model bundle containing the exact ordered SNP manifest, allele semantics, thresholds, frozen prior embeddings, validation fixtures, and SHA256 integrity records. The results demonstrate that population-aware evaluation, negative controls, explicit data provenance, and deployable preprocessing contracts are as important as model architecture when building trustworthy genomic resistance-prediction software.

TABLE OF CONTENTS

COMMITMENT TO RESULTS 4
ABSTRACT 5
TABLE OF CONTENTS 6
LIST OF FIGURES 12
LIST OF TABLES 14
LIST OF ABBREVIATIONS 16
CHAPTER 1. INTRODUCTION 17
1.1. BACKGROUND AND CHALLENGES 17
1.2. LITERATURE SURVEY 17
1.3. OBJECTIVES, SCOPE, AND PIPELINE DESIGN 18
1.4. THESIS STRUCTURE 19
CHAPTER 2. THEORETICAL BACKGROUND 20
2.1. OVERVIEW OF RICE BACTERIAL BLIGHT AND RESISTANCE 20
2.1.1. Rice Bacterial Blight and the Bacterium Xanthomonas oryzae pv. oryzae 20
2.1.2. Resistance Mechanisms and Resistance Genes (the Xa Gene Family) 20
2.1.3. Xoo Strains and the Phenotype Scale 21
2.1.4. Characteristics of Resistance-Gene Regions in the Rice Genome 22
2.2. GENOMIC PHENOTYPE PREDICTION WITH MACHINE LEARNING 22
2.2.1. From Conventional Breeding to Genomic Prediction 22
2.2.2. Principle of a Genomic Prediction Pipeline 23
2.2.3. The Three Designed and Compared Pipelines 23
2.2.4. Domain-Specific Challenges 23
2.3. GENOMIC DATA AND FEATURE-REPRESENTATION METHODS 24
2.3.1. The 3,000 Rice Genomes Project and Consensus Genomes 24
2.3.2. SNPs, Genotype, and Two Feature-Selection Strategies 24
2.3.3. Coordinate Drift and Homology Anchoring 24
2.3.4. One-Hot Encoding of DNA Sequences 25
2.3.5. DNA Language Models, Embeddings, and Biological Priors 25
2.4. MACHINE LEARNING MODELS 26
2.4.1. Classical Machine Learning Models 26
2.4.2. Baseline Deep Learning Models 26
2.4.3. Baseline Deep Learning Models 26
2.4.4. The Enhanced Genomic Transformer (Proposed) 26
2.5. MODEL EVALUATION AND POPULATION STRUCTURE 27

2.5.1. Cross-Validation and StratifiedKFold 27
2.5.2. GroupKFold and Population-Structure Leakage 27
2.5.3. Evaluation Metrics 28
2.5.4. Handling Class Imbalance 28
2.5.5. Negative-Control Experimental Design (Pipeline B) 28
2.5.6. Two Evaluation Protocols Across the Pipelines 29
2.6. RELATED WORK 29
CHAPTER 3. CONSTRUCTION, TRAINING, AND EXPERIMENTAL COMPARISON OF THE THREE PIPELINES 30
3.1. Data acquisition and construction of the BB340 analysis cohort 30
3.1.1. Genomic source: the 3,000 Rice Genomes Project 30
3.1.2. Phenotype source and strain-specific disease response 31
3.1.3. Practical acquisition workflow for the genotype branch 31
3.1.4. Practical acquisition workflow for BAM-derived consensus sequences 32
3.1.5. Resistance-gene and QTL knowledge used by the sequence and EGT branches 33
3.1.6. Three genomic data streams and accession funnel 34
3.2. Shared preprocessing and data-quality controls 35
3.2.1. Phenotype transformation and label semantics 35
3.2.2. Accession matching and prevention of row-order errors 35
3.2.3. Coordinate drift and homology-based sequence extraction 36
3.2.4. %N quality control and why it is biologically relevant 37
3.2.5. Feature construction and dimensions 38
3.2.6. Evaluation leakage controls and reproducibility constraints 39
3.3. Experimental design overview 39
3.4. Pipeline A — gene-centric locus modeling 41
3.4.1. Purpose and processing flow 41
3.4.2. Homology anchoring and clean-gene construction 43
3.4.3. Plant-DNABERT embedding branch 43
3.4.4. Genome-wide SNP and raw-DNA branches 43
3.4.5. Validation design and results 44
3.4.6. Pipeline A interpretation and limitations 45
3.5. Pipeline B — candidate-region and QTL modeling 45
3.5.1. Controlled biological question 45
3.5.2. Candidate and negative-control region construction 47
3.5.3. Candidate/control sequence embedding and SNP baseline 47
3.5.4. Six-row controlled comparison 47
3.5.5. Candidate-versus-control evidence 48
3.5.6. Pipeline B limitations 49
3.6. Pipeline C — Enhanced Genomic Transformer 49

3.6.1. Motivation for multimodal fusion 49
3.6.2. Memory-safe training-only selection of 2,048 SNPs 49
3.6.3. Nucleotide Transformer resistance-gene priors 50
3.6.4. EGT architecture 50
3.6.5. Training protocol and C4 task-weight fine sweep 53
3.6.6. Best-configuration validation performance 54
3.6.7. Architecture and prior ablation 58
3.6.8. Task weighting versus focal gamma 59
3.6.9. Biological interpretation with signed Gradient x Input 59
3.7. Cross-pipeline interpretation 60
3.8. Reproducibility outputs from the experimental pipelines 61
3.9. Chapter summary 61
3.10. Operational data-acquisition and preprocessing checklist 62
3.10.1. Recommended project data layout 62
3.10.2. Genotype acquisition acceptance checks 63
3.10.3. BAM and consensus generation acceptance checks 64
3.10.4. Pipeline-specific reproducibility artifacts 65
3.11. End-to-end data provenance from public source to model-ready input 66
3.11.1. Provenance trace for Pipeline A 67
3.11.2. Provenance trace for Pipeline B 67
3.11.3. Provenance trace for Pipeline C 68
3.11.4. Data-processing traceability checklist for the thesis archive 68
CHAPTER 4. SOFTWARE IMPLEMENTATION AND TESTING 70
4.1. Software architecture and project organization 70
4.1.1. Main modules and responsibilities 70
4.1.2. Model bundle as the deployment boundary 71
4.1.3. Input contract and scientific meaning of the 2,048 features 73
4.2. Building and running the command-line interface 73
4.2.1. Installation and directory preparation 73
4.2.2. CLI verification, demonstration, and batch prediction 74
4.2.3. Expected CLI output and error behavior 74
4.3. Building and operating the bilingual Streamlit dashboard 75
4.3.1. Interface structure and configuration sidebar 75
4.3.2. Run sequence and session-state behavior 76
4.3.3. Prediction tab: per-sample resistance calls 76
4.3.4. Prediction tab: resistance-gene and SNP evidence 77
4.3.5. Batch overview tab 78
4.3.6. Model comparison benchmark tab 79

4.3.7. Enhanced Genomic Transformer information tab 81
4.3.8. Plant-DNABERT tab 82
4.4. User use cases 83
4.4.1. Verify a model bundle before inference 84
4.4.2. Run batch prediction and export a report 85
4.4.3. Run and inspect interactive EGT prediction 85
4.4.4. Compare model families and interpret predictions 86
4.5. Deployment procedures 87
4.5.1. Local desktop deployment 87
4.5.2. Colab-based deployment 87
4.6. Test design and acceptance criteria 88
4.6.1. Golden-sample regression fixtures 89
4.6.2. Golden-sample probability results 89
4.6.3. Detailed test scenarios using at least five real samples 91
4.6.4. Functional and negative test cases 91
4.6.5. Dashboard scenario tests 93
4.7. Interpretation of the software test results 94
4.8. Operational limitations and future software improvements 94
4.9. Implementation consolidation and handoff principles 95
4.10. Detailed CLI command reference 95
4.10.1. Verify command 95
4.10.2. Demo command 96
4.10.3. Predict command 96
4.10.4. Example report schema 97
4.10.5. Plant-DNABERT CLI operation 97
4.11. Detailed dashboard operating workflows 98
4.11.1. Workflow A — verify and demonstrate the eight golden samples 98
4.11.2. Workflow B — predict an uploaded EGT batch 98
4.11.3. Workflow C — run the exported Plant-DNABERT classifiers 99
4.11.4. Dashboard information architecture 99
4.12. Detailed golden-sample verification log 100
4.12.1. Golden sample IRIS_313-10605 (cA(Aus)) 100
4.12.2. Golden sample IRIS_313-8326 101
4.12.3. Golden sample IRIS_313-8386 (XI-adm) 101
4.12.4. Golden sample CX3 (XI-1A) 102
4.12.5. Golden sample IRIS_313-11814 (XI-3) 102
4.12.6. Golden sample IRIS_313-11797 (XI-1A) 103
4.12.7. Golden sample IRIS_313-8454 (XI-1A) 103

4.12.8. Golden sample B097 (XI-adm) 104
4.13. Use-case-to-test traceability 104
4.14. Operator handoff checklist 105
4.16. Internal inference data flow 106
4.16.1. Batch processing and memory behavior 107
4.16.2. Probability, threshold, and class semantics 108
4.17. Error handling and failure-mode design 108
4.17.1. Scientific failure modes not fully solved by the prototype 109
4.18. Dashboard design rationale for research communication 109
4.18.1. Why the sidebar contains model paths and input source 110
4.18.2. Why the Prediction tab shows both cards and explanatory tables 110
4.18.3. Why the dashboard includes model comparison and Plant-DNABERT 110
4.19. Final Chapter 4 summary 111
CHAPTER 5. CONTRIBUTIONS, LIMITATIONS, AND FUTURE WORK 112
5.1. Summary of findings 112
5.2. Main contributions 112
5.2.1. Reproducible multi-representation BB340 benchmark 112
5.2.2. Homology-based extraction with sequence-quality control 112
5.2.3. Candidate-versus-control experimental design 113
5.2.4. Enhanced Genomic Transformer 113
5.2.5. Model interpretation at SNP, region, and prior-token levels 113
5.2.6. Deployable scientific software and model bundle 113
5.3. Limitations 113
5.3.1. Limited phenotype sample size and high dimensionality 113
5.3.2. Different evaluation protocols across pipelines 113
5.3.3. Severe class imbalance for strain V 114
5.3.4. Run-specific supervised SNP panel 114
5.3.5. Missing-genotype encoding 114
5.3.6. Fixed resistance-gene priors 114
5.3.7. Consensus-sequence and mapping uncertainty 114
5.3.8. Attribution is not causal genetics 114
5.3.9. External genotype harmonization is not yet automated 115
5.4. Future work 115
5.4.1. Unified population-aware evaluation 115
5.4.2. Larger and more diverse phenotype panels 115
5.4.3. Stronger biological priors 115
5.4.4. Missingness- and allele-aware genotype preprocessing 115
5.4.5. Robust biological follow-up 115

5.4.6. Production-grade software extension 115
5.5. Final conclusion 116
REFERENCES 117
APPENDICES 119
APPENDIX A. ELECTRONIC SUPPLEMENTARY DATA FILES 119
A.1. Dataset and model summary 120
A.2. Gene-level sequence quality summary 120
A.3. Lu et al. supplementary primers 121
A.4. BB340 phenotype-panel excerpt 122
A.5. Significant-SNP annotation excerpt 123
A.6. Resistance-gene reference manifest excerpt 124
A.7. Matched phenotype-consensus cohort excerpt 125
A.8. Three-branch model-result summary 126
A.9. Lu et al. haplotype supplementary file 126
APPENDIX B. ENHANCED GENOMIC TRANSFORMER SUPPLEMENTARY FIGURES 127

LIST OF FIGURES

Figure 2.1 - Infection cycle of Xoo. 20

Figure 2.2 - Three groups of resistance mechanisms: LRR/NBS-LRR receptors, SWEET promoter mutations blocking TAL effectors, and the xa5/TFIIA variant. 21

Figure 2.3 - Random StratifiedKFold versus subgroup-aware GroupKFold. 28

Figure 3.1 — BB340 experimental workflow: shared data sources, three complementary pipelines, and shared study outputs. 40

Figure 3.2 — Pipeline A workflow: homology-based locus extraction, gene QC, three feature branches, and population-aware evaluation. 42

Figure 3.3 — Pipeline B workflow: population control, genome-wide baseline, candidate/QTL regions, matched random controls, and DNABERT-2 embeddings. 46

Figure 3.4 — EGT multimodal architecture: the SNP branch is fused with eleven frozen NT500M resistance-gene priors through cross-attention. 52

Figure 3.5 — C4 task-weight fine sweep (gamma = 1.5): validation Mean MCC, ROC-AUC, and PR-AUC. 53

Figure 3.6 — Validation Mean MCC across training epochs for the C4-weight configurations. 54

Figure 3.7 — Validation ROC curves for the four EGT strain heads. 55

Figure 3.8 — Validation precision-recall curves for the four EGT strain heads. 56

Figure 3.9 — Confusion matrices for the four EGT strain heads. 57

Figure 3.10 — Validation performance matrix for the selected EGT configuration. 57

Figure 3.11 — EGT ablation at focal gamma = 2.0: validation mean MCC by DNA foundation model and prior configuration. 58

Figure 3.12 — Mean SNP attribution across the 20 predefined resistance/QTL windows, by strain. 60

Figure 3.13 — NT500M resistance-gene prior-token attribution by strain. 60

Figure 4.1 — Dashboard configuration sidebar and main navigation before a prediction run. 76

Figure 4.2 — Per-sample EGT prediction cards for C4, C5, V, and P9a. 77

Figure 4.3 — Resistance-gene/QTL evidence and top contributing SNPs for a selected rice sample. 78

Figure 4.4 — Batch susceptibility-probability heatmap across eight golden rice samples and four Xoo strains. 79

Figure 4.5 — EGT benchmark table comparing Ver4.1 with the original NT500M and AgroNT runs. 80

Figure 4.6 — Dashboard visualization of EGT MCC, PR-AUC, and ROC-AUC by Xoo strain. 80

Figure 4.7 — Plant-DNABERT GroupKFold MCC table and classifier comparison in the dashboard. 81

Figure 4.8 — Enhanced Genomic Transformer tab describing the deployed model and input/output semantics. 81

Figure 4.9 — Live EGT prediction table and batch heatmap generated from the current input batch. 82

Figure 4.10 — Plant-DNABERT inference tab with classifier, threshold, sample-count, and susceptibility-shift controls. 83

Figure 4.11 — Two-dimensional PCA visualization of Plant-DNABERT/XGBoost predictions for the current batch. 83

Figure B.1 - Task-weight sweep validation Mean MCC. 127
Figure B.2 - C4 fine-sweep metrics. 128
Figure B.3 - Validation Mean MCC by epoch. 129
Figure B.4 - Training loss by epoch. 130
Figure B.5 - Teacher-protocol validation ROC curves. 131
Figure B.6 - Teacher-protocol validation precision-recall curves. 132
Figure B.7 - C4 confusion matrix. 133
Figure B.8 - C5 confusion matrix. 133
Figure B.9 - V confusion matrix. 134
Figure B.10 - P9a confusion matrix. 134
Figure B.11 - Top-60 selected-SNP importance heatmap. 135
Figure B.12 - C4 Manhattan-like selected-SNP importance plot. 135
Figure B.13 - C5 Manhattan-like selected-SNP importance plot. 136
Figure B.14 - V Manhattan-like selected-SNP importance plot. 136
Figure B.15 - P9a Manhattan-like selected-SNP importance plot. 137
Figure B.16 - Resistance/QTL-region importance heatmap. 137
Figure B.17 - C4 resistance/QTL-region attribution bar chart. 138
Figure B.18 - C5 resistance/QTL-region attribution bar chart. 138
Figure B.19 - V resistance/QTL-region attribution bar chart. 139
Figure B.20 - P9a resistance/QTL-region attribution bar chart. 139
Figure B.21 - NT500M resistance-gene prior importance heatmap. 140
Figure B.22 - Teacher-protocol validation performance heatmap. 140

LIST OF TABLES

Table 2.1. Distribution of resistant/susceptible samples per strain (actual figures from the dataset, n = 340). 21

Table 2.2. Mean ambiguous-base percentage (%N) and initial gene-level quality-control status. 22

Table 3.1 — The three genomic data streams and their sources 34

Table 3.2 — Accession funnel from the source resources to the usable set per branch..... 34

Table 3.3 — Lesion-length classification and EGT binary grouping derived from Lu et al. (2021) [4]. 35

Table 3.4 — Class distribution per strain in the BB340 cohort (n = 340). 35

Table 3.5 — Alignment stages and their distinct purposes. 36

Table 3.6 — Sequence quality-control levels applied in the two consensus-based pipelines. 37

Table 3.7 — Mean per-gene ambiguous-base percentage and retain/exclude decision in Pipeline A. 37

Table 3.8 — Feature construction and dimensions for the three experimental pipelines. 38

Table 3.9 — Mean GroupKFold MCC by branch and model family, Pipeline A. 44

Table 3.10 — Mean independent-test AUC by branch and model family, Pipeline A. 44

Table 3.11 — GroupKFold MCC by controlled feature row and Xoo strain, Pipeline B. 47

Table 3.12 — Balanced candidate-versus-control comparison under GroupKFold. 48

Table 3.13 — EGT validation performance for gamma = 1.5 and C4 task weight = 1.15. 54

Table 3.14 — Architecture/prior ablation ranked by validation mean MCC (gamma = 2.0). 58

Table 3.15 — Per-strain MCC for the weight-tuned gamma = 1.5 configuration and the gamma = 2.0 configuration. 59

Table 3.16 — Acceptance checks for the genotype acquisition and reduction workflow. 63

Table 3.17 — Core derived artifacts for consensus-sequence processing. 65

Table 3.18 — Minimum reproducibility artifacts required for the three pipelines. 65

Table 3.19 — End-to-end provenance of the principal data objects used by the three pipelines..... 66

Table 4.1 — Software modules and their responsibilities..... 70

Table 4.2 — Twenty-two files in the final EGT deployment bundle..... 72

Table 4.3 — Input formats and validation rules implemented by the local application..... 73

Table 4.4 — Final EGT task thresholds used by the CLI and dashboard. 75

Table 4.5 — User-level use case summary. 84

Table 4.6 — Detailed specification for UC-01: Verify a model bundle before inference 84

Table 4.7 — Detailed specification for UC-02: Run batch prediction and export a report..... 85

Table 4.8 — Detailed specification for UC-03: Run and inspect interactive EGT prediction. 85

Table 4.9 — Detailed specification for UC-04: Compare model families and interpret predictions. 86

Table 4.10 — Supported deployment modes. 88

Table 4.11 — Test layers and acceptance criteria. 88

Table 4.12 — Metadata and expected biological classes of the eight golden samples. 89

Table 4.13 — Expected EGT probabilities for the eight golden samples. 89

Table 4.14 — Thresholded golden-sample predictions and software regression result. 90

Table 4.15 — Six real-sample end-to-end prediction scenarios satisfying the minimum five-sample test requirement. 91
Table 4.16 — Functional and negative test cases for CLI/dashboard operation. 91
Table 4.17 — Dashboard scenario test matrix. 93
Table 4.18 — Verify command syntax table. 95
Table 4.19 — Human-readable fields expected from the CLI demo workflow. 96
Table 4.20 — Predict command syntax table 96
Table 4.21 — Recommended CLI output-report schema. 97
Table 4.22 — Information architecture of the dashboard tabs. 99
Table 4.23 — Golden regression record for IRIS_313-10605. 100
Table 4.24 — Golden regression record for IRIS_313-8326. 101
Table 4.25 — Golden regression record for IRIS_313-8386. 101
Table 4.26 — Golden regression record for CX3. 102
Table 4.27 — Golden regression record for IRIS_313-11814. 102
Table 4.28 — Golden regression record for IRIS_313-11797. 103
Table 4.29 — Golden regression record for IRIS_313-8454. 103
Table 4.30 — Golden regression record for B097. 104
Table 4.31 — Traceability matrix connecting use cases, requirements, and tests. 104
Table 4.32 — Human operator handoff checklist. 106
Table 4.33 — Internal EGT inference data flow shared by CLI and dashboard. 106
Table 4.34 — Operational interpretation rules for dashboard outputs. 108
Table 4.35 — Main software failure modes and explicit handling strategy. 108
Table A.1 - Electronic data-file inventory 119
Table A.2 - Dataset and model summary exported by the study pipeline. 120
Table A.3 - Full gene_quality.csv content. 120
Table A.4 - Lu et al. Table S1 primers used in the bacterial-blight study. 121
Table A.5 - Representative excerpt from Lu et al. Table S2 (complete 340-accession file supplied electronically). 122
Table A.6 - First 25 rows of the significant-SNP annotation file (complete file supplied electronically). 123
Table A.7 - Resistance-gene scientific metadata/reference-sequence manifest excerpt. 124
Table A.8 - First 20 matched accessions (complete 327-row file supplied electronically). 125
Table A.9 - Aggregated summary of results_ALL_3branch.csv (complete 171-row result file supplied electronically). 126
Table A.10 - Compact view of Lu et al. Table S4; the complete haplotype pattern is preserved in the electronic spreadsheet. 126

LIST OF ABBREVIATIONS

3K-RG	3,000 Rice Genomes Project
BB	Bacterial blight
BAM	Binary Alignment/Map
BIM	PLINK marker-information file
CLI	Command-line interface
CNN	Convolutional neural network
EGT	Enhanced Genomic Transformer
FAM	PLINK sample-information file
LL	Lesion length
MAF	Minor allele frequency
MCC	Matthews correlation coefficient
MR	Moderately resistant
MS	Moderately susceptible
PLINK	Whole-genome association/genotype toolset
PR-AUC	Area under the precision-recall curve
QTL	Quantitative trait locus
R	Resistant
ROC-AUC	Area under the receiver operating characteristic curve
S	Susceptible
SNP	Single-nucleotide polymorphism
Xoo	Xanthomonas oryzae pv. oryzae

CHAPTER 1. INTRODUCTION

1.1. BACKGROUND AND CHALLENGES

Rice (Oryza sativa L.) is a staple food that supports more than half of the world's population, making the stability of rice production a cornerstone of global food security [4], [2]. Among the many biotic stresses threatening rice yield, bacterial blight, caused by the vascular pathogen Xanthomonas oryzae pv. oryzae (Xoo), is one of the most destructive bacterial diseases of rice in both tropical and temperate regions [1], [4]. Because the pathogen invades the xylem, chemical control is largely ineffective, and the breeding and deployment of cultivars carrying resistance (R) genes is regarded as the most effective and environmentally friendly management strategy [1], [4].

The advent of large-scale sequencing, in particular the 3,000 Rice Genomes Project (3K RGP) [2], [3], has opened the possibility of predicting resistance phenotypes directly from genomic data. Coupled with machine learning and deep learning, genome-to-phenotype prediction promises a rapid screening tool that complements slow, environment-dependent field evaluation [5]. However, applying these methods to bacterial blight resistance raises several domain-specific challenges that, if overlooked, lead to scientifically misleading conclusions:

- Population structure confounding. Rice accessions belong to distinct subpopulations (for example indica and japonica) that differ both in genome-wide composition and in resistance rates. Under a random data split, a model can achieve high apparent accuracy by recognizing the subpopulation rather than the true genetic basis of resistance, so that prediction accuracies from random cross-validation are strongly inflated [11].
- Coordinate drift. Per-accession consensus genomes are reconstructed with insertions and deletions, so chromosome lengths differ across accessions and fixed reference coordinates no longer point to the same biological locus, invalidating coordinate-based gene extraction.
- Small n, high dimension. The number of phenotyped accessions is on the order of hundreds, whereas the number of features ranges from thousands (gene embeddings) to millions (whole-genome SNPs), creating a high risk of overfitting.
- Class imbalance. Resistance rates differ markedly across Xoo strains, with some strains having very few resistant samples, biasing naively trained classifiers.

This thesis addresses these challenges through a design-and-experimentation approach, constructing and comparing three genomic prediction pipelines, with honest, population-structure-aware evaluation as the central methodological principle.

1.2. LITERATURE SURVEY

Genomic prediction. Selection in breeding has evolved from phenotypic selection to marker-assisted selection, and then to genomic selection, in which genome-wide markers are used to predict genetic value; the foundational framework was introduced by Meuwissen, Hayes, and Goddard [5]. While classical genomic selection targets continuous breeding values, disease resistance governed by major genes is naturally framed as a classification problem, increasingly tackled with machine learning and deep learning.

DNA language models. A recent line of work represents DNA using foundation models adapted from natural language processing. DNABERT introduced a BERT-style encoder pre-trained on genomic sequences [6]; DNABERT-2 improved efficiency using byte-pair encoding for multi-species genomes [7]; and the Nucleotide Transformer scaled this paradigm to build robust foundation models for genomics [8]. Plant-specialized variants such as Plant-DNABERT extend these ideas to plant genomes. These models produce embeddings that can be transferred to downstream tasks with limited labeled data.

Genomics of bacterial blight resistance. More than forty bacterial blight R genes (the Xa family) have been characterized. Directly relevant to this work, Lu et al. conducted a genome-wide association study on a diverse panel of 340 accessions from the 3K RGP, phenotyped against four representative Xoo strains — Z173 (C4), GD1358 (C5), V, and PXO339 (P9a) — dissecting resistance loci from lesion-length data [4]. This panel and its phenotypes are the basis of the present study.

The evaluation pitfall. A critical, often underappreciated issue is that genomic prediction accuracies obtained from random cross-validation can be strongly inflated by population and family structure, so that reported accuracy must be interpreted in light of the prediction scenario and evaluation scheme [11]. This observation motivates the population-structure-aware evaluation and the negative-control design at the heart of this thesis.

1.3. OBJECTIVES, SCOPE, AND PIPELINE DESIGN

The objective of this thesis is to design, implement, and experimentally compare three genomic prediction pipelines for rice bacterial blight resistance, and to draw a methodologically honest conclusion about which feature representations and evaluation schemes yield trustworthy performance. All pipelines share the same 3K RGP data and the four-strain phenotypes of Lu et al. [4].

Pipeline A — Multi-representation comparative benchmark. This pipeline contrasts three feature representations: whole-genome SNPs (filtered by minor allele frequency and reduced by variance) [15], embeddings of homology-anchored resistance-gene regions produced by a frozen Plant-DNABERT encoder, and raw one-hot DNA. Gene regions are extracted by homology anchoring with minimap2 to overcome coordinate drift [12]. On each representation,

classical models (Logistic Regression, Random Forest [13], XGBoost [14], SVM) are compared against deep models (MLP, CNN, and a multi-window CNN). Each representation is evaluated under two schemes — random StratifiedKFold and population-structure-aware GroupKFold — to expose inflation caused by population structure [11].

Pipeline B — Controlled negative-control experiment. This pipeline is designed to answer a sharper scientific question: do learned gene-region embeddings (here from DNABERT-2 [7]) capture genuine biological signal, or merely re-encode population structure? It contrasts embeddings of candidate resistance regions (QTL from Lu et al. [4] plus cloned R genes) against embeddings of randomly chosen control regions of matched size on the same chromosomes, alongside a whole-genome SNP baseline, a subgroup-only baseline, and a %N-quality control. The primary target is the continuous lesion length, which avoids the class-imbalance issue.

Pipeline C — Enhanced Genomic Transformer (proposed). This pipeline proposes a model that integrates supervised ANOVA-selected SNPs with frozen Nucleotide-Transformer embeddings of resistance-gene reference sequences used as biological priors [8]. The two modalities are fused through self-attention and cross-attention [9] in a multi-task network that predicts all four strains jointly, trained with a class-imbalance-aware focal loss [10].

The scope is deliberately framed as a software-engineering artifact as well as a scientific study: the pipelines are delivered as a reproducible command-line tool and dashboard, with requirements analysis, design, and a full testing process (Chapter 4).

1.4. THESIS STRUCTURE

The thesis is organized into five chapters:

- Chapter 1. Introduction — introduces the background, literature survey, objectives, scope, and design of the three genomic prediction pipelines.
- Chapter 2. Theoretical Background — presents the biological, genomic, machine-learning, DNA-foundation-model, and evaluation concepts required by the study.
- Chapter 3. Construction, Training, and Experimental Comparison of the Three Pipelines — documents data acquisition, preprocessing, construction, training, evaluation, and comparison of Pipelines A, B, and C.
- Chapter 4. Software Implementation and Testing — describes the xabb CLI, the bilingual web dashboard, deployment procedures, user use cases, and software test scenarios.
- Chapter 5. Contributions, Limitations, and Future Work — summarizes the research contributions, limitations, proposed extensions, and final conclusions.

CHAPTER 2. THEORETICAL BACKGROUND

2.1. OVERVIEW OF RICE BACTERIAL BLIGHT AND RESISTANCE

2.1.1. Rice Bacterial Blight and the Bacterium Xanthomonas oryzae pv. oryzae

Rice bacterial blight is caused by the bacterium Xanthomonas oryzae pv. oryzae (Xoo) and is one of the most serious bacterial diseases of rice, particularly in Asian rice-growing regions [1], [4]. Xoo is a vascular pathogen: it enters rice leaves through the hydathodes at the leaf margins or through mechanical wounds, then multiplies within the xylem vessels, obstructing the transport of water and nutrients. Typical symptoms are water-soaked lesions that begin at the leaf margins and tips, expand, and turn greyish-white as the leaf dries. Because the pathogen colonizes the vascular system, chemical control is largely ineffective [1]. In research, infection severity is quantified by lesion length after artificial clip-inoculation, which is the basis for classifying rice lines as resistant or susceptible.

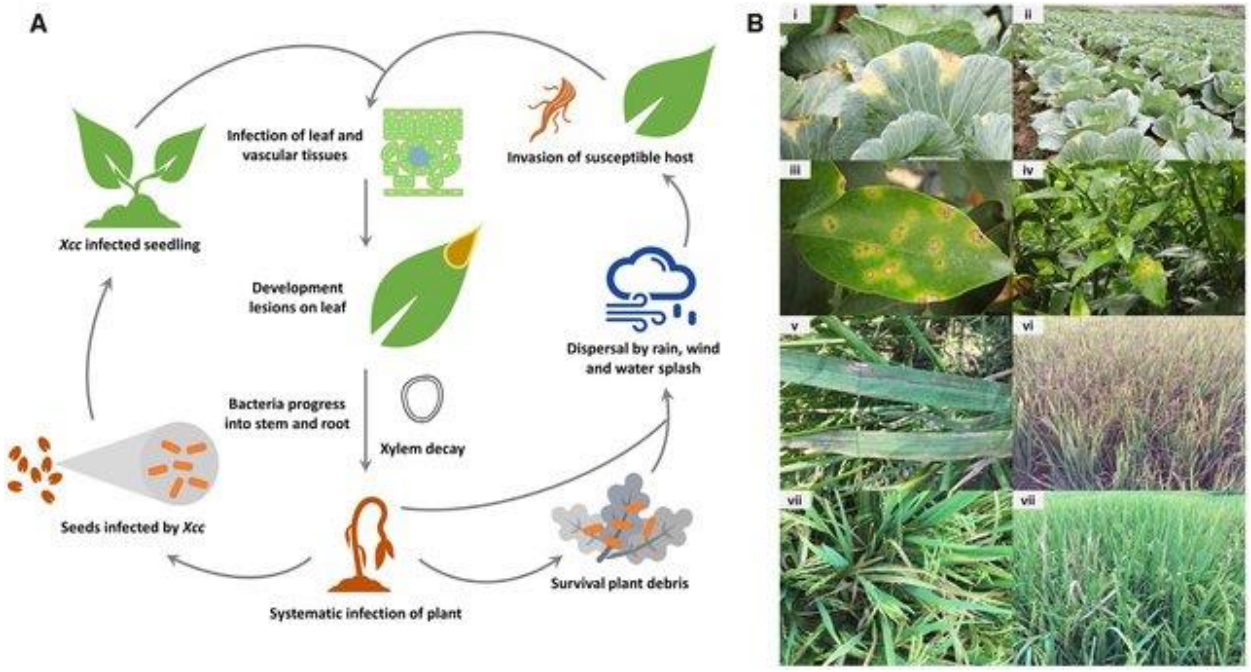


Figure 2.1 - Infection cycle of Xoo.

2.1.2. Resistance Mechanisms and Resistance Genes (the Xa Gene Family)

The most sustainable strategy for controlling bacterial blight is to deploy varieties carrying resistance genes; more than forty bacterial blight resistance genes (the Xa family) have been reported. By convention, dominant genes are capitalized (Xa) and recessive genes are lowercase (xa). Three main mechanistic groups are distinguished. The first group encodes immune receptors or proteins (dominant), for example Xa21 (an LRR receptor kinase), Xa1 (an NBS-LRR protein), and Xa3/Xa26 (LRR-type receptor kinases); these often lie in

repeat-rich NBS-LRR regions. The second group comprises recessive genes related to SWEET sugar transporters: Xoo secretes TAL effectors that bind SWEET promoters to force sugar export, and promoter mutations that block effector binding confer recessive resistance — including xa13 (OsSWEET11) and xa25 (OsSWEET13). The third group comprises other recessive genes such as xa5, which encodes the gamma subunit of transcription factor TFIIA. A key point for this study is that many resistance genes act through promoter variation, so feature extraction must capture the promoter, not only the coding region.

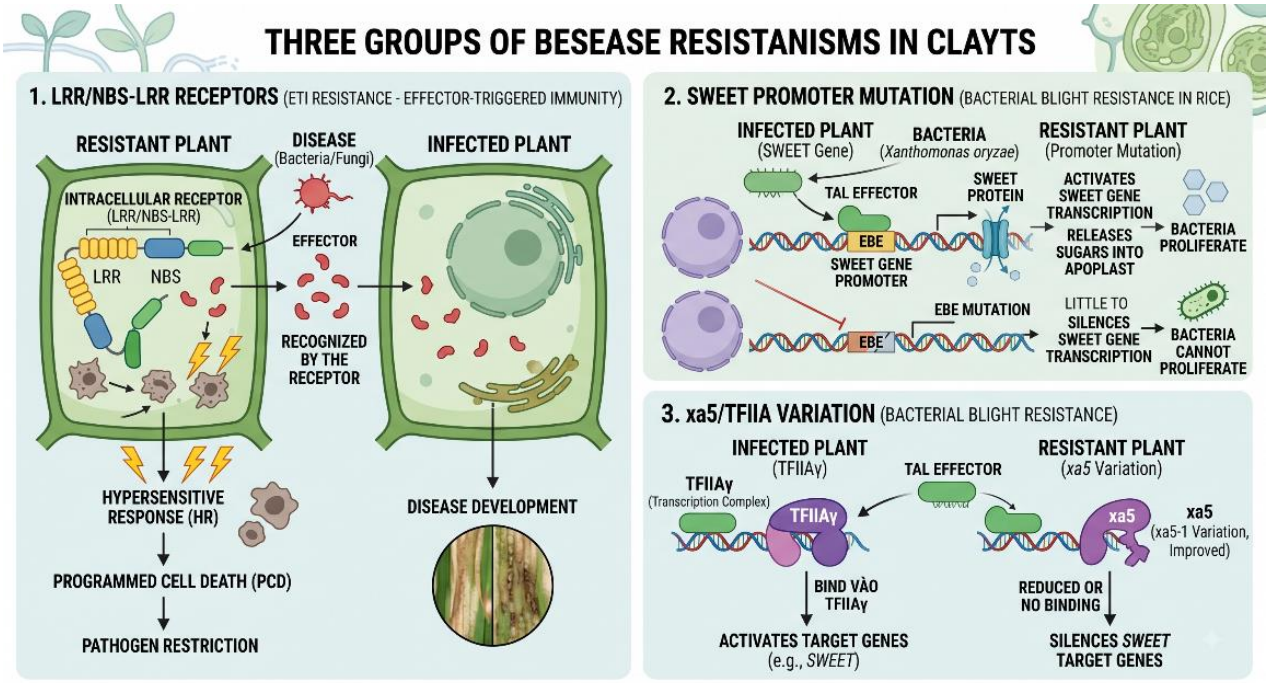


Figure 2.2 - Three groups of resistance mechanisms: LRR/NBS-LRR receptors, SWEET promoter mutations blocking TAL effectors, and the xa5/TFIIA variant.

2.1.3. Xoo Strains and the Phenotype Scale

Xoo exists as many strains, each with its own effector repertoire and virulence spectrum, so a line may resist one strain yet be susceptible to another; resistance must therefore be evaluated separately per strain. Following Lu et al. [4], this study uses phenotypes across four representative Xoo strains — Z173 (C4), GD1358 (C5), V, and PXO339 (P9a) — evaluated on a panel of 340 accessions from the 3,000 Rice Genomes Project. Phenotypes are recorded from lesion length and categorized as R, MR, MS, S, and binarized as R/MR = resistant (1) and MS/S = susceptible (0). Resistance rates differ markedly across strains (Table 2.1); strain V is extremely imbalanced (only 7.9% resistant), motivating the imbalance-aware techniques of Section 2.5.4.

Table 2.1. Distribution of resistant/susceptible samples per strain (actual figures from the dataset, n = 340).

Strain	Resistant(1)	Susceptible(0)	% resistant

C4	171	169	50.3%
C5	66	274	19.4%
V	27	313	7.9%
P9a	173	167	50.9%

2.1.4. Characteristics of Resistance-Gene Regions in the Rice Genome

Predicting resistance from the genome requires accurate extraction of resistance-gene regions from each sample, but these regions pose three difficulties. First, the resistance signal often lies in the promoter, so extraction must include the upstream region. Second, many NBS-LRR genes lie in repeat-rich regions with poor assembly quality and a high proportion of undetermined bases (%N): xa5, xa13, xa25, and Xa21 have low %N (below 8%), whereas Xa1 (~20%), Xa3/Xa26 (~46%), and Xa27 (~51%) are very high and must be excluded. Third, some genes are absent or non-anchorable — for example Xa10 and Xa7 returned no hits across 327 samples, while Xa23 and Xa48 were present only in a subset. The usable panel is therefore much smaller than the theoretical list, and a sequence-based extraction with %N quality control is required (Section 2.3.3).

Table 2.2. Mean ambiguous-base percentage (%N) and initial gene-level quality-control status.

Gene	Mean %N	Initial QC Status
Xa5	0.37%	Retained
xa13 (SWEET11)	0.53%	Retained
xa25 (SWEET13)	3.76%	Retained
Xa21	7.62%	Retained
Xa1	20.35%	Excluded at initial gene-level QC; handled separately at sample level
Xa3/Xa26	45.98%	Excluded
Xa27	51.05%	Excluded

2.2. GENOMIC PHENOTYPE PREDICTION WITH MACHINE LEARNING

2.2.1. From Conventional Breeding to Genomic Prediction

Breeding for resistance has evolved from phenotypic selection — slow, laborious, and confounded by environment — to marker-assisted selection, which screens for a few markers linked to known genes, and then to genomic prediction (genomic selection), in which genome-wide markers predict genetic value; the foundational framework was introduced by Meuwissen, Hayes, and Goddard [5]. Classical genomic selection targets continuous breeding values, but bacterial blight resistance is largely governed by major genes and recorded as discrete levels, so this study frames it as binary classification per strain (and, in Pipeline B, as regression on continuous lesion length).

2.2.2. Principle of a Genomic Prediction Pipeline

A machine-learning genomic prediction pipeline has four stages: data collection (pairing genomic data with a reliable phenotype label); feature representation (converting raw genomic information into numerical form — SNPs, one-hot DNA, or embeddings from DNA language models); model training (learning the map from features to label); and prediction and evaluation. The evaluation scheme is decisive: it must ensure that reported performance reflects genuine predictive ability rather than confounding from data structure, an issue examined in Section 2.5.

2.2.3. The Three Designed and Compared Pipelines

This thesis follows a design-and-experimentation approach with three pipelines that share the 3K RGP data and the four-strain phenotypes of Lu et al. [4]. Pipeline A is a multi-representation benchmark contrasting whole-genome SNPs, Plant-DNABERT embeddings of homology-anchored gene regions, and raw one-hot DNA across classical and deep models, evaluated under both random and population-structure-aware cross-validation. Pipeline B is a controlled negative-control experiment that tests whether gene-region embeddings (DNABERT-2) capture genuine biological signal rather than population structure, by contrasting candidate resistance regions against randomly chosen control regions, with lesion length as the primary target. Pipeline C is the proposed Enhanced Genomic Transformer, which fuses supervised ANOVA-selected SNPs with frozen Nucleotide-Transformer embeddings of resistance-gene reference sequences via cross-attention in a multi-task network.

2.2.4. Domain-Specific Challenges

Four challenges shape all three designs. Population structure confounding: subpopulations differ in both genome-wide composition and resistance rate, so random splits inflate apparent accuracy [11]. Coordinate drift: per-accession consensus genomes carry indels, so fixed reference coordinates no longer point to the same locus. Small n, high dimension: hundreds of samples versus thousands-to-millions of features, a strong overfitting risk. Class imbalance: some strains have very few resistant samples (Section 2.1.3). These motivate, respectively, the population-structure-aware evaluation and negative controls

(Section 2.5), homology anchoring (Section 2.3.3), dimensionality reduction (Section 2.3.2), and imbalance-aware objectives (Section 2.5.4).

2.3. GENOMIC DATA AND FEATURE-REPRESENTATION METHODS

2.3.1. The 3,000 Rice Genomes Project and Consensus Genomes

The genomic data originates from the 3,000 Rice Genomes Project (3K RGP) [2], [3], an international effort that resequenced over three thousand accessions (this study uses 3,024) and provides a public catalogue of genome-wide variation, including roughly 29 million SNPs and 2.4 million small indels across diverse subpopulations [2]. For each accession, a consensus genome is reconstructed by applying that accession-s variants — SNPs and indels — onto the IRGSP-1.0 (Nipponbare) reference, producing a per-accession FASTA over the twelve chromosomes. Because indels are applied, chromosome lengths differ across accessions — a fact with direct consequences for gene extraction (Section 2.3.3).

2.3.2. SNPs, Genotype, and Two Feature-Selection Strategies

A single-nucleotide polymorphism (SNP) is a genomic position where a single base varies among individuals; at a biallelic SNP a genotype is encoded as an allele count (0, 1, 2). The genotype data is stored in PLINK format [15] and contains about 4.82 million SNPs across 3,024 accessions. With only a few hundred phenotyped samples, using millions of features directly is impractical, so the pipelines apply two contrasting selection strategies, both after a minor-allele-frequency filter (MAF > 0.05).

The first strategy is unsupervised, variance-based selection: the SNPs with the highest variance across accessions are kept (top 100,000 in Pipeline A). Because the ranking uses only genotypes and never the labels, it introduces no feature-selection leakage into evaluation. The second strategy is supervised, ANOVA-based selection: for each strain, an ANOVA F-test scores each SNP by how strongly it separates resistant from susceptible samples, and the top-ranked SNPs are kept (2,048 in Pipeline C). Because ANOVA selection uses the labels, it must be computed on the training split only, and this leakage risk must be disclosed when interpreting results.

2.3.3. Coordinate Drift and Homology Anchoring

Extracting a gene by fixed reference coordinates assumes all sequences share one coordinate system. In consensus genomes this fails: indels make chromosome lengths differ, and drift accumulates along a chromosome (near-zero at the start, largest at the end), so a fixed coordinate points to shifted, biologically meaningless regions. The solution is homology anchoring: a reference query sequence for each gene is aligned to each accessions genome with minimap2 [12] (via its Python binding mappy), and the matched region — extended by a flanking margin to include the promoter — is extracted from that accession. Because the query is defined by sequence rather than position, the correct region is recovered

regardless of length changes. In repeat-rich, N-rich regions where edge anchoring fails, a robust variant scatters several short probes across the flanking window and takes the median offset. Each alignment then passes a two-tier %N quality control at the gene and sample levels; regions with excessive %N are excluded, because %N correlates with mapping quality, which in turn correlates with phylogenetic distance — i.e., with population structure itself.

2.3.4. One-Hot Encoding of DNA Sequences

One-hot encoding converts each nucleotide into an orthogonal binary vector — A=(1,0,0,0), C=(0,1,0,0), G=(0,0,1,0), T=(0,0,0,1) — with undetermined positions (N) as an all-zero vector, turning a length-L sequence into a 4xL matrix that preserves the exact order and identity of every base. It is well suited to convolutional networks, whose filters learn short, position-invariant motifs directly from raw bases. In Pipeline A, gene regions are padded or truncated to a fixed length and concatenated into a raw-DNA tensor for the multi-window CNN. The representation discards no sequence information and needs no external model, but requires the model to learn all relevant biology from scratch — demanding when labeled samples are few.

2.3.5. DNA Language Models, Embeddings, and Biological Priors

A more recent approach represents DNA with foundation models adapted from natural language processing: pre-trained on large genomic corpora, they map tokenized DNA into high-dimensional vectors capturing learned sequence patterns. Three such models appear in this thesis. DNABERT [6] is a BERT-style encoder pre-trained on the human genome. DNABERT-2 [7] improves efficiency with byte-pair encoding for multi-species genomes and is used in Pipeline B. The Nucleotide Transformer [8] is a large foundation model (used in Pipeline C), and Plant-DNABERT is a plant-specialized variant used in Pipeline A. In all cases the encoder is kept frozen — with only a few hundred labeled samples, fine-tuning a large model risks severe overfitting, whereas a frozen encoder gives stable general-purpose representations at low cost.

The pipelines use embeddings in two distinct ways. In Pipelines A and B, the model embeds each accessions own extracted gene region (via a sliding window with mean/max pooling) to obtain an individual-level feature vector; in Pipeline A this yields a 768-dimension vector per gene, concatenated with quality indicators into a 3,843-dimension matrix. In Pipeline C, by contrast, the Nucleotide Transformer embeds a small set of resistance-gene reference sequences once, producing fixed embeddings (an 11 x 1,280 matrix) that serve as biological priors — encoded knowledge of what resistance genes look like — injected into the model through cross-attention (Section 2.4.4). The advantage of embeddings is that they inject knowledge learned from vast unlabeled data at far lower dimensionality than raw SNPs; their value depends on the relevance of the pre-trained model to the target organism, which the negative controls of Pipeline B are designed to test.

2.4. MACHINE LEARNING MODELS

2.4.1. Classical Machine Learning Models

Classical models form the baselines of Pipeline A and the reference rows of Pipeline B. Logistic Regression models the probability of resistance as a logistic function of a linear combination of features — a simple, interpretable linear reference. Random Forest [13] is an ensemble of decision trees, each trained on a bootstrap sample with a random feature subset per split; averaging decorrelated trees reduces variance and yields feature-importance estimates. XGBoost [14] is a scalable gradient-boosted tree method in which each new tree corrects the residual errors of the ensemble, widely regarded as state-of-the-art for tabular data. Support Vector Machine [16] separates classes by the maximum-margin hyperplane and, via kernels, models non-linear boundaries, remaining effective when features outnumber samples.

2.4.2. Baseline Deep Learning Models

Pipeline A also evaluates deep models to test whether automatic feature learning outperforms classical models here. The Multi-Layer Perceptron (MLP) is a fully connected feed-forward network operating on flat feature vectors — the simplest deep baseline. Convolutional Neural Networks apply learnable filters that detect local, position-invariant patterns: a 1D-CNN scans along the sequence axis (suited to one-hot DNA), a 2D-CNN treats the input as a grid, and the multi-window CNN (mCNN) uses several kernel sizes in parallel to capture motifs of different lengths simultaneously. With only a few hundred labeled samples, these high-capacity models are prone to overfitting — precisely why the comparison against classical models and the population-structure-aware evaluation (Section 2.5) are essential.

2.4.3. Baseline Deep Learning Models

The proposed model in Pipeline C is built on the attention mechanism of the Transformer [9]. Each element is projected into a query (Q), a key (K), and a value (V); relevance is the scaled dot product of query and key, normalized by softmax into weights that aggregate the values: Attention(Q,K,V) = softmax(QK^T / sqrt(d_k)) V, where d_k is the key dimension. Self-attention is the case where Q, K, V come from the same sequence (each position attends to all others within it); cross-attention is where the query comes from one source and keys/values from another (one representation reads another); and multi-head attention runs several attention functions in parallel in different subspaces. These three variants are the building blocks of the next subsection.

2.4.4. The Enhanced Genomic Transformer (Proposed)

The Enhanced Genomic Transformer (EGT) is the model proposed in Pipeline C. Its central idea is to fuse two complementary sources: individual-level variation (supervised ANOVA-selected SNPs) and biological prior knowledge (frozen Nucleotide-Transformer [8]

embeddings of resistance-gene reference sequences). The architecture proceeds in four stages. First, a multi-scale convolutional encoder processes the SNP profile: following the Inception design [17], parallel convolutional branches with different kernel sizes (plus a dilated branch) capture patterns at multiple scales, and their outputs are recalibrated by a Squeeze-and-Excitation block [18] that weights the most informative channels, turning the flat SNP vector into a sequence of genomic tokens. Second, a learnable positional embedding is added and self-attention [9] lets different genomic regions interact. Third, the resistance-gene prior embeddings are projected into the same space and injected through cross-attention — genomic tokens act as queries, gene priors as keys and values — so the model selectively reads which priors are relevant to each pattern; a feed-forward block with residual connections and layer normalization refines the fused representation, pooled into one genomic latent vector. Fourth, this shared vector feeds four task-specific heads (one per strain), training the model in a multi-task fashion that predicts all strains jointly with a class-imbalance-aware objective (Section 2.5.4). The design goal is not merely higher accuracy but injecting domain knowledge, with cross-attention providing an interpretable link between genomic patterns and known resistance genes.

2.5. MODEL EVALUATION AND POPULATION STRUCTURE

2.5.1. Cross-Validation and StratifiedKFold

Cross-validation estimates generalization by splitting the data into k folds, training on k-1 folds and testing on the held-out fold, then averaging. StratifiedKFold preserves the class ratio in each fold, which stabilizes estimates under imbalance. However, a purely random split lets accessions from the same subpopulation appear in both training and test folds; because subpopulation is predictive of resistance, this yields an optimistic upper bound on performance rather than a realistic estimate for new material.

2.5.2. GroupKFold and Population-Structure Leakage

To obtain an honest estimate, GroupKFold assigns each accession a group — here its population subgroup — and guarantees that no group appears in both training and test folds, so the model is always evaluated on subpopulations it has not seen. This directly targets the central pitfall of genomic prediction: accuracies from random cross-validation can be strongly inflated by population and family structure, so reported accuracy must be interpreted in light of the prediction scenario [11]. The gap between random-CV and GroupKFold performance quantifies how much apparent accuracy is due to leakage rather than genuine resistance signal; a large gap for a representation indicates that it partly encodes population structure. This contrast between the optimistic (random) and honest (GroupKFold) schemes is a central methodological device of Pipelines A and B.

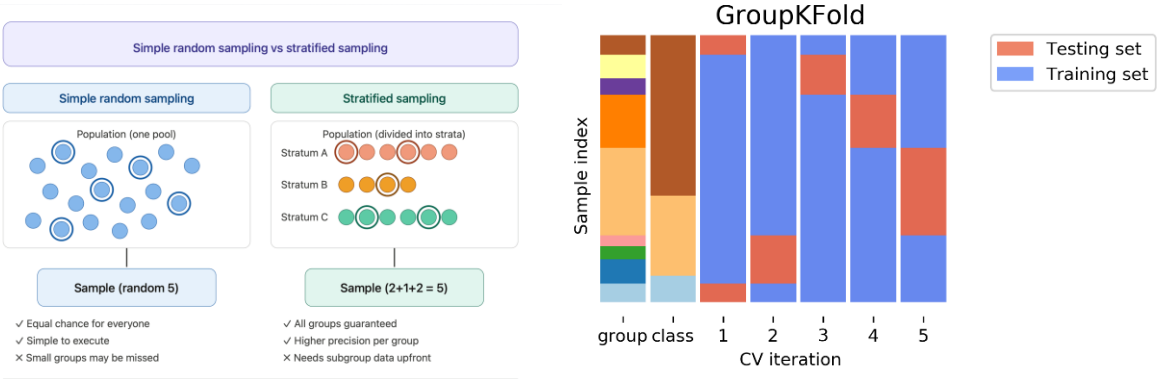


Figure 2.3 - Random StratifiedKFold versus subgroup-aware GroupKFold.

2.5.3. Evaluation Metrics

For binary classification the pipelines report Accuracy, Precision, Recall, and F1, together with the Matthews Correlation Coefficient (MCC) and the Area Under the ROC Curve (AUC). Under severe imbalance, accuracy is misleading — a model can score high by always predicting the majority class — so MCC and AUC, which account for all confusion-matrix cells and are robust to imbalance, are the primary metrics, especially for strain V. Where a decision threshold is needed, it is tuned to maximize MCC. For the continuous lesion-length target of Pipeline B, regression is evaluated by correlation between predicted and observed values (and related error measures), which sidesteps the imbalance issue entirely.

2.5.4. Handling Class Imbalance

Two complementary techniques address class imbalance. Class weighting (via a class weight or positive-class weight in a weighted cross-entropy loss) increases the penalty for misclassifying the minority class, so the rare resistant samples of strain V are not ignored. Focal loss [10] goes further by down-weighting well-classified easy examples through a modulating factor, concentrating training on the hard, minority cases. Pipeline C uses a weighted focal loss combined with per-task weights so that the multi-task objective is not dominated by the easier strains. These techniques change the training objective, not the evaluation metrics, which remain MCC and AUC.

2.5.5. Negative-Control Experimental Design (Pipeline B)

Pipeline B is designed around negative controls to answer a sharp question: do gene-region embeddings capture genuine biological signal, or merely re-encode population structure and assembly quality? The design contrasts several rows per strain: a subgroup-only baseline (population structure alone), a %N-only control (mapping-quality signal alone), whole-genome SNPs and candidate-region SNPs (genotype baselines immune to %N because they are called from 3K RGP data), the embedding of candidate resistance regions (QTL from Lu et al. [4] plus cloned genes), and — crucially — the embedding of

randomly chosen control regions of matched size on the same chromosomes. The strongest conclusion follows only if the candidate-region embedding beats both the random-control-region embedding and the %N-only control; otherwise the embedding is chiefly encoding structure or assembly quality rather than resistance biology. This control-based logic is what allows the pipeline to make a defensible predictive biological claim about the embedding.

2.5.6. Two Evaluation Protocols Across the Pipelines

The three pipelines deliberately use different evaluation protocols, and this must be stated transparently. Pipelines A and B adopt population-structure-aware cross-validation (GroupKFold, contrasted against random cross-validation) as the standard of honest evaluation. Pipeline C, following the protocol of the reference architecture, uses a single 80/20 train/validation split, trains for a fixed number of epochs, and selects epoch and decision thresholds on the validation set; the final numbers are reported on that same validation set and are therefore not an independent test. Consequently, results from Pipeline C are not directly comparable to the GroupKFold results of Pipelines A and B, and any cross-pipeline comparison must account for this difference rather than equating the two kinds of numbers.

2.6. RELATED WORK

This work builds on several strands of prior research. The phenotype data and candidate loci come from Lu et al. [4], who dissected bacterial blight resistance loci in a 340-accession panel from the 3,000 Rice Genomes Project [2], [3]; that project is the source of the genome-wide variation used throughout. The prediction task is framed within the genomic selection paradigm introduced by Meuwissen, Hayes, and Goddard [5]. For feature representation, the pipelines employ DNA foundation models — DNABERT [6], DNABERT-2 [7], and the Nucleotide Transformer [8] — which represent genomic sequences as transferable embeddings. The proposed EGT architecture draws on the Transformer and its attention mechanism [9], the Inception multi-scale convolutional design [17], Squeeze-and-Excitation channel recalibration [18], and focal loss for class imbalance [10]; classical baselines use Random Forest [13], XGBoost [14], and Support Vector Machines [16]. Finally, the methodological core of this thesis — that random cross-validation inflates genomic prediction accuracy under population structure — is grounded in the analysis of Werner et al. [11], which the negative-control design of Pipeline B extends from evaluation into experimental design.

CHAPTER 3. CONSTRUCTION, TRAINING, AND EXPERIMENTAL COMPARISON OF THE THREE PIPELINES

This chapter documents the complete experimental path from public genomic resources to the three resistance-prediction pipelines. The emphasis is deliberately placed on data provenance and preprocessing before model construction because the three pipelines do not consume the same representation of the rice genome. Pipeline A uses genome-wide SNPs, accession-specific resistance-gene loci, and raw one-hot DNA; Pipeline B focuses on candidate resistance/QTL regions and matched negative-control regions; Pipeline C, the proposed Enhanced Genomic Transformer (EGT), uses a study-derived panel of 2,048 selected SNPs together with eleven frozen resistance-gene sequence priors. The chapter therefore treats dataset acquisition, sample linking, sequence reconstruction, quality control, feature construction, model training, evaluation protocol, and reproducibility as parts of one continuous experimental workflow.

All experimental values, tables, and plots reported in this chapter are derived from the project notebooks and exported artifacts. Where a method is inherited from the public literature, the original source is cited. Where a value is generated by this study, it is described explicitly as study-derived rather than as a property of the original 3K-RG release. This distinction is important for the 2,048-SNP EGT panel, the accession-specific consensus sequences, the candidate/control region sets, and the learned model checkpoints.

3.1. Data acquisition and construction of the BB340 analysis cohort

3.1.1. Genomic source: the 3,000 Rice Genomes Project

The genomic backbone of the study is the 3,000 Rice Genomes Project (3K-RG). The project resequenced thousands of Oryza sativa accessions representing broad geographic and population diversity and aligned the reads to the Nipponbare IRGSP-1.0 reference genome [2], [3]. For this thesis, 3K-RG is not treated as a single monolithic file. Instead, the analysis uses two complementary public data products derived from the same sequencing effort: a genome-wide genotype representation and accession-level read alignments. Using both products makes it possible to compare marker-based prediction with sequence-based modeling while keeping the biological cohort as consistent as possible.

The genotype route begins from the filtered 3K-RG SNP collection distributed in PLINK-compatible form. The source used by the notebooks contains 4,817,964 SNPs across 3,024

accessions. The PLINK trio has three complementary responsibilities: BED stores the compact genotype matrix, BIM stores marker metadata including chromosome, physical position, SNP identifier and allele coding, and FAM stores sample identifiers. The BIM file is especially important for later interpretation because every selected EGT SNP must be mapped back to chromosome, physical position, A1/A2 alleles, and the exact model-input order.

The sequence route begins from accession-specific BAM alignments. The original 3K-RG processing aligned sequencing reads to IRGSP-1.0 using BWA-MEM [20]. BAM is the binary representation of the SAM alignment format and stores aligned reads, mapping locations, flags, and alignment metadata [19]. For sequence analyses in this study, these alignments are converted to reference-guided consensus FASTA files with SAMtools. The conversion transforms read-level evidence into an accession-level nucleotide sequence while preserving uncertainty: locations with insufficient evidence can remain N rather than being forced to an unsupported A, C, G, or T call.

3.1.2. Phenotype source and strain-specific disease response

Bacterial-blight phenotype data are taken from Lu et al. (2021), who evaluated a diverse 3K-RG rice panel against four Xanthomonas oryzae pv. oryzae (Xoo) strains: Z173 (C4), GD1358 (C5), V, and PXO339 (P9a) [4]. Disease severity was recorded as lesion length (LL, cm). The original experiment categorized plants as resistant when LL < 5 cm, moderately resistant when 5 <= LL < 10 cm, moderately susceptible when 10 <= LL < 15 cm, and susceptible when LL >= 15 cm. These four biological categories are retained as the provenance of the target definition even when a downstream model uses a binary label.

The phenotyped panel contains 340 accessions and forms the BB340 cohort used throughout the thesis. Because these accessions originate from the same 3K-RG resource, genotype, alignment, and phenotype information can be connected by accession identifier rather than by approximate name matching. The project nevertheless performs explicit identifier checks because accession strings can differ by formatting, aliases, or punctuation between files. Only records that resolve to a unique 3K-RG identifier are propagated into a modeling branch.

3.1.3. Practical acquisition workflow for the genotype branch

The genotype branch is designed so that the original high-dimensional matrix remains

immutable and all study-specific reductions are represented by index lists and exported manifests. The researcher first downloads or stages the filtered 3K-RG PLINK dataset, verifies that the BED, BIM, and FAM files share the same prefix, and checks that the number of markers reported by BIM matches the BED reader. The FAM identifiers are then joined to the 340 phenotyped accessions. This produces the row index used by all later genotype reads. Keeping the row index explicit prevents accidental reordering when the phenotype table is filtered or when subsets are created for training and validation.

1. Stage the PLINK BED/BIM/FAM files from the 3K-RG filtered genotype release in a read-only data directory.

2. Read FAM sample identifiers and normalize the identifier strings used by the phenotype spreadsheet.

3. Join phenotype records to FAM identifiers by 3K_ID and retain only one-to-one matches.

4. Read BIM metadata and preserve original chromosome, physical position, SNP identifier, A1 and A2 coding.

5. For Pipeline A, apply unsupervised SNP filtering; for Pipeline C, defer label-based ANOVA selection until after the train/validation split.

6. Write a manifest for any reduced SNP matrix so that every model input remains traceable to the original BIM row.

3.1.4. Practical acquisition workflow for BAM-derived consensus sequences

The sequence branches require accession-specific nucleotide context rather than only genotype dosages. For each available BB340 accession, the corresponding 3K-RG BAM alignment is obtained and converted to consensus FASTA. The consensus generation step must use a consistent reference coordinate system and must not silently invent bases at low-support positions. The sequence pipelines therefore preserve N characters as an explicit quality signal. This choice later enables the %N quality-control analysis and makes sequence uncertainty visible to both the gene-centric and candidate-region pipelines.

A key practical detail is that the consensus FASTA is not identical in length to IRGSP-1.0. Insertions and deletions accumulated in the accession relative to the reference alter the sequence

length and progressively shift downstream coordinates. Consequently, the project does not use a fixed reference coordinate to cut out a resistance gene from every accession. Reference-gene or region queries are instead re-anchored by sequence homology against each consensus assembly. This design turns the consensus file into an accession-specific search space rather than treating it as a coordinate-preserving copy of Nipponbare.

1. Obtain the accession BAM generated from reads aligned to IRGSP-1.0.

2. Create or use the associated BAM index and confirm that the reference sequence names are consistent.

3. Generate an accession-level consensus FASTA with SAMtools, allowing unsupported positions to remain N.

4. Check the FASTA for the expected chromosome/contig records and record chromosome lengths.

5. Match the consensus accession to the phenotype table.

6. Use homology anchoring, not fixed coordinates, for all downstream gene or region extraction.

3.1.5. Resistance-gene and QTL knowledge used by the sequence and EGT branches

Biological prior information enters the study in two distinct ways. Pipelines A and B use resistance-gene or candidate-region sequences as search queries for homology-based extraction from accession consensus genomes. Pipeline C uses a predefined set of 20 resistance-gene/QTL windows to seed SNP feature selection and eleven fixed reference resistance-gene sequence snippets to construct a frozen Nucleotide Transformer prior. These uses should not be conflated: a genomic window is a coordinate interval used to prioritize SNPs, whereas an NT500M prior token is an embedding derived from a fixed sequence provided to the model as shared biological context.

The EGT resistance/QTL set contains eleven named cloned-gene regions and nine additional resistance/QTL windows. Within each region the selector can seed up to twelve markers before combining them with strain-specific ANOVA selections. The eleven Nucleotide Transformer sequences are encoded once, producing an 11 x 1,280 matrix that is shared across samples. They are therefore reference priors rather than accession-specific gene

reconstructions, and their role is to condition the genomic representation through cross-attention rather than to replace SNP evidence.

3.1.6. Three genomic data streams and accession funnel

The resulting study uses three data streams from a common genomic origin. The genome-wide SNP stream supports marker-based baselines and the EGT selector. The BAM-derived consensus stream supports sample-specific gene and candidate-region extraction. The selected-SNP plus resistance-gene context stream is constructed specifically for the EGT after training-only selection. Table 3.1 summarizes the provenance, content, and coverage of these streams.

Table 3.1 — The three genomic data streams and their sources

Data stream	Source	Content	Coverage
Genome-wide SNPs	3K-RG Filtered SNP Dataset [2]	4,817,964 SNPs	3,024 accessions
Consensus sequences	3K-RG BAM (BWA-MEM to IRGSP-1.0) via SAMtools [19],[20]	Per-accession consensus FASTA	324 usable
Selected-SNP + R-gene context	Study-derived from BB340 + reference R-genes	2,048 ANOVA SNPs + 11 NT500M gene embeddings	340 (BB340)

The three branches do not retain the same number of accessions because their technical requirements differ. SNP and EGT experiments can use all 340 phenotyped accessions because every sample has a genotype row in the working PLINK set. Sequence branches additionally require a usable consensus FASTA, and the Plant-DNABERT branch applies further gene-level quality requirements. The resulting funnel is therefore a methodological consequence of the data representation rather than an arbitrary deletion of samples.

Table 3.2 — Accession funnel from the source resources to the usable set per branch.

Stage	Retained	Used by
3K-RG accessions in genotype release	3,024	Source genotype pool
Phenotyped accessions (BB340)	340	SNP / EGT branches
Matched to usable consensus FASTA	324–327	Sequence branches
Clean after gene-level %N QC	303	Pipeline A embedding branch

3.2. Shared preprocessing and data-quality controls

3.2.1. Phenotype transformation and label semantics

The four lesion-length categories are collapsed into a binary resistance problem by grouping R and MR together and MS and S together. The biological grouping is constant across pipelines; only the numeric polarity differs. Pipeline A historically codes the resistant group as the positive class, whereas the EGT code uses label 1 for susceptibility so that sigmoid outputs can be interpreted directly as P(Susceptible). The thesis therefore reports the biological class names alongside numeric labels whenever thresholds or probabilities are discussed.

Table 3.3 — Lesion-length classification and EGT binary grouping derived from Lu et al. (2021) [4].

Class	Lesion length (cm)	Biological group	EGT binary label
Resistant (R)	LL < 5	Resistant	0
Moderately resistant (MR)	5 <= LL < 10	Resistant	0
Moderately susceptible (MS)	10 <= LL < 15	Susceptible	1
Susceptible (S)	LL >= 15	Susceptible	1

Class balance differs strongly by strain. C4 and P9a are close to balanced, C5 is dominated by susceptible samples, and V has very few resistant accessions. This imbalance changes the interpretation of accuracy and motivates the use of MCC, ROC-AUC, PR-AUC, class weighting, and focal loss rather than relying on accuracy alone. It also explains why some sequence-based experiments exclude V while the multitask EGT retains it with explicit imbalance handling.

Table 3.4 — Class distribution per strain in the BB340 cohort (n = 340).

Strain	Resistant	Susceptible	% resistant	Minority / fold (5-fold)
C4	171	169	50.3%	33.8
C5	66	274	19.4%	13.2
V	27	313	7.9%	5.4
P9a	173	167	50.9%	33.4

3.2.2. Accession matching and prevention of row-order errors

Accession matching is treated as a reproducibility-critical preprocessing step. The

phenotype spreadsheet, FAM sample list, consensus FASTA names, and intermediate feature tables are all reduced to a stable accession identifier before any matrix is assembled. The resulting mapping is saved or kept in memory as an explicit index rather than inferred from file order. This is essential because genotype readers return rows in FAM order, whereas phenotype tables may be sorted by strain response or spreadsheet order.

After the join, every branch checks for duplicate identifiers, unmatched phenotypes, and mismatched sample counts. For supervised feature selection, the training indices refer to positions in this merged cohort, and those positions are converted back to the original BED row indices before genotype chunks are read. This two-level indexing is one of the safeguards that prevents a feature selector from accidentally using the wrong samples.

3.2.3. Coordinate drift and homology-based sequence extraction

Consensus-sequence extraction introduces a coordinate problem that does not occur in the PLINK branch. Reference-guided consensus sequences preserve reference ordering, but insertions and deletions alter the total length of a chromosome. A reference coordinate that correctly locates a gene in Nipponbare can therefore land upstream or downstream of the homologous locus in a particular accession. The error becomes larger toward the end of the chromosome as indel differences accumulate.

The solution used in the sequence pipelines is homology anchoring with minimap2 through the mappy Python interface [12]. A reference resistance-gene or region query is aligned directly to the accession consensus. The best compatible hit or a robust multi-anchor estimate is used to transfer the interval, after which flanking sequence can be added to retain promoter context. This makes locus identity depend on sequence homology rather than on the assumption of a globally fixed coordinate.

Table 3.5 — Alignment stages and their distinct purposes.

Alignment stage	Query	Reference	Tool	Purpose
Upstream (3K-RG)	Accession sequencing reads	IRGSP-1.0	BWA-MEM [20]	BAM generation
In-study	R-gene / candidate-region query	Per-accession consensus	minimap2 / mappy [12]	Homology-based coordinate transfer

3.2.4. %N quality control and why it is biologically relevant

Consensus positions without sufficient read evidence are represented by N. The fraction of N bases in an extracted gene or region is therefore a practical indicator of how much sequence information is genuinely available for that accession. A high %N can arise from low read depth, poor mapping, structural divergence relative to the reference, or failure of the homology anchor. Using such a region without filtering would allow a sequence embedding model to learn missingness patterns rather than nucleotide biology.

Pipeline A applies quality control at three levels. First, individual hits are screened using alignment identity, coverage, %N, and secondary-hit information to avoid obvious paralog ambiguity. Second, genes are evaluated across accessions, and the initial clean-gene set requires an average %N of approximately 10% or less together with sufficient usable samples. Third, Xa1 is handled conditionally at the sample level because the gene-wide average fails the clean-gene threshold even though a subset of accessions has usable Xa1 sequence. Pipeline B applies an analogous rule at the regional level and uses subgroup + %N as a control feature set to test whether an embedding advantage is merely reflecting assembly quality.

Table 3.6 — Sequence quality-control levels applied in the two consensus-based pipelines.

Pipeline	QC level	Threshold / evidence	Action
A	Per-hit	identity / coverage / %N / secondary hits	Flag low-quality or paralog-ambiguous extraction
A	Gene-level	mean %N <= ~10% and sufficient clean accessions	Retain clean gene for embedding
A	Sample-level	per-sample %N	Conditional handling of Xa1
B	Regional	mean %N <= MAX_PCTN	Retain region for embedding

Table 3.7 — Mean per-gene ambiguous-base percentage and retain/exclude decision in Pipeline A.

Gene	Mean %N	Decision
xa5	0.37%	Retained
xa13 / SWEET11	0.53%	Retained
xa25 / SWEET13	3.76%	Retained

Gene	Mean %N	Decision
Xa21	7.62%	Retained
Xa1	20.4%	Sample-level handling
Xa3 / Xa26	46.0%	Excluded
Xa27	51.0%	Excluded
Xa10, Xa7	No hits (0 / 327)	Excluded
Xa23, Xa48	Partial presence	Presence/absence indicator

3.2.5. Feature construction and dimensions

After shared preprocessing, each pipeline constructs a representation suited to its scientific question. Pipeline A deliberately compares three modalities: a broad genome-wide SNP baseline, Plant-DNABERT embeddings of the retained resistance-gene loci, and one-hot raw DNA tensors. Pipeline B narrows the comparison to candidate and negative-control regions so that sequence embeddings can be tested against a matched biological null. Pipeline C combines a small, ordered SNP panel with frozen resistance-gene priors and therefore has a much smaller genomic input dimension than the genome-wide baselines.

Table 3.8 — Feature construction and dimensions for the three experimental pipelines.

Pipeline / branch	Selection / construction	Feature dimension
A — genome-wide SNP	MAF > 0.05, then top 100,000 by variance	303 x 100,000
A — Plant-DNABERT	5 retained gene embeddings + 3 presence/QC indicators	303 x 3,843
A — raw DNA	one-hot A/C/G/T, fixed length	(N, 4, 15,000)
B — candidate embedding	DNABERT-2, 1000/500 sliding windows	~6,144
B — control embedding	matched negative-control regions	~4,608
C — EGT SNP	training-only ANOVA + 20 biological seed windows	340 x 2,048
C — gene priors	NT500M, 11 fixed reference R-gene sequences	11 x 1,280

3.2.6. Evaluation leakage controls and reproducibility constraints

Feature selection is handled differently depending on whether it uses labels. Pipeline A's variance-based SNP reduction is unsupervised and therefore does not use phenotype information. By contrast, the EGT ANOVA F-score explicitly depends on C4, C5, V, and P9a labels. For that reason the F-score scan is restricted to training rows only after the 80/20 split has been fixed. The validation accessions are never used to compute the SNP ranking.

Population structure is a second source of optimistic evaluation. Pipelines A and B therefore report GroupKFold results that keep population subgroups separated across folds in addition to random cross-validation. The EGT experiments retain the teacher-style 80/20 protocol for direct comparison with the historical baseline; because the same validation set is used for checkpoint selection, threshold optimization, and task-weight selection, EGT values are described as validation performance rather than independent test performance. The chapter keeps these protocols separate instead of comparing their numerical scores as if they were interchangeable.

3.3. Experimental design overview

Figure 3.1 summarizes the study as three complementary experiments that share public data sources but answer different questions. Pipeline A asks whether gene-centric sequence representations improve prediction relative to genome-wide SNPs and raw DNA. Pipeline B asks whether the sequence signal is specific to biologically nominated BB regions by comparing candidate embeddings against matched control regions. Pipeline C asks whether a compact SNP representation can be fused with resistance-gene priors in a multitask neural architecture while retaining interpretability.

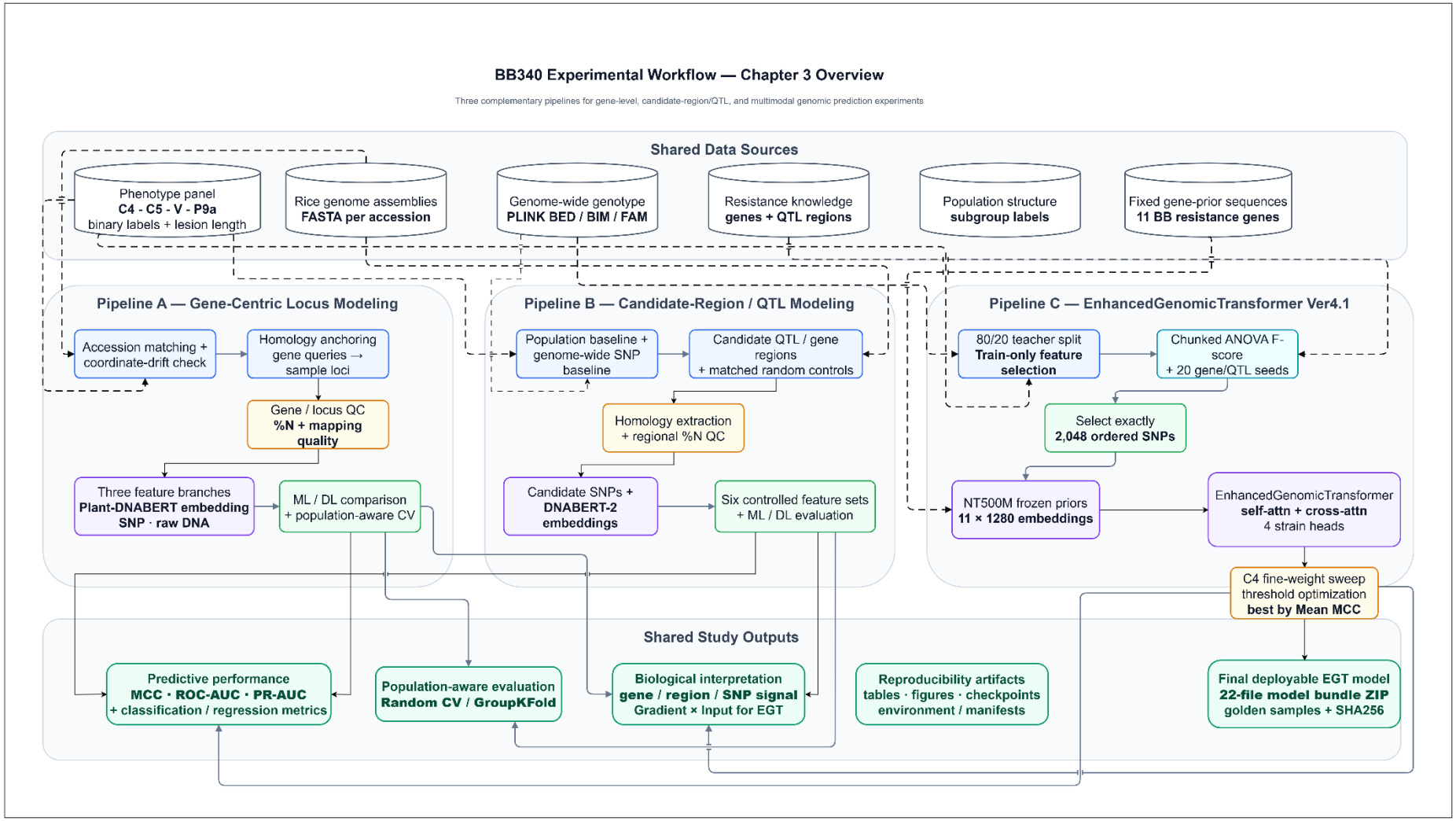


Figure 3.1 — BB340 experimental workflow: shared data sources, three complementary pipelines, and shared study outputs.

The three pipelines also use different standards of evidence. For A and B, GroupKFold is the most conservative measure because it limits population leakage. For B, candidate-versus-control performance is additionally required to support a region-specific biological signal. For C, the controlled ablation of the prior architecture and the signed Gradient x Input analysis provide evidence about how the model uses SNP and resistance-gene context. These differences are intentional and are revisited in the cross-pipeline discussion.

3.4. Pipeline A — gene-centric locus modeling

3.4.1. Purpose and processing flow

Pipeline A is the broadest sequence-versus-marker comparison in the study. It begins from accession-matched phenotype, consensus FASTA, and PLINK genotype data, then reconstructs resistance-gene loci by homology anchoring. The sequence branch applies the %N quality controls described above and produces two representations of the retained loci: Plant-DNABERT embeddings and raw one-hot DNA. In parallel, the genotype branch produces a reduced genome-wide SNP matrix.

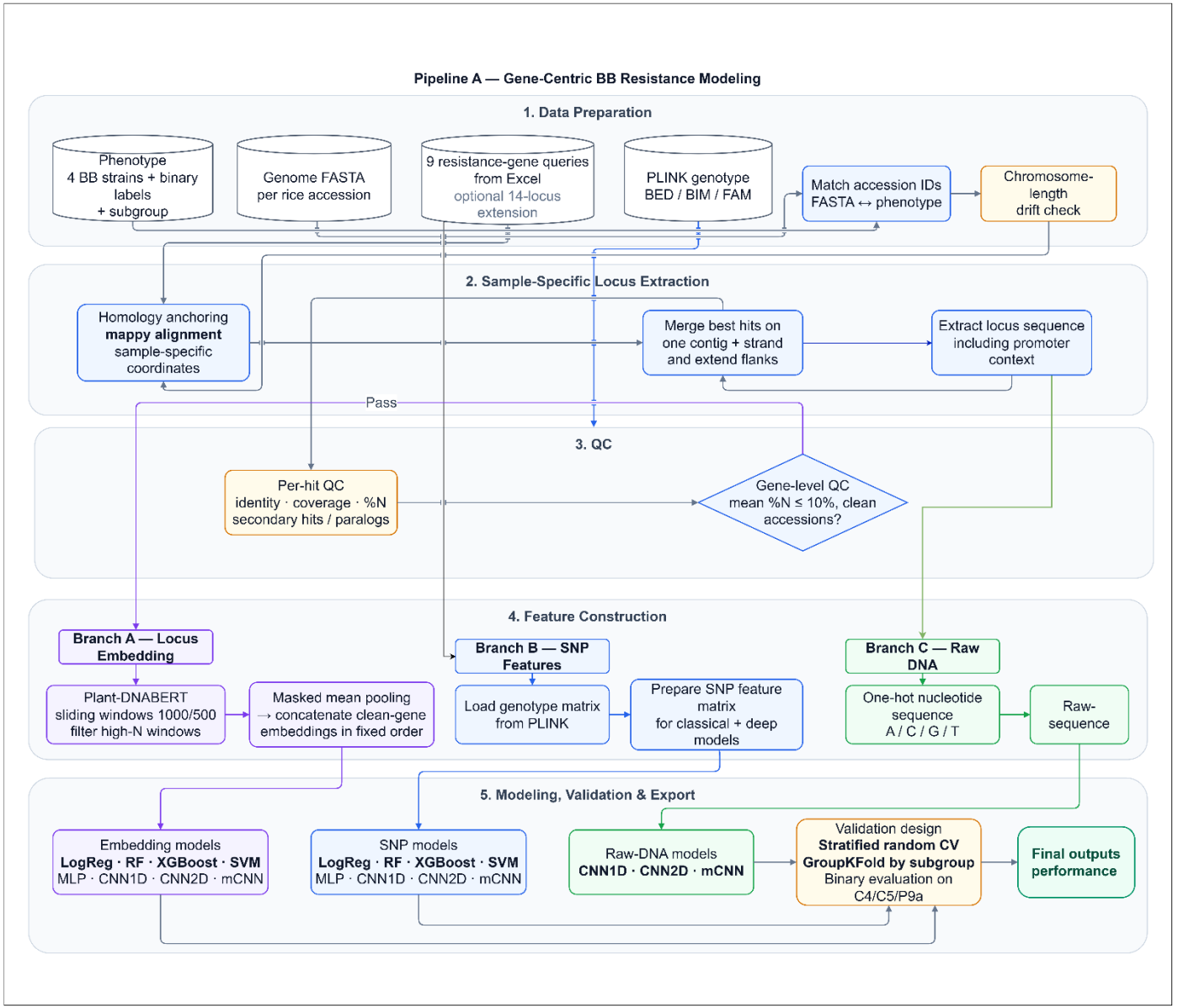


Figure 3.2 — Pipeline A workflow: homology-based locus extraction, gene QC, three feature branches, and population-aware evaluation.

3.4.2. Homology anchoring and clean-gene construction

For each matched consensus genome, reference resistance-gene queries are aligned against the accession sequence. The pipeline records the best compatible hit, orientation, coverage, sequence identity, and %N. Where multiple related hits are possible, secondary-hit information is used to avoid treating a likely paralog as the target locus. The extracted sequence is extended with flanking context where required by the experiment.

The gene-level quality summary is then formed by averaging %N across successful accession-gene extractions and counting the number of usable accessions. The clean set contains xa5, xa13/SWEET11, xa25/SWEET13, and Xa21 under the initial threshold. Xa1 is not treated as uniformly clean; instead, clean Xa1 instances can be used while poor or absent instances are handled separately. This distinction is retained in the report because describing Xa1 as simply 'excluded' would not match the downstream feature construction.

3.4.3. Plant-DNABERT embedding branch

The clean sequence loci are encoded with the Plant-DNABERT model used by the notebook. Long sequences are divided into overlapping windows, high-N windows are removed, and token-level hidden states are pooled into one vector per locus. The five retained/conditionally retained gene embeddings contribute 5 x 768 = 3,840 dimensions. Three presence/absence or QC variables extend the final vector to 3,843 dimensions. This fixed order is treated as part of the model contract: a classifier exported from this branch must receive the same gene blocks and indicator columns in the same order used during training.

The embedding branch is evaluated with Logistic Regression, Random Forest, XGBoost, SVM, and MLP/classifier variants. The classical models provide a useful counterpoint to the deeper sequence models because the dataset has only a few hundred accessions. A stronger architecture is not automatically superior when the effective sample size is small and population structure is strong.

3.4.4. Genome-wide SNP and raw-DNA branches

The SNP branch begins from all 4,817,964 markers. Markers with minor-allele frequency <= 0.05 are removed, leaving 3,315,874 SNPs, and the 100,000 highest-variance markers are retained. Because this ranking does not use the phenotype, it can be computed before cross-validation without label leakage. The resulting matrix is used by classical models and neural

architectures as a genome-wide marker baseline.

The raw-DNA branch preserves nucleotide identity explicitly by one-hot encoding A/C/G/T over fixed-length sequence inputs. It therefore removes the pretrained language-model prior and tests whether convolutional architectures can learn useful motifs directly from the available BB340 sequence data. The raw branch is high-dimensional and sample-limited, so its main role is comparative rather than as the expected best production model.

3.4.5. Validation design and results

Pipeline A is evaluated under both random stratified cross-validation and subgroup-aware GroupKFold, with an additional held-out test analysis in the notebook. The GroupKFold result is the more conservative estimate for the population-structured rice panel because accessions from the same subgroup are less likely to appear on both sides of a split [11]. The gap between random CV and GroupKFold is therefore interpreted as evidence of population-related leakage rather than as random instability alone.

Table 3.9 — Mean GroupKFold MCC by branch and model family, Pipeline A.

Branch	Classical models: mean GroupKFold MCC	Deep models: mean GroupKFold MCC
SNP (genome-wide)	0.200	0.144
Embedding (Plant-DNABERT)	0.183	0.149
Raw one-hot DNA	—	0.182

Table 3.10 — Mean independent-test AUC by branch and model family, Pipeline A.

Branch	Classical models: mean independent-test AUC	Deep models: mean independent-test AUC
SNP (genome-wide)	0.828	0.744
Embedding (Plant-DNABERT)	0.758	0.686
Raw one-hot DNA	—	0.689

The main result is that classical marker models remain strong on this small cohort. The best single GroupKFold result is obtained by a Random Forest on the SNP representation, whereas the sequence embedding and raw-DNA branches are competitive but do not

consistently exceed the genome-wide marker baseline. The result does not imply that the sequence embeddings contain no biological signal; rather, it motivates the more controlled experiment in Pipeline B, where candidate-region embeddings are compared with matched control regions under the same representation.

3.4.6. Pipeline A interpretation and limitations

Pipeline A demonstrates that BB prediction can be learned from multiple genomic representations, but it also exposes the cost of evaluating population-structured genomic data with random folds. The marker branch benefits from genome-wide ancestry information, which can be predictive even when it is not specific to causal resistance mechanisms. The sequence embedding branch is narrower and biologically motivated, but its feature dimension is still large relative to the number of accessions. The branch should therefore be interpreted as a representation benchmark rather than as proof that a particular R-gene sequence causes resistance in the evaluated sample.

3.5. Pipeline B — candidate-region and QTL modeling

3.5.1. Controlled biological question

Pipeline B addresses the principal ambiguity left by Pipeline A: a model can perform well because a genomic representation encodes population structure rather than resistance biology. To separate these effects, the pipeline defines candidate BB regions from QTLs and cloned resistance genes and constructs matched random negative-control regions on the same chromosomes with comparable sizes. Candidate and control regions are then processed through the same extraction, quality-control, embedding, and evaluation procedures.

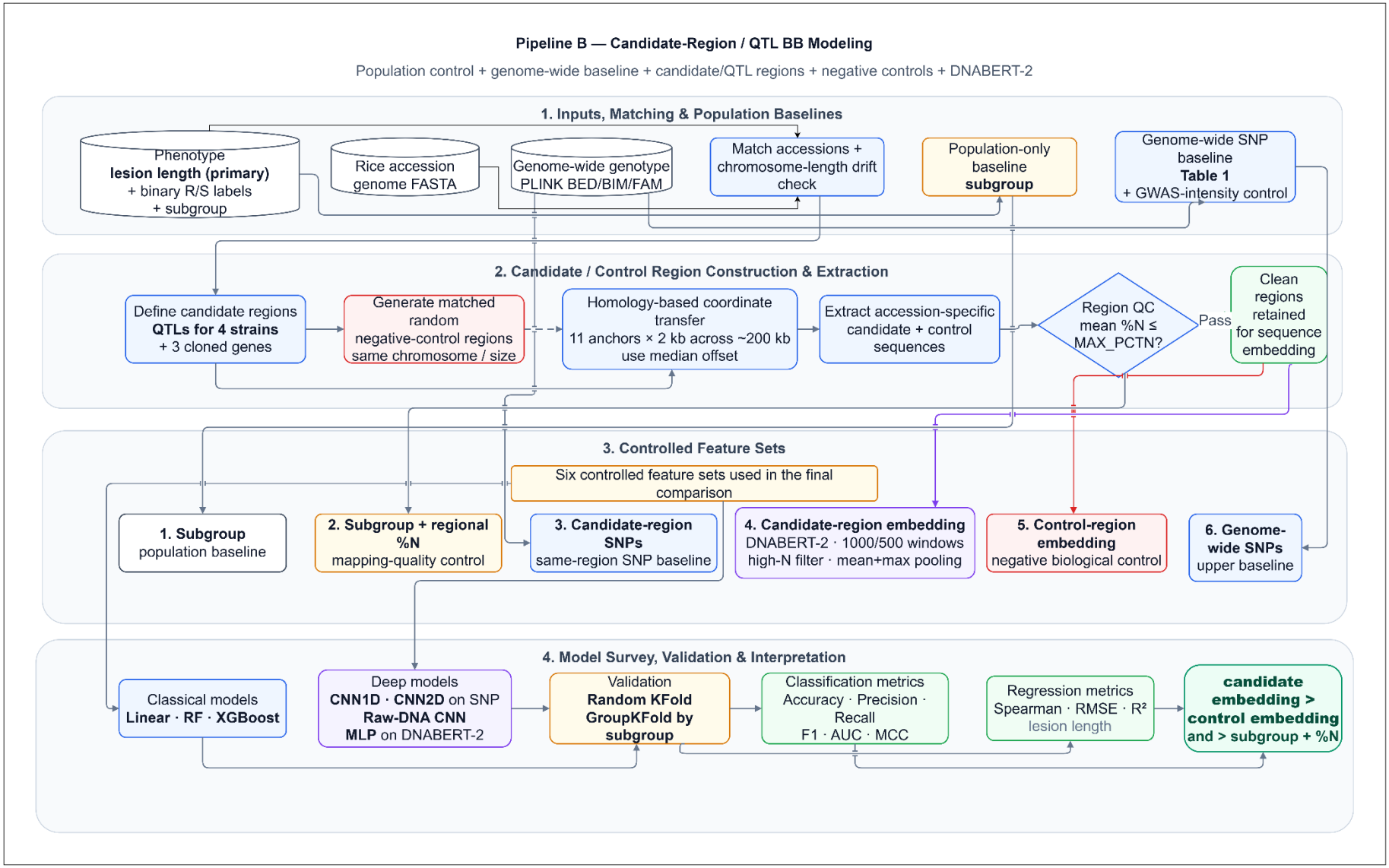


Figure 3.3 — Pipeline B workflow: population control, genome-wide baseline, candidate/QTL regions, matched random controls, and DNABERT-2 embeddings.

3.5.2. Candidate and negative-control region construction

Candidate regions include strain-relevant QTL intervals together with cloned resistance-gene regions. The control generator samples intervals from the same chromosome and matches region size while avoiding the nominated candidate locus. This preserves broad chromosome-level properties while removing the biological nomination. As a result, a candidate-versus-control comparison asks a sharper question than a candidate-versus-random-genome comparison.

For wide intervals, direct edge anchoring can be unstable in repetitive or N-rich sequence. The robust transfer routine scatters multiple short probes across the reference interval, aligns them independently, and uses a median offset to estimate the accession-specific coordinate shift. This reduces the influence of a single failed or paralogous anchor and makes the extraction more tolerant of local sequence gaps.

3.5.3. Candidate/control sequence embedding and SNP baseline

Clean candidate and control sequences are encoded with DNABERT-2 [7]. Long regions are partitioned into overlapping windows, high-N windows are excluded, and window representations are pooled into one fixed-length feature vector per region set. In parallel, candidate-region SNP features are obtained from the genotype matrix so that an embedding can be compared with a marker representation from the same biological region.

The experiment also includes a genome-wide SNP baseline and two population-oriented controls: subgroup alone and subgroup plus regional %N. The subgroup + %N row is particularly important because sequence missingness can correlate with distance from the Nipponbare reference, and therefore with rice population structure. A candidate embedding should outperform this control if it is to be interpreted as carrying sequence information beyond assembly quality.

3.5.4. Six-row controlled comparison

Table 3.11 — GroupKFold MCC by controlled feature row and Xoo strain, Pipeline B.

Feature row	C4	C5	V	P9a
1 — subgroup (baseline)	—	—	—	—

Feature row	C4	C5	V	P9a
2 — subgroup + %N	0.177	0.184	0.122	-0.009
3 — candidate-region SNPs	0.226	0.058	0.122	0.190
4 — candidate-region embedding	0.334	0.301	0.252	0.256
5 — control-region embedding	0.100	0.223	0.225	0.113
6 — genome-wide SNPs	0.239	0.233	0.358	0.451

The candidate-region embedding is the strongest of the region-specific representations for C4, C5, and P9a and is competitive for V. The genome-wide SNP baseline remains strongest for V and P9a, showing that broad genomic information still has predictive value. However, the core biological test is not whether the candidate embedding beats every genome-wide model; it is whether it consistently exceeds an equivalently processed matched control region.

3.5.5. Candidate-versus-control evidence

Table 3.12 — Balanced candidate-versus-control comparison under GroupKFold.

Strain	Candidate embedding: GroupKFold MCC	Control embedding: GroupKFold MCC	Candidate - control
C4	0.300	0.100	+0.201
C5	0.285	0.223	+0.062
V	0.234	0.225	+0.010
P9a	0.244	0.113	+0.131

The candidate-minus-control gap is positive for all four strains. The largest effect occurs for C4, whereas V shows only a small difference. This pattern supports the conclusion that the nominated regions contain resistance-related signal beyond a matched genomic background, but the size of that signal is strain dependent. The result is appropriately framed as predictive biological evidence rather than causal proof, because the embedding aggregates correlated

sequence variation across a region.

Per-region analysis further identifies the xa5 cluster as the strongest candidate region under the reported GroupKFold evaluation. This region-level signal is consistent with the known role of xa5 in BB resistance and provides an interpretable link between the controlled benchmark and established rice resistance biology. Nevertheless, the model attribution and GroupKFold performance do not establish that a particular variant is causal in the studied accession.

3.5.6. Pipeline B limitations

The candidate/control design reduces one major confounder but does not eliminate every source of dependence. Candidate intervals are biologically selected and can differ from random intervals in gene density, local recombination, mapping uniqueness, or allele-frequency spectrum. The matched-control experiment should therefore be read as stronger evidence than an unpaired baseline, not as a randomized causal experiment. The limited BB340 sample size also constrains how finely the candidate intervals can be subdivided without creating unstable estimates.

3.6. Pipeline C — Enhanced Genomic Transformer

3.6.1. Motivation for multimodal fusion

Pipeline C combines two complementary information sources. The SNP branch carries accession-specific genotype variation, whereas the resistance-gene branch provides a fixed biological prior derived from characterized BB resistance sequences. The fusion is implemented with cross-attention so that the genomic representation can query the prior tokens rather than simply concatenating one large context vector. This architecture is intended to retain the predictive value of marker data while making the model explicitly aware of resistance-gene context.

3.6.2. Memory-safe training-only selection of 2,048 SNPs

The original genotype matrix is too large to load as a dense 340 x 4.8-million array in a typical Colab session. The selector therefore scans the BED file in chunks of 100,000 markers and only for the training rows. Within each chunk, missing genotypes are replaced by zero dosage for compatibility with the trained pipeline, very low-variance markers are ignored, and an ANOVA F-score is computed separately for C4, C5, V, and P9a.

Feature selection combines a biological seed and an adaptive statistical allocation. Up to

twelve markers are sampled from each of 20 predefined resistance/QTL windows. In parallel, the highest F-score markers are taken separately for the four tasks with allocations of 550 for C4, 200 for C5, 550 for V, and 200 for P9a. The union is trimmed or filled using the maximum F-score across strains until exactly 2,048 unique markers remain. Finally, markers are sorted by chromosome and physical position. The phrase 'ordered by physical position' is important: this is not an LD-block construction.

After selection, only the 2,048 columns needed by the model are read for the 272 training and 68 validation accessions. A1 allele dosage is normalized by dividing by two. The exact input order, BIM index, chromosome, physical position, SNP ID, allele coding, F-scores, selection reason, and resistance/QTL membership are exported to the final model manifest so that inference does not depend on reopening the training notebook.

3.6.3. Nucleotide Transformer resistance-gene priors

The prior branch uses eleven fixed resistance-gene sequence snippets. Each sequence is tokenized with the Nucleotide Transformer 500M human-reference model using a maximum length of 512 and fixed-length padding, and the final hidden state is mean-pooled to obtain a 1,280-dimensional vector [8]. The resulting 11 x 1,280 matrix is frozen and shared across all accessions. Thus, the prior contributes a stable set of biological tokens while sample-specific information continues to come from the SNP branch.

Although the Nucleotide Transformer was developed as a general DNA foundation model, the thesis uses it only as a frozen encoder for these fixed prior sequences. The foundation model is not fine-tuned on BB340. For deployable inference, the exact 11 x 1,280 matrix is bundled with the downstream EGT, so a local user does not need to download or run the foundation model again.

3.6.4. EGT architecture

The 2,048 SNP dosages are first processed by a one-dimensional convolutional stem and two Inception-style blocks [17]. Each Inception block uses multiple kernel sizes, including a dilated branch, so the network can aggregate local genotype patterns at different receptive-field scales. Squeeze-and-Excitation modules recalibrate channel importance [18]. Pooling compresses the representation to 128 genomic tokens of 64 dimensions, to which a learnable positional embedding is added.

A four-head self-attention layer then mixes information among genomic tokens [9]. In the prior branch, each 1,280-dimensional gene embedding is projected through a small multilayer perceptron to 64 dimensions. Cross-attention uses the genomic tokens as queries and the eleven projected gene priors as keys and values. The fused sequence passes through a residual feed-forward block and is mean-pooled to a 64-dimensional sample representation. Four independent heads produce logits for C4, C5, V, and P9a; after sigmoid transformation these are interpreted as susceptibility probabilities.

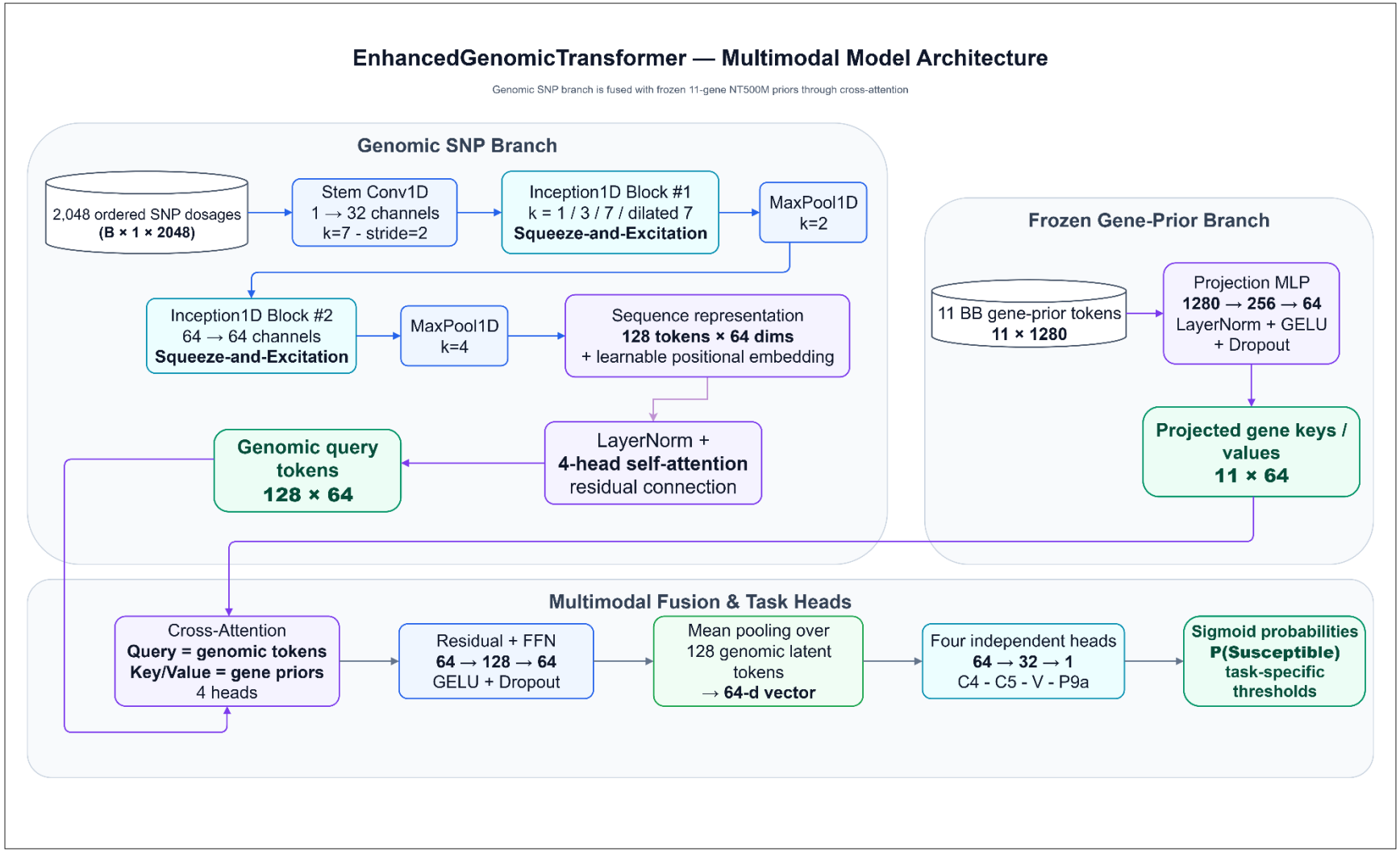


Figure 3.4 — EGT multimodal architecture: the SNP branch is fused with eleven frozen NT500M resistance-gene priors through cross-attention.

3.6.5. Training protocol and C4 task-weight fine sweep

The EGT experiments follow the teacher-style 80/20 protocol: 272 accessions are used for training and 68 for validation with random_state = 42. All configurations are trained for the full 60 epochs without early stopping. Optimization uses AdamW with a learning rate of 3e-4, weight decay 0.02, batch size 8, cosine learning-rate scheduling to 1e-6, and gradient clipping at norm 1. The loss is a task-weighted focal binary cross-entropy with gamma = 1.5 and per-task positive-class weights [10].

To evaluate the supervisor's suggestion to increase task weight, Ver4.1 performs a controlled sweep of the C4 task coefficient while C5, V, and P9a remain at 1.0. The tested C4 values are 1.05, 1.075, 1.10, 1.125, 1.15, 1.175, and 1.20. For every configuration, the validation threshold for each strain is searched from 0.15 to 0.85 in increments of 0.02, choosing the threshold that maximizes MCC. The primary checkpoint and configuration criterion is validation mean MCC.

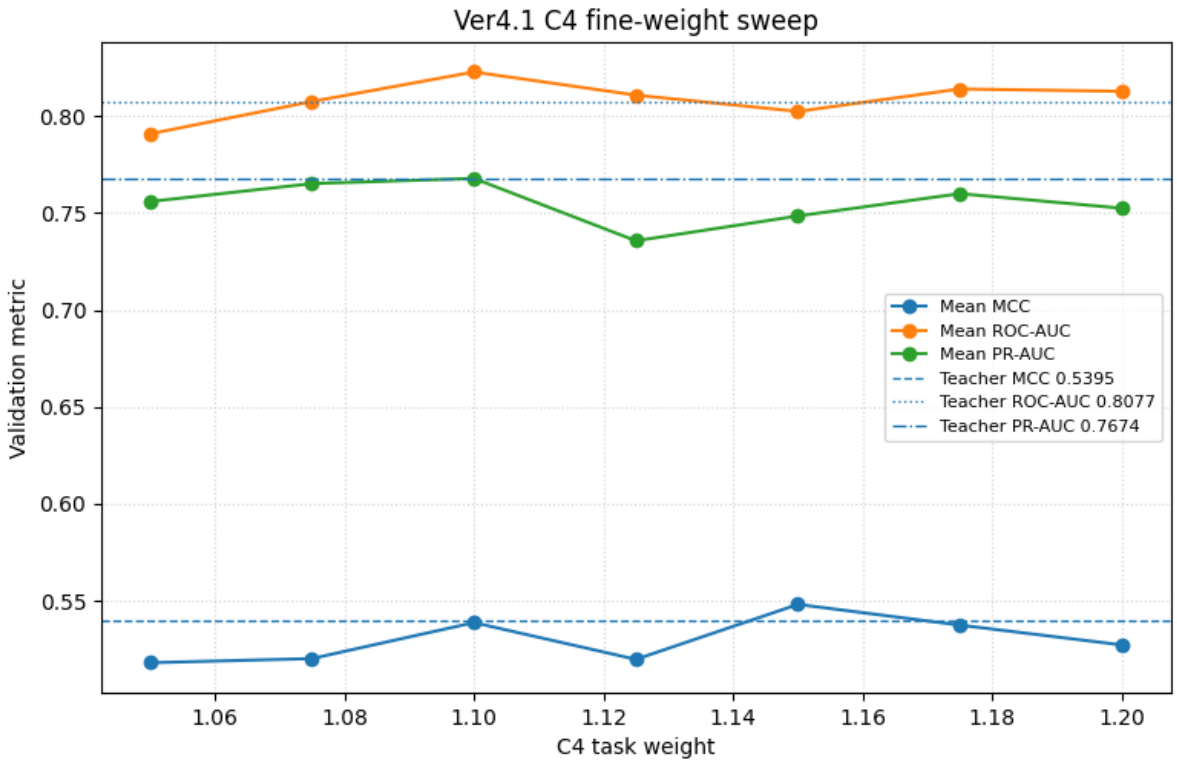


Figure 3.5 — C4 task-weight fine sweep (gamma = 1.5): validation Mean MCC, ROC-AUC, and PR-AUC.

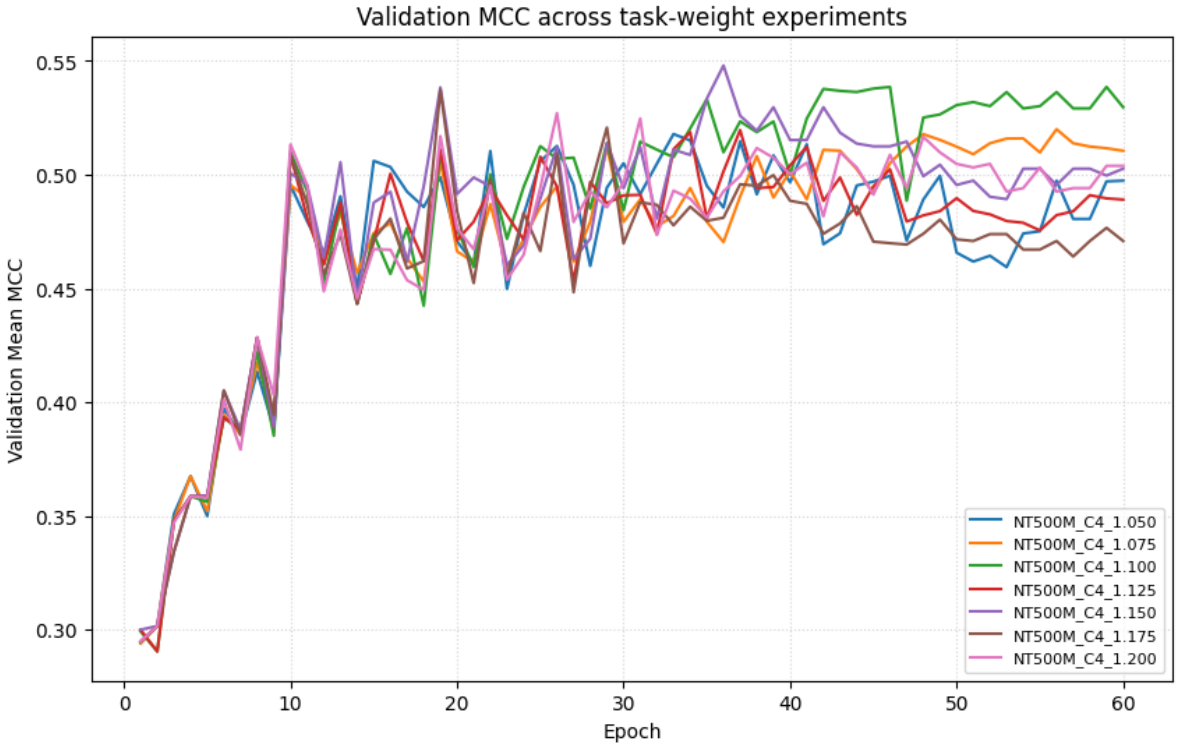


Figure 3.6 — Validation Mean MCC across training epochs for the C4-weight configurations.

The curve is relatively flat across the fine sweep, which indicates that task weighting is primarily a re-balancing mechanism rather than a large source of overall accuracy gain. The best mean MCC is obtained at C4 weight 1.15, epoch 36. Because the same validation set is used to choose the epoch, thresholds, and weight, the result should be reported as the selected validation configuration, not as an unbiased estimate of future generalization.

3.6.6. Best-configuration validation performance

Table 3.13 — EGT validation performance for gamma = 1.5 and C4 task weight = 1.15.

Strain	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC	PR-AUC	MCC
C4	0.74	0.78	0.74	0.73	0.76	0.74	0.73	0.47
C5	0.81	0.97	0.76	0.94	0.85	0.88	0.96	0.63
V	0.90	0.60	0.38	0.97	0.46	0.75	0.43	0.42
P9a	0.84	0.82	0.89	0.77	0.86	0.85	0.87	0.67

The mean MCC of the selected configuration is approximately 0.548. C5 and P9a are the strongest tasks, while V remains the most difficult despite its high accuracy. For V, the high specificity and low recall show why accuracy alone is misleading: a classifier can correctly

label many susceptible samples while still missing a large fraction of the rare resistant class. MCC is therefore used as the primary threshold-based metric, with ROC-AUC and PR-AUC used as complementary ranking measures.

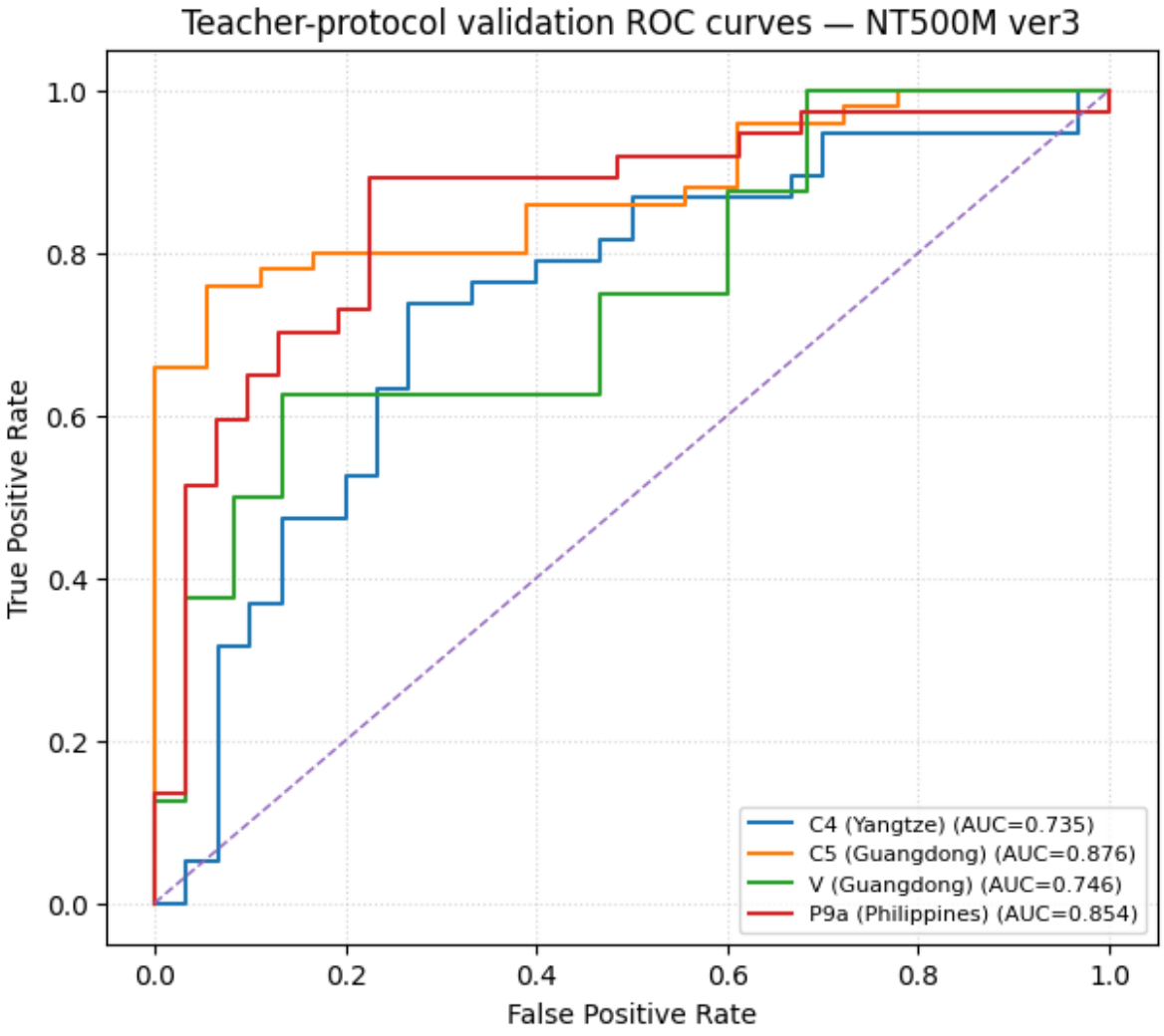


Figure 3.7 — Validation ROC curves for the four EGT strain heads.

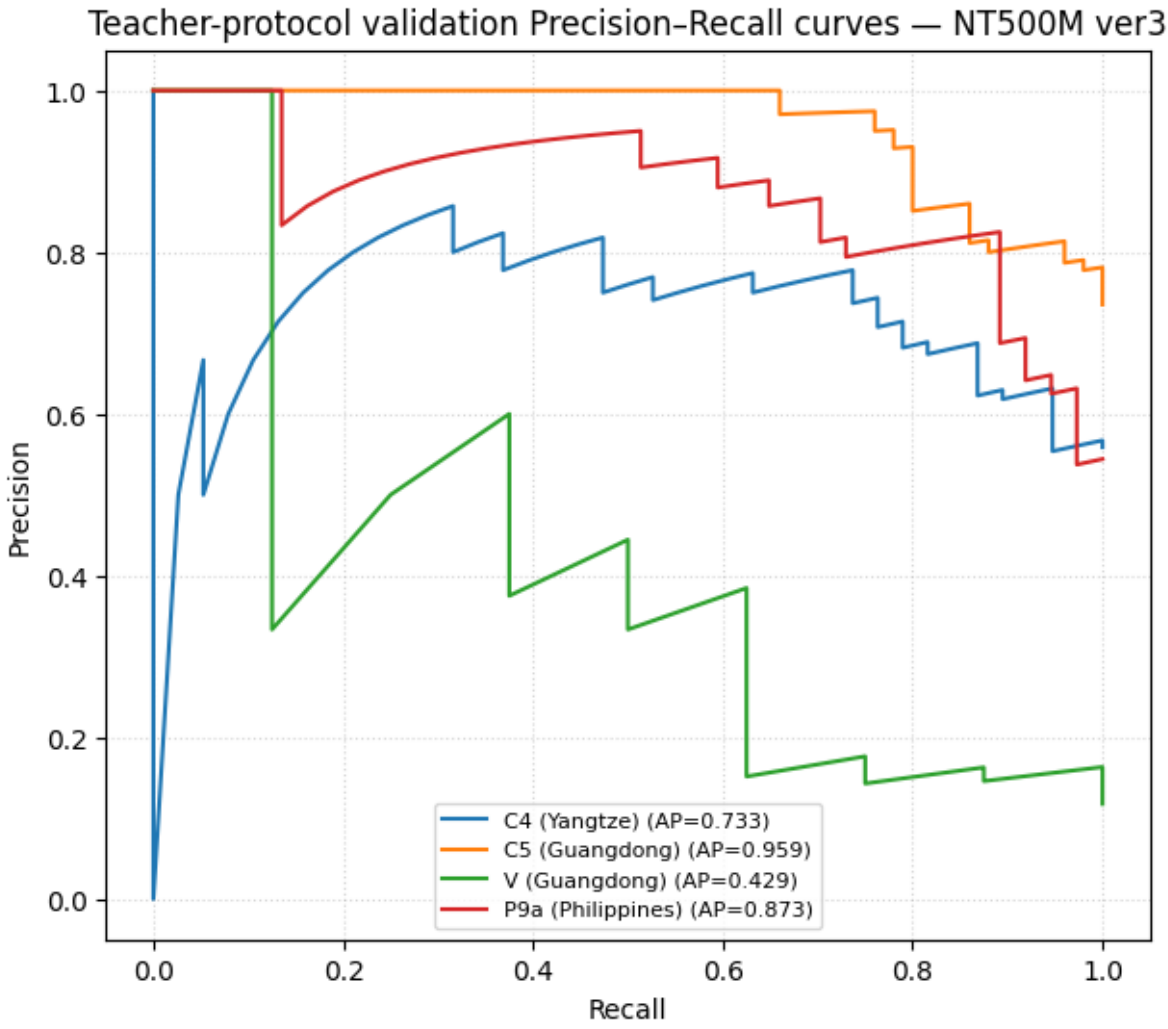


Figure 3.8 — Validation precision-recall curves for the four EGT strain heads.

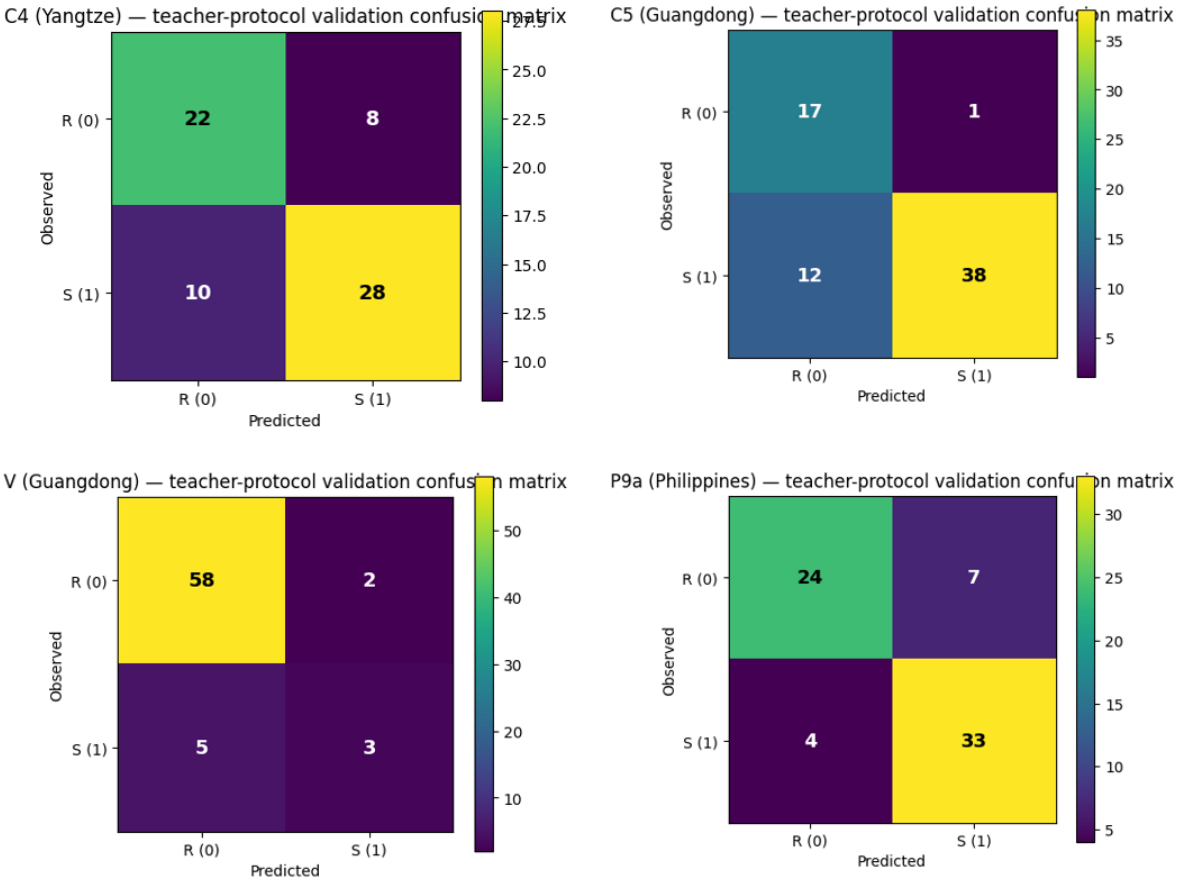


Figure 3.9 — Confusion matrices for the four EGT strain heads.

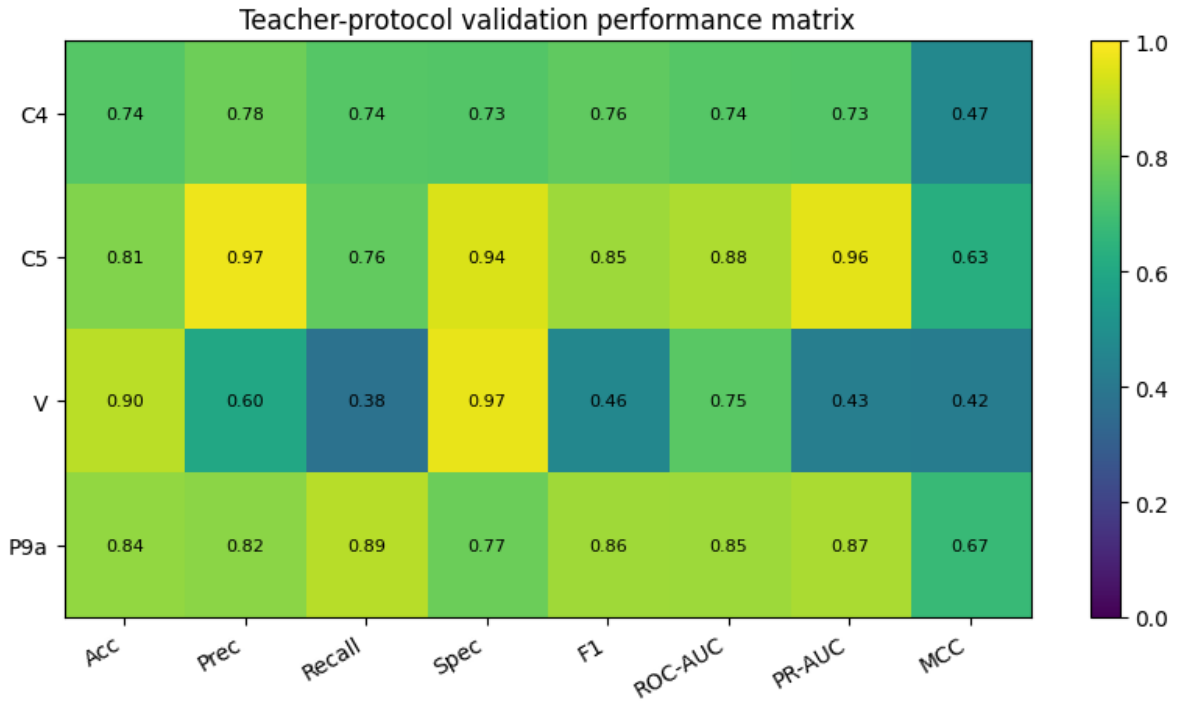


Figure 3.10 — Validation performance matrix for the selected EGT configuration.

3.6.7. Architecture and prior ablation

A separate gamma = 2 experiment compares the full 11-prior cross-attention model with reduced or alternative prior designs under the same 2,048-SNP input. The full NT500M prior model has the highest reported mean MCC, a five-prior version is slightly lower, and concatenating a single context embedding performs worse than cross-attending to individual prior tokens. DNABERT-based alternatives perform below the NT500M variants. Adding ten population-structure principal components also reduces mean MCC, which is used as a negative control against the idea that explicit population components should automatically improve the genomic model.

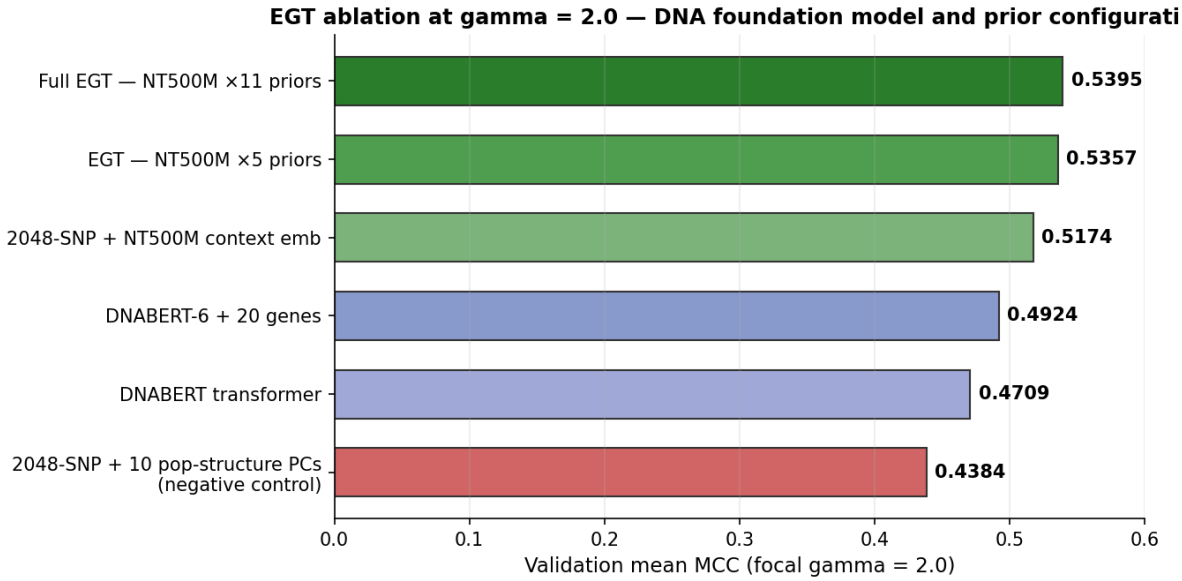


Figure 3.11 — EGT ablation at focal gamma = 2.0: validation mean MCC by DNA foundation model and prior configuration.

Table 3.14 — Architecture/prior ablation ranked by validation mean MCC (gamma = 2.0).

Variant (gamma = 2)	Prior	Mean MCC
Full EGT — 2,048 SNP (20 regions) + 11 priors	NT500M x11	0.5395
EGT — 5 gene priors	NT500M x5	0.5357
2,048 SNP + context embedding (concatenation)	NT500M	0.5174
DNABERT-6 + 20 genes	DNABERT	0.4924
DNABERT transformer	DNABERT	0.4709

Variant (gamma = 2)	Prior	Mean MCC
2,048 SNP + 10 population PCs (control)	NT500M	0.4384

3.6.8. Task weighting versus focal gamma

Task weighting and focal gamma influence the model in different ways. A modestly larger C4 coefficient changes the relative contribution of the C4 loss while preserving the focal treatment of easy and hard examples. Increasing gamma changes the focal weighting of examples within every task. The available experiments indicate that both levers mainly redistribute performance across strains rather than producing a large increase in average MCC.

Table 3.15 — Per-strain MCC for the weight-tuned gamma = 1.5 configuration and the gamma = 2.0 configuration.

Strain	gamma = 1.5, C4 weight = 1.15	gamma = 2.0, uniform task weights	Delta
C4	0.470	0.507	+0.037
C5	0.630	0.583	-0.047
V	0.420	0.493	+0.073
P9a	0.670	0.576	-0.094
Mean	0.548	0.540	-0.008

Interpretation caveat: this is a configuration-level comparison, not a pure one-factor gamma ablation, because the gamma = 1.5 run uses C4 weight 1.15 while the gamma = 2.0 run uses uniform task weights.

3.6.9. Biological interpretation with signed Gradient x Input

The trained EGT is interpreted with Gradient x Input attribution computed separately for each strain. For SNPs, the gradient of the strain output with respect to each normalized SNP input is multiplied by the input value, yielding a signed contribution for the current prediction. Absolute magnitude describes model importance, while the sign is interpreted according to the EGT label polarity: positive values point toward susceptibility (label 1) and negative values toward resistance (label 0).

SNP attributions can be aggregated over the predefined resistance/QTL windows to identify which biological regions contribute most strongly for each strain. Attribution can also be computed over the eleven frozen gene-prior tokens. These plots answer an interpretability

question—what information the trained model is relying on—not a causal genetics question. Accordingly, the report refers to model-important, candidate, or prediction-associated SNPs and regions rather than claiming that the highest attribution is a causal resistance mutation.

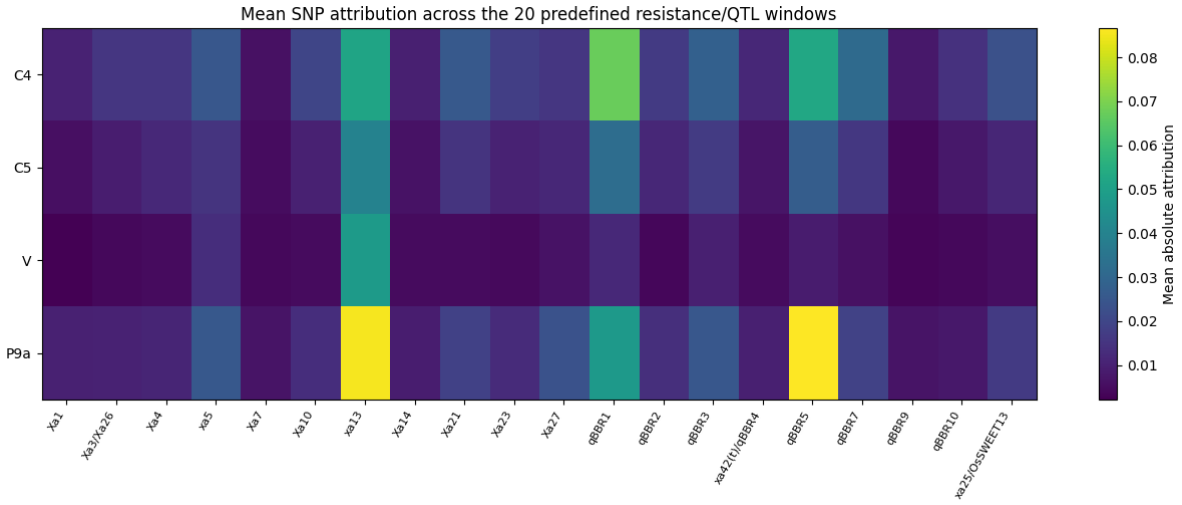


Figure 3.12 — Mean SNP attribution across the 20 predefined resistance/QTL windows, by strain.

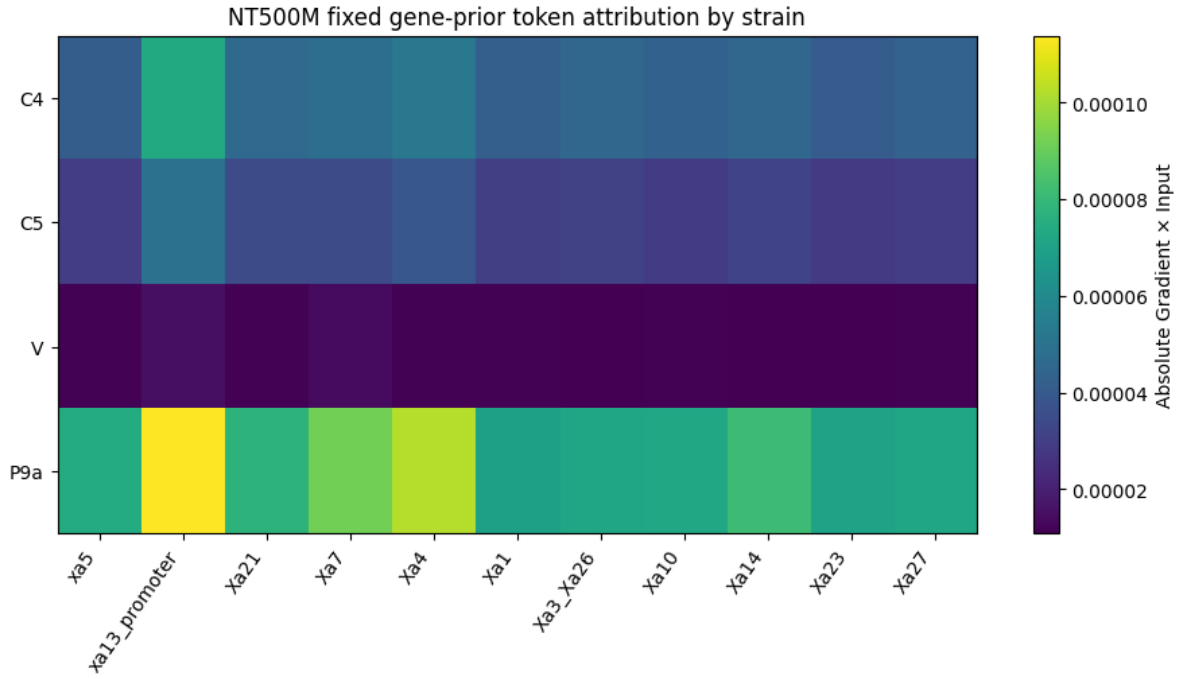


Figure 3.13 — NT500M resistance-gene prior-token attribution by strain.

3.7. Cross-pipeline interpretation

The three pipelines should be read as complementary experiments rather than as a single leaderboard. Pipeline A demonstrates how strongly broad genomic prediction can be influenced by population structure and shows that classical SNP models are difficult to beat on a few hundred samples. Pipeline B then controls the biological question more tightly and finds a

positive candidate-minus-control embedding gap for every strain. Pipeline C moves from representation benchmarking to an integrated multitask model that combines ordered SNP evidence with resistance-gene priors.

The evaluation protocols are not identical. GroupKFold scores from Pipelines A and B are intentionally stricter with respect to population structure, whereas the EGT numbers come from the historical 80/20 teacher protocol. It would therefore be incorrect to conclude that an EGT mean MCC of 0.548 is categorically superior to a GroupKFold value from another pipeline. A future unified benchmark should evaluate all three under the same locked split or repeated subgroup-aware design.

Within their own protocols, the experiments provide a coherent picture. Genome-wide SNP information is a strong baseline; candidate region embeddings show resistance-specific signal beyond matched controls; and the EGT can use a compact 2,048-SNP representation while retaining an explicit biological prior and sample-level interpretability. The strongest methodological contribution is therefore the combination of controlled baselines, leakage-aware preprocessing, and reproducibility assets rather than a single isolated metric.

3.8. Reproducibility outputs from the experimental pipelines

Every pipeline exports intermediate tables instead of leaving important decisions only in notebook state. Examples include accession-match tables, chromosome-length checks, gene/region quality reports, candidate/control definitions, model comparison tables, and EGT feature manifests. The EGT final handoff goes further by packaging the checkpoint, exact ordered SNP manifest, preprocessing contract, fixed prior sequences and embeddings, thresholds, training split, training history, per-sample validation predictions, calibration information, population-aware summaries, golden samples, and hashes.

This export strategy is necessary because a deployable prediction program should not have to infer hidden assumptions from a training notebook. The local CLI and dashboard described in Chapter 4 consume the exported assets directly and therefore treat feature order, allele orientation, missing-value handling, threshold semantics, and biological priors as explicit model metadata.

3.9. Chapter summary

Chapter 3 established the full experimental chain from 3K-RG data acquisition to three

complementary BB resistance pipelines. The data-processing sections showed why genotype and sequence branches require different preprocessing and why homology anchoring and %N control are essential for consensus-derived loci. Pipeline A compared broad marker, embedding, and raw-DNA representations; Pipeline B demonstrated that candidate-region embeddings outperform matched negative controls; and Pipeline C fused 2,048 training-selected SNPs with eleven NT500M resistance-gene priors in the proposed EGT. The next chapter moves from experimental notebooks to software operation: packaging the trained assets, building a command-line interface and bilingual web dashboard, and verifying inference with real golden samples.

3.10. Operational data-acquisition and preprocessing checklist

The scientific descriptions in Sections 3.1–3.2 can be translated into a practical acquisition checklist that makes the study reproducible for another researcher. The principle is to retain the original public resources unchanged and place every study-specific transformation in a separate derived-data directory. This avoids a common source of ambiguity in computational genomics, where a reduced marker matrix or a consensus FASTA is later mistaken for an original consortium release.

3.10.1. Recommended project data layout

A reproducible local workspace separates source data, derived sequence data, model inputs, and experiment outputs. The PLINK source prefix is kept intact; phenotype spreadsheets are stored as source documents; BAM-derived consensus FASTA files are treated as derived resources; and every pipeline writes to its own timestamped or versioned output directory. The exact directory names may differ between machines, but the logical separation should remain.

```
project_root/
  data_source/
    genotype/
      base_filtered.bed
      base_filtered.bim
      base_filtered.fam
    phenotype/
      Lu_2021_supplementary.xlsx
    bam/
      <accession>.bam
      <accession>.bam.bai
  data_derived/
    consensus_fasta/
    gene_loci/
```

```
  candidate_regions/
pipeline_A/
pipeline_B/
pipeline_C_EGT/
deployment/
```

The source/derived distinction also simplifies integrity checking. If a derived file is deleted, it can be regenerated from the documented source and preprocessing parameters. If a source file changes, the project should treat the change as a new data version because marker indices, accession availability, or consensus sequences may differ.

3.10.2. Genotype acquisition acceptance checks

1. Confirm that BED, BIM, and FAM share the same prefix and that all three files are non-empty.

2. Read the FAM file and verify that accession identifiers are unique before joining to phenotype records.

3. Read the BIM file and verify chromosome, base-pair position, SNP ID, A1, and A2 fields. These values become part of the EGT deployment manifest.

4. Open the BED matrix with the intended dosage orientation and record whether dosage counts A1. Do not switch allele orientation after model training.

5. Confirm the marker count against the expected filtered release and record the exact count in the run configuration.

6. Perform all label-based marker selection only after the training split is fixed; preserve the selected original BIM indices.

7. For any reduced matrix, export a mapping from reduced column order back to the original BIM row so the model can be reconstructed without relying on notebook state.

Table 3.16 — Acceptance checks for the genotype acquisition and reduction workflow.

Check	Expected evidence	Failure consequence
PLINK trio present	BED/BIM/FAM with common prefix	Genotype matrix cannot be interpreted reliably
Unique sample IDs	No duplicate FAM accession identifiers	Phenotype row may map to wrong genotype
BIM metadata available	chr, bp, ID, A1, A2	Selected SNP cannot be reproduced or harmonized

Check	Expected evidence	Failure consequence
Dosage orientation fixed	A1-count semantics recorded	0/1/2 values may be biologically inverted
Marker count recorded	Run config stores exact SNP count	Different dataset versions become indistinguishable
Training-only ANOVA	Selector receives train rows only	Validation leakage inflates model selection
Reduced-column manifest	Model order -> original BIM index	Local deployment must guess feature order

3.10.3. BAM and consensus generation acceptance checks

For sequence modeling, the most important reproducibility issue is not simply whether a FASTA file exists, but whether it was created from the intended accession alignment and reference naming convention. The consensus-generation record should therefore retain the accession identifier, source BAM path or source URL, reference version, SAMtools version, command parameters, and output checksum. A sequence that cannot be traced back to the correct BAM should be excluded rather than assigned to an accession by filename similarity.

1. Verify that the BAM header contains the expected IRGSP-1.0 reference sequence names and lengths.

2. Index the BAM when required and confirm that random access succeeds.

3. Generate consensus using the documented SAMtools procedure while allowing unsupported sites to remain N.

4. Record the total length of each consensus chromosome/contig and compare it with the reference to quantify coordinate drift.

5. Normalize the output FASTA header so that the accession can be linked unambiguously to BB340.

6. Run homology anchoring for a small set of known loci as a smoke test before processing all genes/regions.

7. Compute %N at the extracted locus/region level and retain the QC table as an experiment artifact.

Table 3.17 — Core derived artifacts for consensus-sequence processing.

Artifact	Minimum fields to retain	Used by
consensus manifest	accession, BAM source, reference, sequence file, checksum	Pipelines A and B
chromosome-length check	accession, chromosome, reference length, consensus length, delta	Coordinate-drift diagnosis
homology extraction report	accession, target, contig, strand, identity, coverage, %N, status	Gene/region extraction
gene_quality.csv	target gene, mean %N, usable accession count, clean flag	Pipeline A
region_quality.csv	candidate/control region, mean %N, clean status	Pipeline B

3.10.4. Pipeline-specific reproducibility artifacts

A final reproducibility review should be able to answer three questions without executing a notebook: which samples were used, which features were used, and which model configuration produced the reported result. The following table lists the minimum artifacts that satisfy these questions for each pipeline.

Table 3.18 — Minimum reproducibility artifacts required for the three pipelines.

Pipeline	Sample-definition artifact	Feature-definition artifact	Model/evaluation artifact
A	matched accession table and train/test/CV splits	100k SNP index; 3,843-d feature order; raw-DNA locus order	model/scaler files, GroupKFold results, independent-test results
B	matched consensus/phenotype table and subgroup labels	candidate/control region definitions, quality report, embedding order	six-row benchmark, candidate-control comparison, per-region scores
C / EGT	exact 272/68 split and sample IDs	2,048 SNP manifest + 11 fixed prior sequences/embeddings	checkpoint, thresholds, history, scorecard, predictions, hashes, golden fixtures

This artifact-centered approach also supports later software development. The Chapter 4 application does not need the experimental notebooks because the EGT handoff already contains the model state, ordered marker contract, fixed priors, thresholds, and regression fixtures. The same principle can be applied to Pipeline A by exporting a similarly complete 3,843-dimensional Plant-DNABERT bundle.

3.11. End-to-end data provenance from public source to model-ready input

A central requirement of this thesis is that a reader can distinguish original consortium data from study-derived representations. Table 3.19 makes that provenance explicit by following each data object from its external source through the transformation performed in this study to the final object consumed by a model. The same accession can therefore appear as a genotype row, a consensus FASTA, a gene-locus sequence, a candidate-region embedding, or a reduced 2,048-SNP vector without implying that these objects were all distributed by 3K-RG.

Table 3.19 — End-to-end provenance of the principal data objects used by the three pipelines

Source object	Study transformation	Derived object	Consumed by
3K-RG filtered PLINK genotype	ID matching + MAF/variance reduction	100,000-SNP matrix	Pipeline A SNP models
3K-RG filtered PLINK genotype	80/20 split + train-only ANOVA + 20-region seeding	2,048 ordered SNP matrix	Pipeline C EGT
3K-RG accession BAM	SAMtools reference-guided consensus	Per-accession consensus FASTA	Pipelines A and B
Consensus FASTA + R-gene query	minimap2/mappy homology anchoring + flank extraction	Accession-specific gene locus	Pipeline A embedding/raw-DNA
Consensus FASTA + candidate/QTL interval	multi-anchor coordinate transfer + %N QC	Accession-specific candidate/control region	Pipeline B
Clean gene loci	Plant-DNABERT embedding + pooling	3,843-d feature vector	Pipeline A exported classifiers

Source object	Study transformation	Derived object	Consumed by
Clean candidate/control regions	DNABERT-2 window embedding + pooling	Candidate/control embedding vector	Pipeline B
11 fixed reference R-gene sequences	NT500M tokenization + frozen mean pooling	11 x 1,280 prior matrix	Pipeline C cross-attention

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3.11.1. Provenance trace for Pipeline A

Pipeline A starts with two independent source branches that meet only after accession matching. The marker branch reads genotype from the original filtered PLINK dataset. The sequence branch begins from BAM-derived consensus FASTA and does not reuse the PLINK matrix to reconstruct nucleotide sequence. Homology anchoring then converts a reference resistance-gene query into an accession-specific locus, and %N QC determines whether that locus is suitable for embedding or raw-DNA encoding.

The 3,843-dimensional Plant-DNABERT input is therefore a derived feature table, not a raw sequence file. It is reproducible only if the gene order, embedding model, pooling method, and presence/absence variables are preserved. The deployment code correctly requires exactly 3,843 features for this reason. A future user who wants to classify a new accession must either reproduce the upstream homology-and-embedding procedure or provide a feature table generated by an equivalent validated pipeline.

3.11.2. Provenance trace for Pipeline B

Pipeline B adds one more layer of study-derived design: candidate and control regions. The regions are not an external benchmark dataset. Candidate intervals are assembled from resistance/QTL knowledge, while control intervals are sampled by the study to match chromosome and approximate length. Both interval sets are transferred to each accession consensus and pass through the same %N filter and DNABERT-2 embedding process.

Because the candidate and control embeddings share the same sequence source and model architecture, their performance difference can be interpreted more cleanly than a comparison between unrelated feature pipelines. The provenance table should therefore be retained with any exported Pipeline B result so that the negative-control intervals cannot later be mistaken

for published BB loci.

3.11.3. Provenance trace for Pipeline C

Pipeline C has the strictest feature provenance requirement because its 2,048 columns are selected with phenotype information. The marker panel exists only after the fixed 80/20 split and the training-only F-score calculation. The exact list is consequently tied to the run configuration, sample split, label conversion, biological seed windows, and tie-breaking/order rules. Re-running selection after changing any of these inputs can produce a different 2,048-SNP panel even when the starting PLINK dataset is unchanged.

For deployment, this study avoids re-running the selector. The selected panel is exported once and becomes part of the model identity. External data must be harmonized to that panel rather than using a newly selected set. The same logic applies to the eleven NT500M prior embeddings: the bundle carries the exact frozen matrix used by the checkpoint, so the local application is insulated from later changes to the online foundation-model repository.

3.11.4. Data-processing traceability checklist for the thesis archive

1. Archive the original phenotype spreadsheet without editing it in place.

2. Record the exact PLINK prefix, marker count, and sample count used by the genotype experiments.

3. Archive the accession-matching table that defines BB340 row order.

4. Archive the consensus-generation manifest or the final consensus FASTA checksums used by sequence branches.

5. Archive gene_quality.csv and region_quality.csv together with the thresholds used to produce their clean flags.

6. Archive every reduced-feature manifest: 100k SNP indices, 3,843-d feature order, candidate/control region order, and 2,048-SNP EGT order.

7. Archive model hyperparameters, seeds, split indices, thresholds, and the selection criterion used to name the best checkpoint.

8. Archive per-sample predictions and not only aggregate metrics so that figures and tables can be regenerated.

9. Archive the deployable bundle and its SHA256 so that Chapter 4 can be reproduced independently of the notebooks.

This checklist turns the experimental chapter into an auditable computational record. It also clarifies the boundary between data acquisition, preprocessing, model development, and deployment, which is essential when the thesis is later converted into software documentation or a reproducibility package.

CHAPTER 4. SOFTWARE IMPLEMENTATION AND TESTING

This chapter describes how the trained research artifacts are converted into a reproducible local application. The implementation has two user-facing interfaces: a command-line interface (CLI) intended for scripted and batch operation, and a bilingual Streamlit web dashboard intended for interactive inspection, demonstration, and thesis defense. Both interfaces are designed around the same principle: inference must use exported model assets directly and must not retrain the model or reconstruct hidden preprocessing rules from the training notebook.

The primary live model is the BB340 EnhancedGenomicTransformer. The dashboard also exposes the Plant-DNABERT branch as a second model family for benchmark and, when its exported artifacts are available, direct inference. The software therefore separates interface logic, inference services, model artifacts, test-data generation, and visualization. This separation reduces the risk that a UI change alters scientific preprocessing and makes it possible to test the inference core independently from the dashboard.

4.1. Software architecture and project organization

4.1.1. Main modules and responsibilities

The project is organized as a thin interface layer above self-contained inference services. The dashboard code in app.py is responsible for language selection, input selection, visualization, and user interaction. The EGTService module loads and verifies the final model bundle, preprocesses the ordered SNP dosages, runs the PyTorch model, applies strain-specific thresholds, and produces attribution records. The CLI invokes the same service functions without requiring a browser. make_test_data.py supplies deterministic golden samples or synthetic demonstration inputs, while the Colab launcher starts the Streamlit application behind a tunnel when a local browser session is not available.

Table 4.1 — Software modules and their responsibilities.

Project component	Role in the software	Scientific responsibility
app.py	Bilingual Streamlit dashboard	Presents predictions, metrics, attribution, benchmark views, and input validation
egt_cli.py	Command-line interface	Verify bundle, run demos, batch predict, write CSV reports

Project component	Role in the software	Scientific responsibility
egt_service.py	Self-contained EGT inference core	Load model, enforce preprocessing, predict four strains, compute explainability
make_test_data.py	Test-data generator	Load golden fixtures or generate synthetic dosage matrices
run_dashboard_colab.py	Colab launcher	Starts Streamlit and exposes a tunnel URL
model2_service.py / model2_cli.py	Plant-DNABERT service and CLI	Use exported 3,843-d model artifacts without retraining
model_comparison.csv	Benchmark table	Provides EGT/teacher/AgroNT comparison values shown by dashboard
04_selected_2048_snps.csv	Model-input manifest	Defines exact SNP order, positions, alleles, and biological region mapping

4.1.2. Model bundle as the deployment boundary

The software treats the final EGT bundle as the deployment boundary. The application does not require the training notebook, the 4.8-million-SNP PLINK source, or the Nucleotide Transformer weights during routine inference. Instead, it requires the exact 2,048 ordered SNP inputs for a sample and the frozen 11 x 1,280 resistance-gene prior matrix already stored in the bundle. This design reduces runtime dependencies and ensures that the biological prior used locally is numerically identical to the one used by the selected model.

Integrity checking is performed before prediction. The dashboard run sequence explicitly initializes the service by checking the model bundle and SHA256 records, prepares the selected input source, then runs inference and explainability. If the bundle is incomplete or an input file violates the required schema, the application stops rather than silently reshaping or guessing the missing fields.

Table 4.2 — Twenty-two files in the final EGT deployment bundle.

Bundle file	Purpose
00_README.md	Human-readable handoff and inference requirements
01_best_model.pt	Exact selected PyTorch state_dict
02_model_runtime.py	Standalone architecture, preprocessing, inference and golden verification
03_model_spec.json	Architecture, hyperparameters, seeds, loss, optimizer and selection criterion
04_selected_2048_snps.csv	Exact ordered SNP manifest with A1/A2 and selection provenance
05_preprocessing_and_selection.json	Genotype, label and feature-selection contract
06_r_gene_sequences.fasta	Exact eleven fixed resistance-gene prior sequences
07_foundation_model_provenance.json	Foundation-model identifier and immutable provenance
08_frozen_r_gene_embeddings.npy	Exact 11 x 1,280 frozen prior matrix
09_thresholds_and_labels.json	Task thresholds and probability/label semantics
10_training_splits.csv	Exact train/validation membership
11_training_history.csv	Per-epoch training history
12_fine_sweep_scorecard.csv	C4 weight-sweep scorecard
13_per_sample_predictions.csv	Per-sample probabilities and decisions
14_validation_metrics_by_strain.csv	Final validation metrics by strain
15_roc_pr_curves.npz	Numerical ROC/PR curve arrays
16_confusion_and_calibration.json	Confusion matrices and calibration diagnostics
17_population_aware_validation.csv	Validation results stratified by population/subgroup
18_golden_samples.npz	Deterministic inference fixtures and expected outputs
19_software_environment.json	Python/package/CUDA environment at export
20_bundle_manifest.json	Machine-readable bundle inventory and critical assertions
21_SHA256SUMS.txt	SHA256 for the other model assets

4.1.3. Input contract and scientific meaning of the 2,048 features

For EGT inference, a row contains exactly 2,048 SNP dosages in the order defined by 04_selected_2048_snps.csv. The dosage counts the BIM A1 allele and therefore takes values 0, 1, or 2 when genotype is observed. Missing values are mapped to zero and the dosage is divided by two before entering the network, matching the training pipeline. The application intentionally exposes this contract in the dashboard because a numerically valid 2,048-column matrix can still be biologically wrong if its SNP order or allele orientation is incorrect.

Table 4.3 — Input formats and validation rules implemented by the local application.

Input format	Required structure	Validation performed
CSV	sample_id + 2,048 dosage columns (f0 ... f2047 or equivalent ordered columns)	One ID column; remaining matrix must represent the exact model order
NPZ	raw_A1_dosage_with_nan with shape (N, 2048); optional sample_id	Required key and exact second dimension checked
Golden fixture	18_golden_samples.npz packaged with model	Known raw dosages, expected probabilities and expected classes
Plant-DNABERT CSV	sample_id optional + exactly 3,843 numeric feature columns	Column count and finite-value checks

4.2. Building and running the command-line interface

4.2.1. Installation and directory preparation

The local command-line workflow is intentionally lightweight. The source directory is prepared with the Python requirements and the unzipped EGT bundle placed beside the application code. The Plant-DNABERT artifacts can be added in a separate models_14locus directory. Because the EGT bundle contains the frozen prior embeddings, local inference does not require downloading Nucleotide Transformer 500M.

```
pip install -r requirements.txt
unzip BB340_EGT_FINAL_MODEL_BUNDLE.zip -d
BB340_EGT_FINAL_MODEL_BUNDLE
```

After installation, the first recommended command is verification rather than prediction. Verification checks the bundle hashes, loads the exact architecture and state_dict, runs the bundled golden samples, and confirms that the local probabilities and binary calls agree with the expected fixtures within the allowed numerical tolerance.

4.2.2. CLI verification, demonstration, and batch prediction

```
# 1. Verify the exported model against the eight golden samples
python egt_cli.py verify --bundle ./BB340_EGT_FINAL_MODEL_BUNDLE

# 2. Run the eight real golden samples and print four-strain
predictions
python egt_cli.py demo --bundle ./BB340_EGT_FINAL_MODEL_BUNDLE

# 3. Generate a synthetic demonstration batch
python make_test_data.py --mode synthetic --n 10 --n-susceptible
3 --out demo_input.csv

# 4. Predict a CSV/NPZ batch and write a report
python egt_cli.py predict --bundle ./BB340_EGT_FINAL_MODEL_BUNDLE
--input demo_input.csv --out report.csv
```

The verify command is a reproducibility gate, the demo command is a human-readable smoke test, and predict is the operational batch interface. All three are thin wrappers around the same inference contract. This prevents the CLI from becoming a second, independently evolving implementation of the scientific model.

4.2.3. Expected CLI output and error behavior

A successful prediction report contains one sample identifier and four strain-specific probability/class pairs for EGT. Probabilities are P(Susceptible), not generic confidence scores. Each probability is compared against its own validation-selected threshold; therefore a value below 0.5 may still be susceptible for a low-threshold task, and a value above 0.5 may still be resistant for a high-threshold task.

Table 4.4 — Final EGT task thresholds used by the CLI and dashboard.

Strain	Decision threshold in final bundle	Decision rule
C4	0.31	Susceptible if P(Susceptible) >= 0.31
C5	0.69	Susceptible if P(Susceptible) >= 0.69
V	0.35	Susceptible if P(Susceptible) >= 0.35
P9a	0.33	Susceptible if P(Susceptible) >= 0.33

Input errors are handled as schema errors rather than model predictions. For NPZ input, the dashboard requires the key raw_A1_dosage_with_nan and checks that its shape is (N, 2048). For the Plant-DNABERT branch, the feature table must contain exactly 3,843 numeric dimensions. Missing model artifacts are surfaced to the user instead of being substituted with placeholder predictions.

4.3. Building and operating the bilingual Streamlit dashboard

4.3.1. Interface structure and configuration sidebar

The web application is implemented in Streamlit and configured for a wide layout. A fixed sidebar contains the language switch, the path to the EGT bundle, the path to the Plant-DNABERT artifact directory, model-loading status, input-source selection, and the main Run prediction action. The main pane uses a green project header and six functional tabs: Prediction, Batch overview, Model comparison benchmark, Enhanced Genomic Transformer, Plant-DNABERT, and About.

English and Vietnamese strings are defined in a shared dictionary and selected from session state, so the scientific variables—strain names, probabilities, thresholds, and metrics—remain identical when the display language changes. This bilingual design is useful for local development and defense presentation without duplicating the scientific computation.

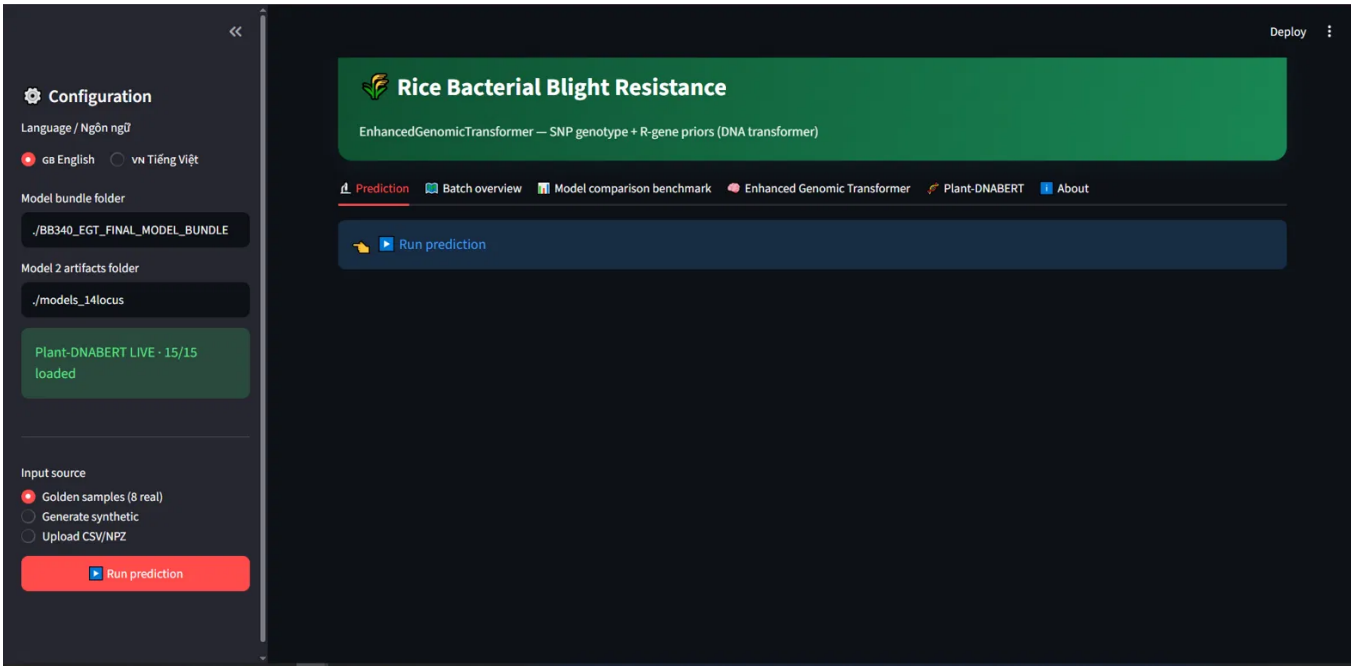


Figure 4.1 — Dashboard configuration sidebar and main navigation before a prediction run.

4.3.2. Run sequence and session-state behavior

When the user presses Run prediction, the dashboard performs three explicit stages. First, it initializes the EGT service and verifies the model bundle and hash records. Second, it prepares the selected input source: bundled golden samples, a synthetic demonstration batch, or an uploaded CSV/NPZ file. Third, it runs inference for all samples with explainability enabled and stores the resulting objects in Streamlit session state. The status panel reports completion or a specific error rather than leaving the user with an empty page.

1. Check the EGT bundle and SHA256 integrity through the service loader.

2. Load golden, synthetic, or uploaded input and validate the input schema.

3. Separate sample identifiers from the 2,048 dosage matrix.

4. Run EGT inference for C4, C5, V, and P9a and request explanation records.

5. Cache the result list and input matrix in session state for the other tabs.

6. Render per-sample, batch, benchmark, and interpretability views from the same cached result.

4.3.3. Prediction tab: per-sample resistance calls

The Prediction tab is designed for the most direct user question: how does one rice accession respond to each of the four Xoo strains? After selecting a sample, the dashboard

displays four cards. Each card shows the strain name, final RESISTANT or SUSCEPTIBLE call, P(Susceptible), and the task-specific decision threshold. Green cards indicate a resistant call and red/orange cards indicate a susceptible call.

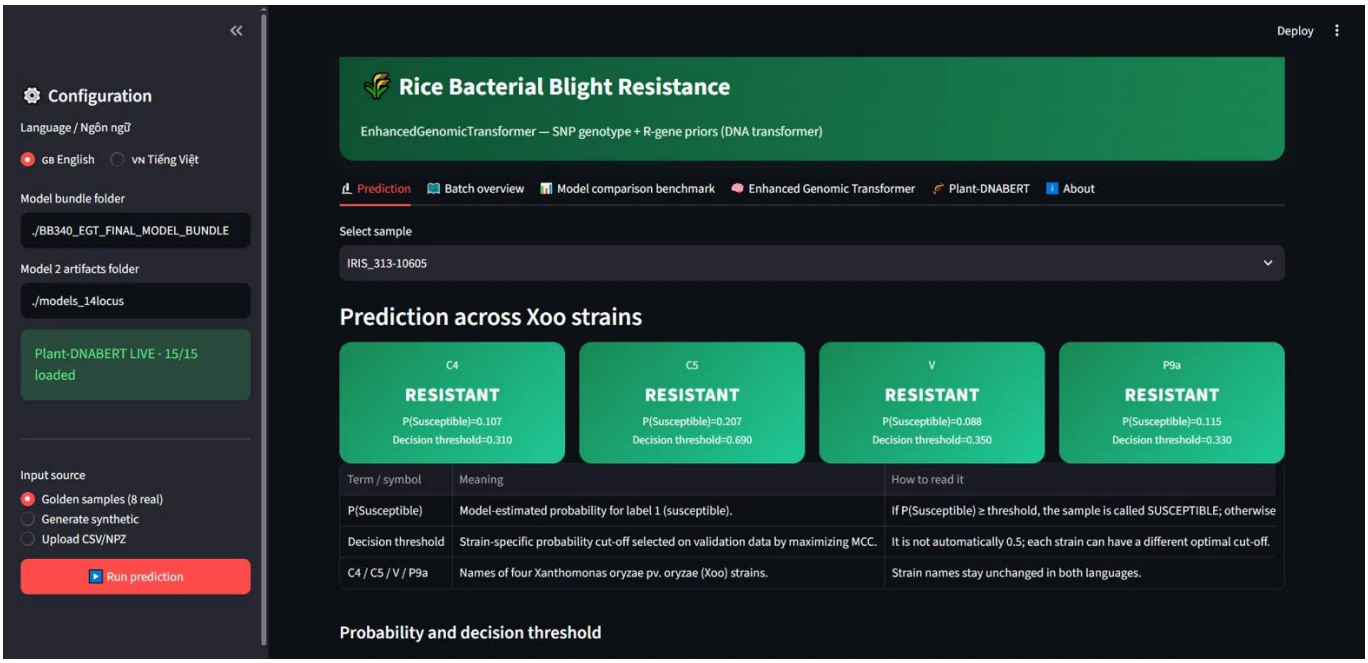


Figure 4.2 — Per-sample EGT prediction cards for C4, C5, V, and P9a.

A probability-versus-threshold chart follows the cards. This is preferable to showing only the final class because it reveals how far a prediction lies from the operational decision boundary. The dashboard explicitly explains that the threshold is not automatically 0.5; it is selected on the validation data by maximizing MCC for the corresponding strain.

4.3.4. Prediction tab: resistance-gene and SNP evidence

The lower portion of the Prediction tab exposes model attribution. Resistance-gene/QTL evidence aggregates signed Gradient x Input values over SNPs assigned to predefined biological windows. A second panel lists the strongest individual SNP contributions, including the SNP identifier, mapped biological region when available, signed contribution, and direction toward resistance or susceptibility.

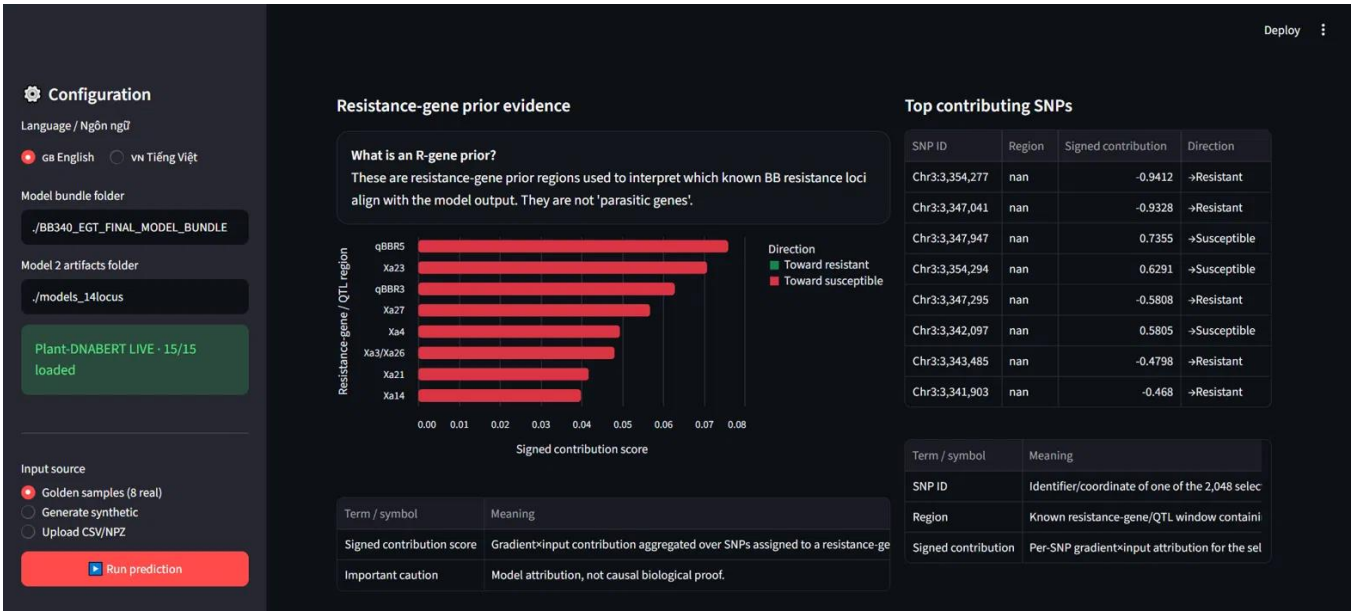


Figure 4.3 — Resistance-gene/QTL evidence and top contributing SNPs for a selected rice sample.

The application intentionally labels these values as model attribution rather than causal biological evidence. A positive contribution means the current input pushes the selected strain output toward susceptibility, while a negative contribution pushes it toward resistance. The genome-wide attribution map orders the 2,048 selected SNPs by chromosome and physical position; its vertical axis is attribution magnitude, not a GWAS p-value.

4.3.5. Batch overview tab

The Batch overview tab summarizes many accessions at once. Each row is a rice sample, each column is a strain, and the heatmap cell contains P(Susceptible). Green cells are closer to zero, red cells are closer to one, and numeric labels make the figure readable even when color reproduction is poor. The accompanying batch table retains both the probability and final thresholded class.

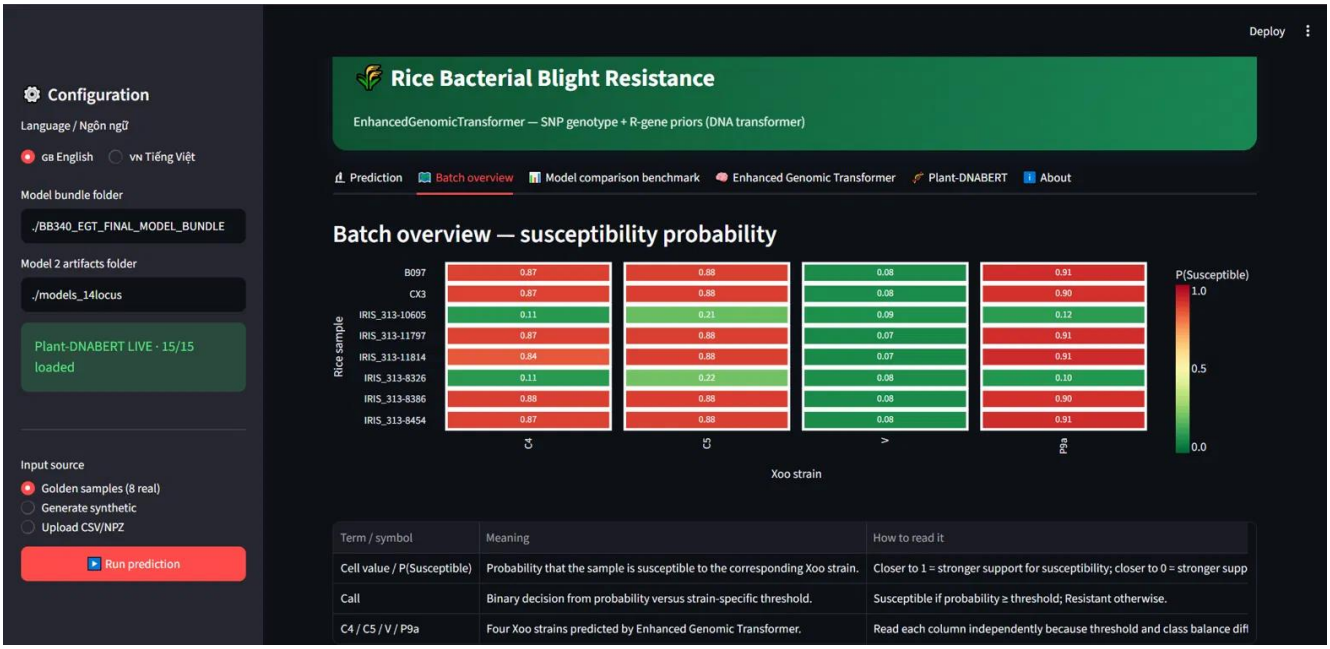


Figure 4.4 — Batch susceptibility-probability heatmap across eight golden rice samples and four Xoo strains.

This view supports two kinds of comparison. First, it reveals samples that are consistently resistant or susceptible across several strains. Second, it reveals strain-specific patterns, such as the golden susceptible samples that remain strongly resistant to V while being susceptible to C4, C5, and P9a. Because each strain has a different threshold and class distribution, the columns are interpreted independently rather than as one four-class softmax.

4.3.6. Model comparison benchmark tab

The Model comparison benchmark tab separates two evaluation families. For EGT, it displays the proposed Ver4.1 results together with the original NT500M teacher run and the AgroNT ablation under the same 80/20 protocol. The comparison reports MCC, ROC-AUC, and PR-AUC by strain and emphasizes MCC as the primary threshold-based metric under class imbalance.

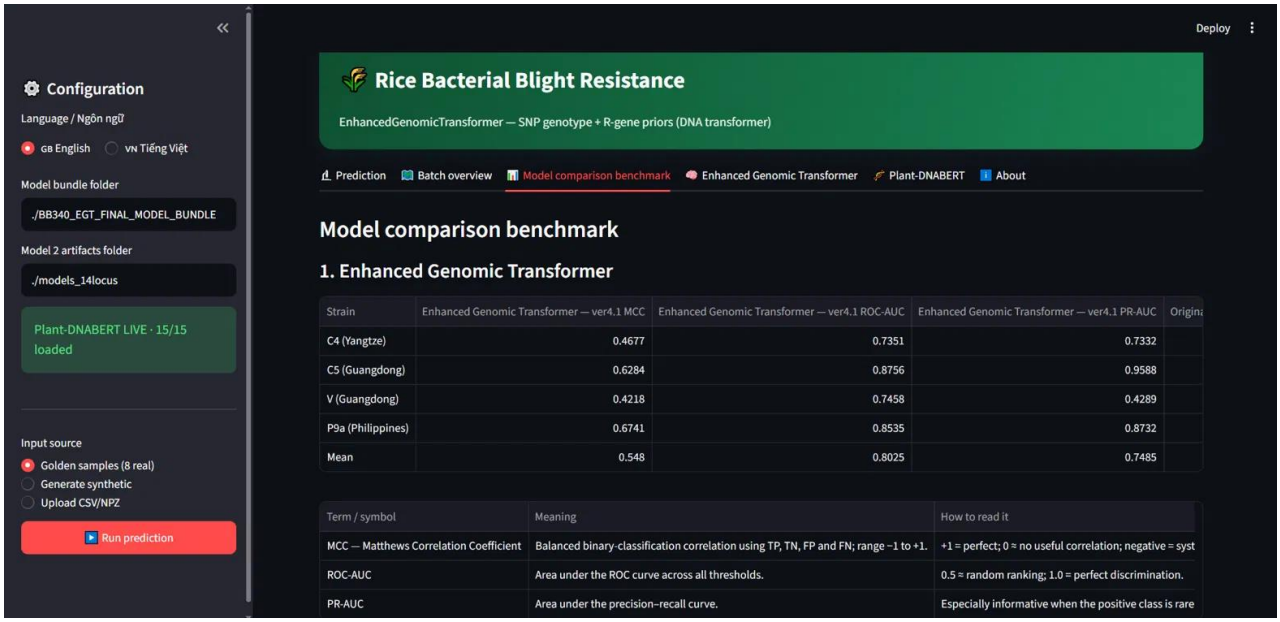


Figure 4.5 — EGT benchmark table comparing Ver4.1 with the original NT500M and AgroNT runs.

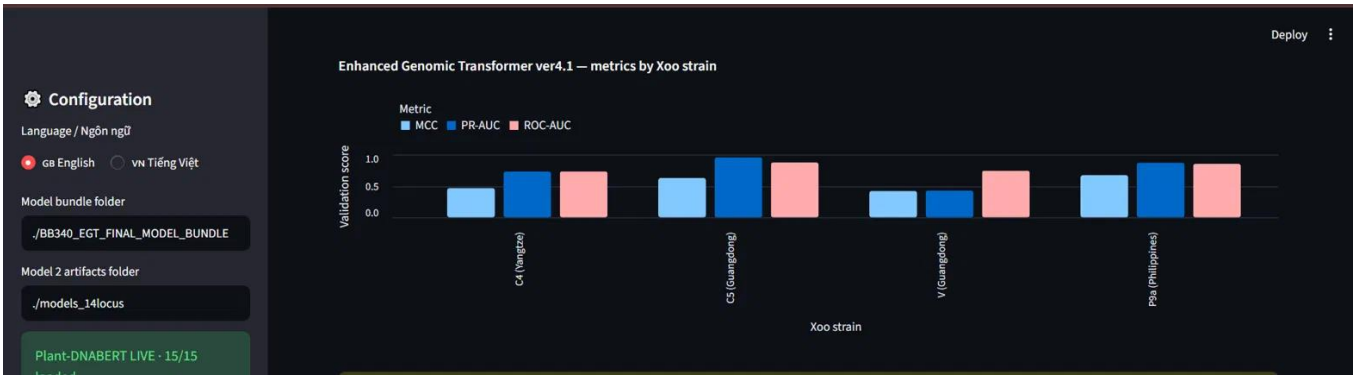


Figure 4.6 — Dashboard visualization of EGT MCC, PR-AUC, and ROC-AUC by Xoo strain.

The second part of the benchmark tab displays Plant-DNABERT GroupKFold MCC for the five exported classifiers and the three trained strains C4, C5, and P9a. V is intentionally absent from this model family because the corresponding sequence-classification branch was not trained for V under the selected setup. The dashboard states this explicitly so that an absent value is not confused with a failed inference.

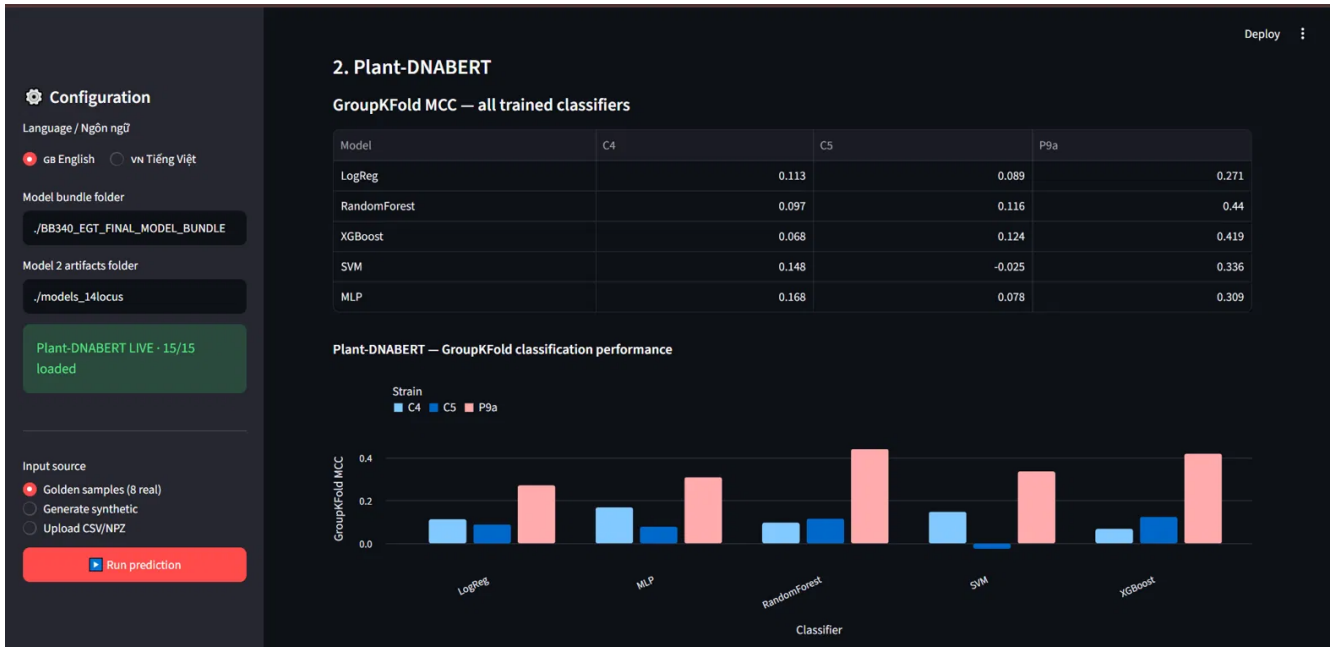


Figure 4.7 — Plant-DNABERT GroupKFold MCC table and classifier comparison in the dashboard.

4.3.7. Enhanced Genomic Transformer information tab

The dedicated EGT tab documents the deployed inference contract inside the application. It states that the primary model consumes 2,048 A1-allele SNP dosages in exact model order, maps missing dosage to zero, divides dosage by two, and fuses the resulting genomic representation with eleven frozen 1,280-dimensional resistance-gene priors. The output is four susceptibility probabilities followed by task-specific thresholding.

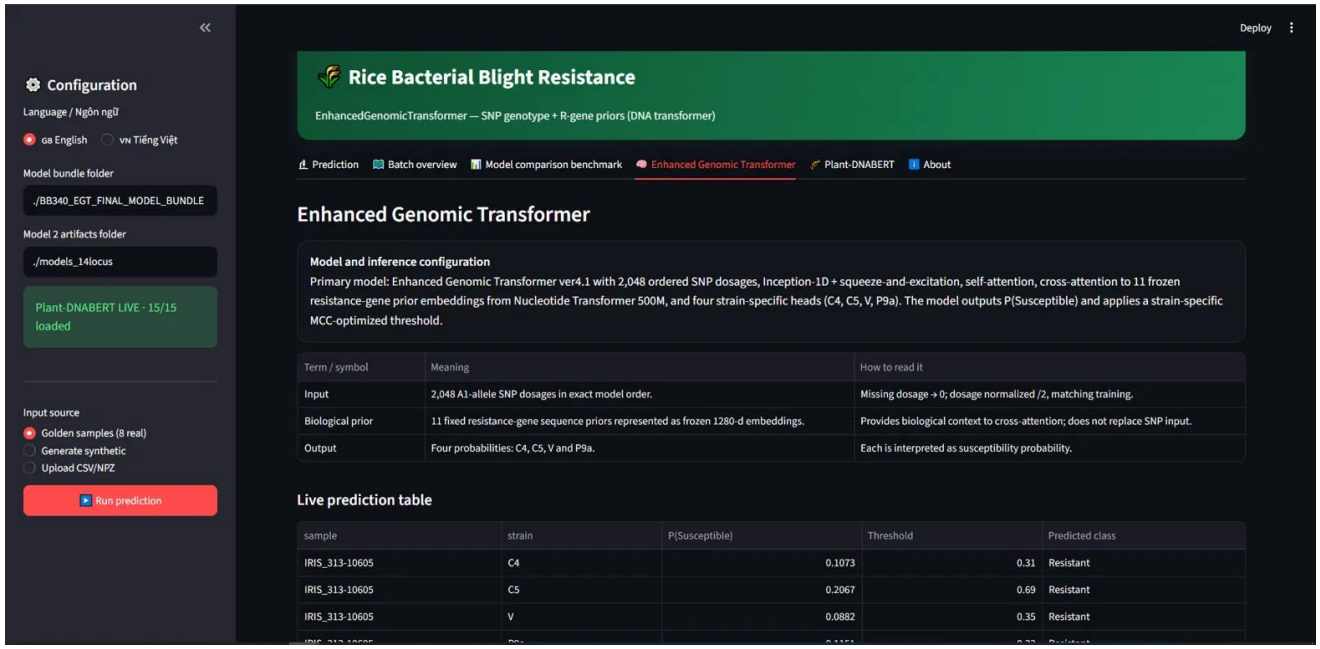


Figure 4.8 — Enhanced Genomic Transformer tab describing the deployed model and input/output semantics.

When predictions are already available, the EGT tab can also render a live prediction table and a two-dimensional PCA projection of the current input batch. The PCA display is deliberately described as a descriptive visualization, not as the exact decision surface of the original high-dimensional network. A local k-nearest-neighbor background may be overlaid only to help visually separate the predicted classes.

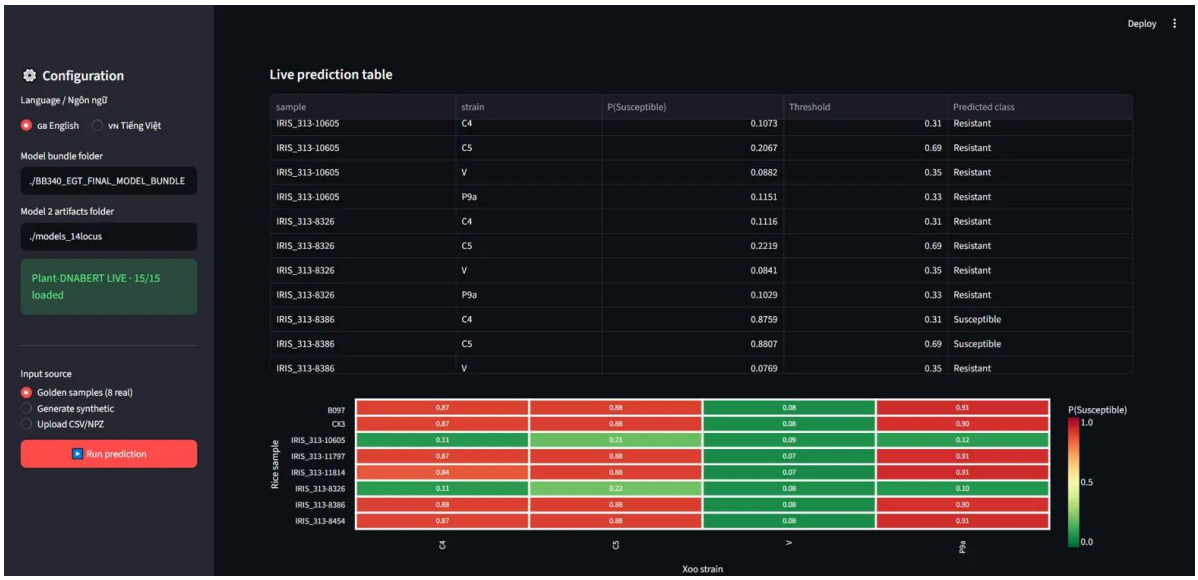


Figure 4.9 — Live EGT prediction table and batch heatmap generated from the current input batch.

4.3.8. Plant-DNABERT tab

The Plant-DNABERT tab supports the second model family using exported 3,843-dimensional artifacts. Its feature vector consists of five 768-dimensional gene embeddings—xa5, xa13/SWEET11, xa25/SWEET13, Xa21, and Xa1—plus three presence/absence variables. The user can choose among Logistic Regression, Random Forest, XGBoost, SVM, and MLP when the corresponding artifacts are present.

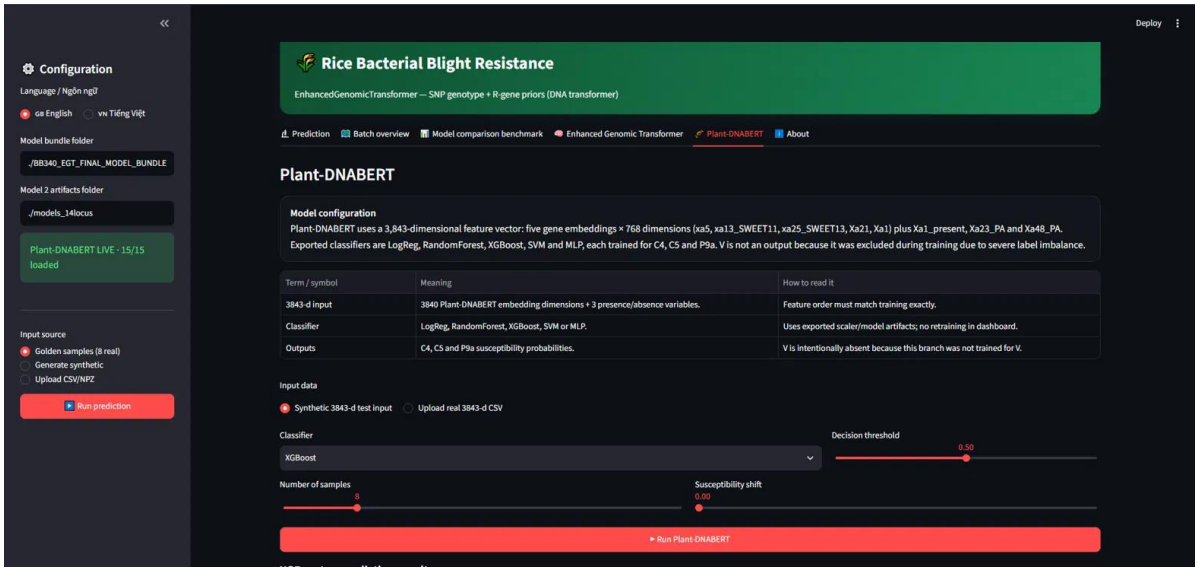


Figure 4.10 — Plant-DNABERT inference tab with classifier, threshold, sample-count, and susceptibility-shift controls.

The tab accepts either a synthetic 3,843-dimensional feature batch sampled for demonstration or an uploaded real feature CSV produced by the upstream Plant-DNABERT notebook. The dashboard does not reconstruct raw FASTA into embeddings itself. This separation keeps the interactive app lightweight and avoids silently running a different sequence preprocessing path from the one validated in Pipeline A.



Figure 4.11 — Two-dimensional PCA visualization of Plant-DNABERT/XGBoost predictions for the current batch.

4.4. User use cases

The system has two interface actors: a technical user who prefers reproducible scripted operation through the CLI and an interactive user who explores predictions and explanations through the dashboard. Four use cases cover the principal user-visible behavior without

splitting minor display actions into separate requirements.

Table 4.5 — User-level use case summary.

Use case	Interface	Primary actor	Goal
UC-01 Verify a model bundle	CLI	Researcher / developer	Confirm integrity and exact golden-sample reproducibility before use
UC-02 Run batch prediction	CLI	Researcher / analyst	Predict C4/C5/V/P9a for a CSV/NPZ batch and export a report
UC-03 Run and inspect interactive prediction	Web dashboard	Researcher / reviewer	Load golden/synthetic/uploaded input and inspect per-strain predictions
UC-04 Compare models and interpret evidence	Web dashboard	Researcher / reviewer	Inspect benchmarks, gene/QTL evidence, SNP attribution, and Plant-DNABERT results

4.4.1. Verify a model bundle before inference

Table 4.6 — Detailed specification for UC-01: Verify a model bundle before inference

Field	Specification
Use case ID	UC-01
Interface	CLI
Primary actor	Researcher or developer preparing a local deployment
Preconditions	Python environment installed; BB340_EGT_FINAL_MODEL_BUNDLE is present and readable.
Trigger	User executes egt_cli.py verify with the bundle path.
Main success flow	The CLI loads the manifest and hashes, verifies model assets, instantiates the exact architecture, loads the state_dict and frozen priors, runs the eight golden fixtures, and compares probabilities/classes with expected outputs.

Field	Specification
Alternative / exception flow	Missing file, hash mismatch, architecture mismatch, or golden-sample disagreement causes verification to fail and no operational prediction is accepted.
Postconditions	A PASS result establishes that the local runtime reproduces the exported inference fixtures within tolerance.

4.4.2. Run batch prediction and export a report

Table 4.7 — Detailed specification for UC-02: Run batch prediction and export a report.

Field	Specification
Use case ID	UC-02
Interface	CLI
Primary actor	Researcher or analyst
Preconditions	Verified bundle is available; input CSV or NPZ obeys the EGT input contract.
Trigger	User executes egt_cli.py predict and provides input/output paths.
Main success flow	The CLI reads sample IDs and 2,048 ordered A1 dosages, applies missing-value and /2 preprocessing, runs all four heads, applies task thresholds, and writes a report containing probabilities and final classes.
Alternative / exception flow	An invalid NPZ key, wrong feature count, non-numeric matrix, or unreadable bundle is reported as an error rather than reshaped automatically.
Postconditions	A machine-readable report is created for downstream analysis or archiving.

4.4.3. Run and inspect interactive EGT prediction

Table 4.8 — Detailed specification for UC-03: Run and inspect interactive EGT prediction.

Field	Specification
Use case ID	UC-03
Interface	Web dashboard
Primary actor	Researcher, supervisor, or thesis reviewer
Preconditions	Dashboard is running; EGT bundle path is configured.
Trigger	User selects golden, synthetic, or upload input and presses Run prediction.
Main success flow	The dashboard verifies the bundle, validates input, runs inference and explainability, then allows a sample to be selected. Four result cards, probabilities, thresholds, resistance/QTL evidence, top SNPs, and batch views become available.
Alternative / exception flow	If no input is supplied or a file violates the schema, the status panel shows an error and results are not cached as successful.
Postconditions	Predictions are stored in session state and can be explored across Prediction, Batch, Comparison, and EGT tabs.

4.4.4. Compare model families and interpret predictions

Table 4.9 — Detailed specification for UC-04: Compare model families and interpret predictions.

Field	Specification
Use case ID	UC-04
Interface	Web dashboard
Primary actor	Researcher, supervisor, or thesis reviewer
Preconditions	Dashboard has access to model_comparison.csv; Plant-DNABERT artifacts may also be available.
Trigger	User opens Model comparison, EGT, or Plant-DNABERT tabs after or independently of an EGT prediction run.
Main success flow	The dashboard displays EGT benchmark metrics, Plant-DNABERT GroupKFold

Field	Specification
	scores, model/input semantics, and when predictions exist, attribution and PCA visualizations. If Plant-DNABERT artifacts are complete, the user can run exported classifiers directly.
Alternative / exception flow	If Plant-DNABERT artifacts are incomplete, the dashboard reports the missing state rather than claiming a live model is loaded.
Postconditions	The user can compare validated model behavior and inspect model evidence without retraining.

4.5. Deployment procedures

4.5.1. Local desktop deployment

For local use, the project directory contains the Python scripts, requirements file, model comparison table, the unzipped EGT bundle, and optionally the Plant-DNABERT artifact directory. After dependency installation, Streamlit is started from the project root so relative paths in the sidebar resolve consistently.

```
streamlit run app.py
# Default local URL:
# http://localhost:8501
```

The browser-based dashboard and CLI can be used from the same environment. The CLI is preferable for automated batch runs, shell scripts, and reproducible test logs; the dashboard is preferable for sample exploration, visual explanation, and presentation. Because both depend on the same exported bundle, a successful CLI verification is a useful precondition for a dashboard demonstration.

4.5.2. Colab-based deployment

When the software is demonstrated from Google Colab, the project directory is uploaded or mounted, the final model bundle is extracted beside the application, and the provided launcher starts Streamlit plus a Cloudflare tunnel. This mode is convenient for a thesis defense because the model can run on a remote notebook while the reviewer interacts with a browser URL. The tunnel affects only access to the interface; inference semantics remain the same as local execution.

Table 4.10 — Supported deployment modes.

Deployment mode	Command / entry point	Typical use	Model retraining?
Local CLI	python egt_cli.py ...	Verification, batch prediction, scripted experiments	No
Local dashboard	streamlit run app.py	Interactive interpretation and presentation	No
Colab dashboard	run_dashboard_colab.py / Streamlit + tunnel	Remote demonstration	No
Plant-DNABERT CLI/dashboard	model2_cli.py or Plant-DNABERT tab	Run exported 3,843-d classifiers	No

4.6. Test design and acceptance criteria

Testing is divided into four layers. Integrity tests verify that the deployed assets have not changed. Golden inference tests verify numerical reproduction of known predictions. Functional tests verify input validation, batch reporting, thresholding, and model-2 artifact detection. Interface tests verify that the dashboard exposes the expected views and that displayed probabilities and classes match the underlying prediction records.

Table 4.11 — Test layers and acceptance criteria.

Test layer	What is checked	Acceptance criterion
Bundle integrity	Required files and SHA256	No missing asset and no hash mismatch
Golden inference	Eight real fixtures x four strain heads	All probabilities within tolerance and all classes identical
Input validation	CSV/NPZ shape, required key, numeric content	Invalid input rejected with explicit error
Threshold logic	P(Susceptible) versus strain threshold	Displayed and exported class agrees with threshold rule
Batch operation	Multiple samples in one run	One complete four-strain record per sample
Explainability	Gene-region/SNP evidence returned	Attribution view renders for valid EGT prediction

Test layer	What is checked	Acceptance criterion
Plant-DNABERT artifacts	5 classifiers x 3 strains	15/15 slots loaded for LIVE status
UI navigation	Prediction/Batch/Benchmark/EGT/Plant-DNABERT tabs	Tabs render without retraining or changing result semantics

4.6.1. Golden-sample regression fixtures

The final bundle contains eight real validation samples selected as deterministic golden fixtures. Each fixture stores raw A1 dosages, the normalized model input, the true four-task labels, expected probabilities, expected thresholded predictions, and the task thresholds. The golden samples are used as a software regression test, not as a new estimate of model performance. Their purpose is to answer a deployment question: does the local runtime reproduce the exact trained model behavior?

Table 4.12 — Metadata and expected biological classes of the eight golden samples.

Sample	Population	True C4	True C5	True V	True P9a
IRIS_313-10605	cA(Aus)	Resistant	Resistant	Resistant	Resistant
IRIS_313-8326	cB(Bas)	Resistant	Resistant	Resistant	Resistant
IRIS_313-8386	XI-adm	Susceptible	Susceptible	Resistant	Susceptible
CX3	XI-1A	Susceptible	Susceptible	Resistant	Susceptible
IRIS_313-11814	XI-3	Susceptible	Susceptible	Resistant	Susceptible
IRIS_313-11797	XI-1A	Susceptible	Susceptible	Resistant	Susceptible
IRIS_313-8454	XI-1A	Susceptible	Susceptible	Resistant	Susceptible
B097	XI-adm	Susceptible	Susceptible	Resistant	Susceptible

4.6.2. Golden-sample probability results

Table 4.13 — Expected EGT probabilities for the eight golden samples.

Sample	P(Susceptible) C4	P(Susceptible) C5	P(Susceptible) V	P(Susceptible) P9a
IRIS_313-10605	0.1073	0.2067	0.0882	0.1151
IRIS_313-8326	0.1116	0.2219	0.0841	0.1029
IRIS_313-8386	0.8759	0.8807	0.0769	0.9022
CX3	0.8724	0.8797	0.0774	0.9021
IRIS_313-11814	0.8361	0.8781	0.0745	0.9128
IRIS_313-11797	0.8714	0.8779	0.0747	0.9070
IRIS_313-8454	0.8714	0.8777	0.0807	0.9100
B097	0.8744	0.8771	0.0786	0.9097

Table 4.14 — Thresholded golden-sample predictions and software regression result.

Sample	C4	C5	V	P9a	Overall
IRIS_313-10605	Resistant (PASS)	Resistant (PASS)	Resistant (PASS)	Resistant (PASS)	PASS
IRIS_313-8326	Resistant (PASS)	Resistant (PASS)	Resistant (PASS)	Resistant (PASS)	PASS
IRIS_313-8386	Susceptible (PASS)	Susceptible (PASS)	Resistant (PASS)	Susceptible (PASS)	PASS
CX3	Susceptible (PASS)	Susceptible (PASS)	Resistant (PASS)	Susceptible (PASS)	PASS
IRIS_313-11814	Susceptible (PASS)	Susceptible (PASS)	Resistant (PASS)	Susceptible (PASS)	PASS
IRIS_313-11797	Susceptible (PASS)	Susceptible (PASS)	Resistant (PASS)	Susceptible (PASS)	PASS
IRIS_313-8454	Susceptible (PASS)	Susceptible (PASS)	Resistant (PASS)	Susceptible (PASS)	PASS
B097	Susceptible (PASS)	Susceptible (PASS)	Resistant (PASS)	Susceptible (PASS)	PASS

All 32 strain-level calls in the eight golden fixtures agree with the stored expected labels. The first two samples are resistant to all four strains. The remaining six are susceptible to C4, C5, and P9a while remaining resistant to V. This pattern is also visible in the dashboard heatmaps and provides an easily recognized visual regression signature during demonstration.

4.6.3. Detailed test scenarios using at least five real samples

Table 4.15 — Six real-sample end-to-end prediction scenarios satisfying the minimum five-sample test requirement.

Test case	Sample	Population	Input source	Observed / expected probabilities	Expected final calls	Result
TC-01	IRIS_313-10605	cA(Aus)	Golden real sample	C4=0.107, C5=0.207, V=0.088, P9a=0.115	C4=R, C5=R, V=R, P9a=R	PASS
TC-02	IRIS_313-8326	cB(Bas)	Golden real sample	C4=0.112, C5=0.222, V=0.084, P9a=0.103	C4=R, C5=R, V=R, P9a=R	PASS
TC-03	IRIS_313-8386	XI-adm	Golden real sample	C4=0.876, C5=0.881, V=0.077, P9a=0.902	C4=S, C5=S, V=R, P9a=S	PASS
TC-04	CX3	XI-1A	Golden real sample	C4=0.872, C5=0.880, V=0.077, P9a=0.902	C4=S, C5=S, V=R, P9a=S	PASS
TC-05	IRIS_313-11814	XI-3	Golden real sample	C4=0.836, C5=0.878, V=0.074, P9a=0.913	C4=S, C5=S, V=R, P9a=S	PASS
TC-06	IRIS_313-11797	XI-1A	Golden real sample	C4=0.871, C5=0.878, V=0.075, P9a=0.907	C4=S, C5=S, V=R, P9a=S	PASS

These scenarios exercise both balanced and strain-specific outcomes. IRIS_313-10605 and IRIS_313-8326 test the all-resistant path, including the unusually high C5 threshold of 0.69. IRIS_313-8386, CX3, IRIS_313-11814, and IRIS_313-11797 test the mixed path in which C4, C5, and P9a are susceptible while V remains resistant. The probability margins are large for these golden fixtures, which makes them stable regression tests across CPU environments.

4.6.4. Functional and negative test cases

Table 4.16 — Functional and negative test cases for CLI/dashboard operation.

ID	Scenario	Input / precondition	Expected behavior	Status
FT-01	Bundle verification	Valid final bundle	SHA256 and golden verification complete	PASS
FT-02	Golden batch load	Bundled 18_golden_samples.npz	Eight sample IDs and (8,2048) dosage matrix loaded	PASS
FT-03	CSV batch prediction	Valid sample_id + 2048 dosage columns	Four probabilities/classes per sample and report CSV	Expected behavior
FT-04	NPZ shape validation	raw_A1_dosage_with_nan shape != (N,2048)	Input rejected with explicit shape error	Expected behavior
FT-05	NPZ key validation	Archive missing raw_A1_dosage_with_nan	Input rejected and present keys reported	Expected behavior
FT-06	No input selected	Upload mode without file	Status reports no input; prediction not run	Expected behavior
FT-07	Batch visualization	Eight golden samples	Heatmap and batch table contain 8 x 4 predictions	PASS
FT-08	Threshold logic	Golden probabilities	Displayed class equals probability >= strain threshold	PASS
FT-09	Plant-DNABERT artifact inspection	Complete models_14locus directory	LIVE status reports 15/15 loaded	PASS in supplied UI
FT-10	Plant-DNABERT invalid CSV	Feature count != 3843	Input rejected before classifier call	Expected behavior
FT-11	Language switch	Toggle English / Vietnamese	Labels change while numerical predictions remain unchanged	Expected behavior
FT-12	Download report	Completed EGT prediction batch	egt_report.csv generated from cached results	Expected behavior

4.6.5. Dashboard scenario tests

The supplied dashboard screenshots document the principal user paths. Before inference, the application shows the configured bundle directories and the selected input source. After inference, the same batch can be inspected at three levels: per-sample decisions, multi-sample heatmaps, and model-level benchmark/interpretability views. The following scenario table maps the UI evidence to the use cases.

Table 4.17 — Dashboard scenario test matrix.

UI test	Scenario	Evidence	Acceptance criterion	Result
UI-01	Open dashboard and verify configuration	Figure 4.1	Bundle path, model-2 path, language switch, input radio, Run button visible	PASS
UI-02	Inspect a selected sample	Figure 4.2	Four strain cards show probability, threshold and class	PASS
UI-03	Inspect EGT explanation	Figure 4.3	Resistance/QTL contribution and top-SNP table visible	PASS
UI-04	Inspect batch prediction	Figure 4.4	Eight-sample x four-strain heatmap rendered	PASS
UI-05	Inspect EGT benchmark	Figures 4.5–4.6	Per-strain metrics and model comparison rendered	PASS
UI-06	Inspect Plant-DNABERT benchmark	Figure 4.7	Five classifiers x three strains displayed	PASS
UI-07	Inspect live EGT table	Figures 4.8–4.9	Model semantics, live probabilities, batch heatmap available	PASS
UI-08	Run Plant-DNABERT interface	Figures 4.10–4.11	Classifier and input controls plus 2D	PASS

UI test	Scenario	Evidence	Acceptance criterion	Result
			descriptive projection available	

4.7. Interpretation of the software test results

The golden tests demonstrate reproducibility of the exported EGT on a local runtime. They do not constitute an additional independent validation set because the fixtures originate from the experiment's validation cohort. Their role is analogous to a regression test in software engineering: when the model code, environment, or packaging changes, the eight fixtures provide an immediate warning if probabilities, threshold logic, input ordering, or label semantics have drifted.

The dashboard tests confirm that the application exposes model uncertainty and thresholds rather than only categorical calls. This is particularly important for genomic decision support because the four strain heads have different imbalance and threshold characteristics. The batch heatmap also makes the six susceptible-for-C4/C5/P9a but resistant-for-V golden samples visually distinct from the two all-resistant samples.

The software also preserves methodological caveats from Chapter 3. The benchmark tab explicitly describes EGT values as results under the 80/20 validation protocol rather than a locked independent test. Attribution plots are labeled as model explanations, not causal biological proof. The Plant-DNABERT branch states when V is outside its trained output space, and the dashboard does not fabricate a V result for that model family.

4.8. Operational limitations and future software improvements

The current application is a research prototype rather than a clinical or breeding-decision system. It assumes that the user provides genotype dosages corresponding exactly to the model's 2,048 SNP manifest. A production deployment should add a genotype harmonization layer that maps external VCF/PLINK variants to the manifest, resolves strand/allele orientation, detects missing markers, and produces an auditable conversion report before inference.

The EGT model can run locally without the foundation model because the frozen prior embeddings are bundled, but this also means the prior sequences are fixed at deployment time.

A future version could support a second mode that re-encodes verified rice-reference or accession-specific resistance-gene sequences under an explicitly pinned DNA foundation-model revision. Such a mode should remain separate from the existing frozen-prior model so that predictions remain reproducible.

The dashboard currently emphasizes interpretability and demonstration. Future engineering work can add user authentication, project histories, server-side job queues for large genotype batches, persistent prediction records, model version selection, downloadable PDF reports, and automated ingestion from standard VCF/PLINK files. These additions should be implemented around the inference service rather than by duplicating scientific preprocessing inside UI callbacks.

4.9. Implementation consolidation and handoff principles

The implementation and test results above establish the core handoff principle of the project: the research model is deployable only when the scientific preprocessing contract travels with the checkpoint. The following sections expand this principle into detailed CLI operation, dashboard workflows, sample-level verification logs, and a traceability matrix that connects the four user use cases to executable tests.

4.10. Detailed CLI command reference

The CLI is intentionally small enough to be used as both an operator interface and a test harness. Each command has a single scientific responsibility and accepts the model directory explicitly, which avoids silently loading an unintended checkpoint from the current working directory.

4.10.1. Verify command

Verify is the mandatory pre-inference command when a bundle has been copied between machines, extracted from an archive, or modified. Its acceptance result is binary: the local runtime must reproduce the integrity manifest and the golden fixtures. A successful verification does not claim that the model is accurate on new breeding material; it claims that this machine is executing the exported model consistently.

Table 4.18 — Verify command syntax table.

Argument	Meaning	Example
verify	Select bundle-verification mode	python egt_cli.py verify ...
--bundle	Path to unzipped final bundle	--bundle ./BB340_EGT_FINAL_MODEL_BUNDLE
Exit status	Success only when integrity and golden checks pass	0 on PASS; non-zero on failure

4.10.2. Demo command

Demo loads the eight real golden samples bundled with the model and prints a human-readable four-strain prediction for each. This command is useful during a thesis defense or installation check because it does not require the user to prepare an external input file. The results should reproduce the same resistant/susceptible pattern shown in Figures 4.2 and 4.4.

Table 4.19 — Human-readable fields expected from the CLI demo workflow.

Output item	Meaning
sample identifier	Golden rice accession
C4/C5/V/P9a probability	P(Susceptible) from each EGT head
threshold	Task-specific MCC-selected cutoff stored in bundle
final call	Susceptible when probability >= threshold, otherwise Resistant
optional evidence	Region/SNP attribution when explanation is requested by the service

4.10.3. Predict command

Predict is the operational batch mode. It accepts a CSV or NPZ input, validates the EGT feature contract, predicts all samples in one service call, and writes a report. A report row is self-contained: it stores the sample ID together with C4, C5, V, and P9a probabilities and calls. This structure is convenient for downstream spreadsheet analysis and for archiving test evidence.

Table 4.20 — Predict command syntax table.

Argument	Required?	Description
predict	Yes	Select batch inference mode
--bundle	Yes	Path to verified EGT bundle
--input	Yes	CSV/NPZ batch in the EGT feature contract
--out	Yes	Destination CSV report

4.10.4. Example report schema

The report schema keeps probability and class together. This prevents a downstream user from applying a generic 0.5 threshold and reproduces the model's actual operational decision. The following schema is recommended even when a separate application consumes the predictions.

Table 4.21 — Recommended CLI output-report schema.

Column	Type	Meaning
sample_id	string	Rice accession identifier
C4_prob	float	P(Susceptible) for C4
C4_call	string	Thresholded C4 class
C5_prob / C5_call	float / string	C5 probability and class
V_prob / V_call	float / string	V probability and class
P9a_prob / P9a_call	float / string	P9a probability and class

4.10.5. Plant-DNABERT CLI operation

The second model family has a separate CLI because its input contract is fundamentally different. It does not accept the 2,048 SNP dosage matrix. It accepts the 3,843-dimensional Plant-DNABERT feature vector produced by the sequence embedding pipeline and can run one of the exported classifiers for C4, C5, and P9a.

```
unzip models_14locus.zip -d models_14locus
python model2_cli.py demo   --dir ./models_14locus   --model SVM
--n 6   --bias 0.5
python model2_cli.py predict   --dir ./models_14locus   --model
XGBoost   --input features.csv   --out model2_report.csv
```

Keeping the two CLIs separate prevents accidental interchange of feature representations. A raw 2,048-SNP EGT input cannot be sent to Plant-DNABERT, and a 3,843-dimensional

embedding vector cannot be sent to EGT.

4.11. Detailed dashboard operating workflows

4.11.1. Workflow A — verify and demonstrate the eight golden samples

1. Start Streamlit from the project root and confirm that the EGT bundle path resolves.

2. Select English or Vietnamese. The language choice affects labels only, not the inference values.

3. Select Golden samples (8 real) as the input source.

4. Press Run prediction. The status panel checks the bundle, prepares the input, then runs inference plus explainability.

5. Open Prediction and select IRIS_313-10605. Confirm that all four result cards are Resistant.

6. Switch to IRIS_313-8386. Confirm Susceptible for C4, C5 and P9a and Resistant for V.

7. Open Batch overview and confirm the 8 x 4 probability heatmap.

8. Open Enhanced Genomic Transformer and compare the live table with the batch heatmap.

This workflow is the preferred defense demonstration because every expected output is already fixed by the bundle. It demonstrates real model inference while avoiding external file-format uncertainty.

4.11.2. Workflow B — predict an uploaded EGT batch

1. Prepare CSV with sample_id and exactly 2,048 ordered A1-dosage columns, or prepare NPZ with raw_A1_dosage_with_nan.

2. Select Upload CSV/NPZ in the dashboard sidebar.

3. Upload the file and allow the dashboard to validate the required key and matrix shape.

4. Press Run prediction and confirm the number of completed samples in the status message.

5. Inspect per-sample calls in Prediction and the entire batch in Batch overview.

6. Download egt_report.csv and archive it together with the input file and bundle version.

For a real external genotype file, the critical operation occurs before the dashboard: the user's variants must be harmonized to the exact manifest, including allele orientation. The current prototype intentionally does not guess this mapping.

4.11.3. Workflow C — run the exported Plant-DNABERT classifiers

1. Place models_14locus beside the application and confirm the sidebar status reports 15/15 loaded.

2. Open the Plant-DNABERT tab and choose synthetic or real 3,843-dimensional input.

3. Choose one of LogReg, RandomForest, XGBoost, SVM, or MLP.

4. Set the demonstration threshold and, for synthetic mode, number of samples and susceptibility shift.

5. Run the classifier and inspect C4/C5/P9a probability heatmap and table.

6. Select a strain for the descriptive PCA projection; if LogReg is used, inspect the direct coefficient-based per-gene contribution view.

The dashboard makes the absence of V explicit for this branch. This is a model-definition constraint, not missing software output.

4.11.4. Dashboard information architecture

Table 4.22 — Information architecture of the dashboard tabs.

Tab	Primary question answered	Requires current EGT prediction?	Main outputs
Prediction	What is the prediction for one sample?	Yes	4 cards, thresholds, gene/QTL evidence, top SNPs, attribution track
Batch overview	How do all samples compare across strains?	Yes	Heatmap and batch table
Model comparison benchmark	How do trained models compare?	No	EGT benchmark, Plant-DNABERT GroupKFold table/charts

Tab	Primary question answered	Requires current EGT prediction?	Main outputs
Enhanced Genomic Transformer	What exactly is the deployed EGT and what did it predict?	Partly	Input/output contract, live table, batch heatmap, PCA projection
Plant-DNABERT	Can the exported 3,843-d classifiers be run?	No EGT dependency	Classifier controls, results, heatmap, PCA, LogReg attribution
About	What does the application demonstrate?	No	Project and interface notes

4.12. Detailed golden-sample verification log

The following per-sample tables document probability, threshold, expected biological label, predicted class, and probability margin from the threshold. The margin is not a calibrated confidence interval; it is simply an operational measure of how far the current probability is from the binary cut-off. Large margins are desirable for regression fixtures because small numerical differences between CPU/GPU environments are unlikely to change the class.

4.12.1. Golden sample IRIS_313-10605 (cA(Aus))

IRIS_313-10605 is one of the deterministic validation fixtures exported with the final bundle. The table below records the exact expected inference outputs used to verify the local runtime. A PASS requires the thresholded class to match the fixture for all four strains and the probability vector to remain within the bundle tolerance.

Table 4.23 — Golden regression record for IRIS_313-10605.

Strain	P(Susceptible)	Threshold	|P-threshold|	Expected class	Predicted class	Result
C4	0.1073	0.31	0.2027	Resistant	Resistant	PASS
C5	0.2067	0.69	0.4833	Resistant	Resistant	PASS
V	0.0882	0.35	0.2618	Resistant	Resistant	PASS
P9a	0.1151	0.33	0.2149	Resistant	Resistant	PASS

The expected operational pattern for this sample is all four strains resistant. This pattern can be cross-checked against the Prediction cards and the batch heatmap without inspecting the

numerical fixture file directly.

4.12.2. Golden sample IRIS_313-8326

IRIS_313-8326 is one of the deterministic validation fixtures exported with the final bundle. The table below records the exact expected inference outputs used to verify the local runtime. A PASS requires the thresholded class to match the fixture for all four strains and the probability vector to remain within the bundle tolerance.

Table 4.24 — Golden regression record for IRIS_313-8326.

Strain	P(Susceptible)	Threshold	|P-threshold|	Expected class	Predicted class	Result
C4	0.1116	0.31	0.1984	Resistant	Resistant	PASS
C5	0.2219	0.69	0.4681	Resistant	Resistant	PASS
V	0.0841	0.35	0.2659	Resistant	Resistant	PASS
P9a	0.1029	0.33	0.2271	Resistant	Resistant	PASS

The expected operational pattern for this sample is all four strains resistant. This pattern can be cross-checked against the Prediction cards and the batch heatmap without inspecting the numerical fixture file directly.

4.12.3. Golden sample IRIS_313-8386 (XI-adm)

IRIS_313-8386 is one of the deterministic validation fixtures exported with the final bundle. The table below records the exact expected inference outputs used to verify the local runtime. A PASS requires the thresholded class to match the fixture for all four strains and the probability vector to remain within the bundle tolerance.

Table 4.25 — Golden regression record for IRIS_313-8386.

Strain	P(Susceptible)	Threshold	|P-threshold|	Expected class	Predicted class	Result
C4	0.8759	0.31	0.5659	Susceptible	Susceptible	PASS
C5	0.8807	0.69	0.1907	Susceptible	Susceptible	PASS
V	0.0769	0.35	0.2731	Resistant	Resistant	PASS
P9a	0.9022	0.33	0.5722	Susceptible	Susceptible	PASS

The expected operational pattern for this sample is C4/C5/P9a susceptible with V resistant. This pattern can be cross-checked against the Prediction cards and the batch heatmap without

inspecting the numerical fixture file directly.

4.12.4. Golden sample CX3 (XI-1A)

CX3 is one of the deterministic validation fixtures exported with the final bundle. The table below records the exact expected inference outputs used to verify the local runtime. A PASS requires the thresholded class to match the fixture for all four strains and the probability vector to remain within the bundle tolerance.

Table 4.26 — Golden regression record for CX3.

Strain	P(Susceptible)	Threshold	|P-threshold|	Expected class	Predicted class	Result
C4	0.8724	0.31	0.5624	Susceptible	Susceptible	PASS
C5	0.8797	0.69	0.1897	Susceptible	Susceptible	PASS
V	0.0774	0.35	0.2726	Resistant	Resistant	PASS
P9a	0.9021	0.33	0.5721	Susceptible	Susceptible	PASS

The expected operational pattern for this sample is C4/C5/P9a susceptible with V resistant. This pattern can be cross-checked against the Prediction cards and the batch heatmap without inspecting the numerical fixture file directly.

4.12.5. Golden sample IRIS_313-11814 (XI-3)

IRIS_313-11814 is one of the deterministic validation fixtures exported with the final bundle. The table below records the exact expected inference outputs used to verify the local runtime. A PASS requires the thresholded class to match the fixture for all four strains and the probability vector to remain within the bundle tolerance.

Table 4.27 — Golden regression record for IRIS_313-11814.

Strain	P(Susceptible)	Threshold	|P-threshold|	Expected class	Predicted class	Result
C4	0.8361	0.31	0.5261	Susceptible	Susceptible	PASS
C5	0.8781	0.69	0.1881	Susceptible	Susceptible	PASS
V	0.0745	0.35	0.2755	Resistant	Resistant	PASS
P9a	0.9128	0.33	0.5828	Susceptible	Susceptible	PASS

The expected operational pattern for this sample is C4/C5/P9a susceptible with V resistant. This pattern can be cross-checked against the Prediction cards and the batch heatmap without

inspecting the numerical fixture file directly.

4.12.6. Golden sample IRIS_313-11797 (XI-1A)

IRIS_313-11797 is one of the deterministic validation fixtures exported with the final bundle. The table below records the exact expected inference outputs used to verify the local runtime. A PASS requires the thresholded class to match the fixture for all four strains and the probability vector to remain within the bundle tolerance.

Table 4.28 — Golden regression record for IRIS_313-11797.

Strain	P(Susceptible)	Threshold	|P-threshold|	Expected class	Predicted class	Result
C4	0.8714	0.31	0.5614	Susceptible	Susceptible	PASS
C5	0.8779	0.69	0.1879	Susceptible	Susceptible	PASS
V	0.0747	0.35	0.2753	Resistant	Resistant	PASS
P9a	0.9070	0.33	0.5770	Susceptible	Susceptible	PASS

The expected operational pattern for this sample is C4/C5/P9a susceptible with V resistant. This pattern can be cross-checked against the Prediction cards and the batch heatmap without inspecting the numerical fixture file directly.

4.12.7. Golden sample IRIS_313-8454 (XI-1A)

IRIS_313-8454 is one of the deterministic validation fixtures exported with the final bundle. The table below records the exact expected inference outputs used to verify the local runtime. A PASS requires the thresholded class to match the fixture for all four strains and the probability vector to remain within the bundle tolerance.

Table 4.29 — Golden regression record for IRIS_313-8454.

Strain	P(Susceptible)	Threshold	|P-threshold|	Expected class	Predicted class	Result
C4	0.8714	0.31	0.5614	Susceptible	Susceptible	PASS
C5	0.8777	0.69	0.1877	Susceptible	Susceptible	PASS
V	0.0807	0.35	0.2693	Resistant	Resistant	PASS
P9a	0.9100	0.33	0.5800	Susceptible	Susceptible	PASS

The expected operational pattern for this sample is C4/C5/P9a susceptible with V resistant. This pattern can be cross-checked against the Prediction cards and the batch heatmap without

inspecting the numerical fixture file directly.

4.12.8. Golden sample B097 (XI-adm)

B097 is one of the deterministic validation fixtures exported with the final bundle. The table below records the exact expected inference outputs used to verify the local runtime. A PASS requires the thresholded class to match the fixture for all four strains and the probability vector to remain within the bundle tolerance.

Table 4.30 — Golden regression record for B097.

Strain	P(Susceptible)	Threshold	|P-threshold|	Expected class	Predicted class	Result
C4	0.8744	0.31	0.5644	Susceptible	Susceptible	PASS
C5	0.8771	0.69	0.1871	Susceptible	Susceptible	PASS
V	0.0786	0.35	0.2714	Resistant	Resistant	PASS
P9a	0.9097	0.33	0.5797	Susceptible	Susceptible	PASS

The expected operational pattern for this sample is C4/C5/P9a susceptible with V resistant. This pattern can be cross-checked against the Prediction cards and the batch heatmap without inspecting the numerical fixture file directly.

4.13. Use-case-to-test traceability

A traceability matrix links each user-facing requirement to one or more executable tests. This prevents the test chapter from becoming a collection of unrelated screenshots: every test exists because it verifies a use case, a scientific input contract, or an explicit failure condition.

Table 4.31 — Traceability matrix connecting use cases, requirements, and tests.

Requirement / use case	Primary tests	Evidence	Acceptance
UC-01 Verify model bundle	FT-01 + eight golden records	SHA256 verification and Tables 4.23–4.30	All assets valid; all golden fixtures reproduce
UC-02 CLI batch prediction	FT-02, FT-03, FT-04, FT-05, FT-08, FT-12	Schema checks and batch report	Valid batches predicted; invalid batches rejected
UC-03 Interactive EGT prediction	UI-01 to UI-04, UI-07	Figures 4.1–4.4 and 4.8–4.9	Cards, heatmap, table and explanation agree with results

Requirement / use case	Primary tests	Evidence	Acceptance
UC-04 Model comparison and interpretation	UI-05, UI-06, UI-08, FT-09, FT-10	Figures 4.5–4.7 and 4.10–4.11	Benchmarks and model-2 artifact state displayed correctly
Exact EGT input contract	FT-04, FT-05, threshold logic	Input errors + manifest/bundle checks	No silent reshaping or generic thresholding
Bilingual presentation	FT-11	Language switch in sidebar	Numbers unchanged while labels switch language

4.14. Operator handoff checklist

Before the software is handed to another researcher, partner, or defense machine, the operator should archive not only the source code but also the exact model and test assets. The checklist below is intentionally redundant with the model manifest because it is written for a human operator rather than a parser.

1. Copy BB340_EGT_FINAL_MODEL_BUNDLE.zip and record the SHA256 of the ZIP itself.

2. Extract the bundle and confirm that exactly 22 expected files are present.

3. Run CLI verify and retain the PASS transcript or log.

4. Run the eight-sample demo and retain the generated CSV report.

5. Start the dashboard and confirm the EGT bundle path resolves before the defense.

6. If Plant-DNABERT is required live, confirm models_14locus reports 15/15 loaded.

7. Record the Python environment or install from requirements.txt in a clean environment.

8. Keep the exact 04_selected_2048_snps.csv beside any external genotype-harmonization procedure.

9. Do not replace thresholds with 0.5; use the thresholds stored in the bundle.

10. Do not describe attribution as causal genetic evidence.

11. State clearly that the EGT benchmark is validation performance under the 80/20

protocol.

Table 4.32 — Human operator handoff checklist.

Handoff item	Why it matters	Ready when
Final ZIP + ZIP SHA256	Identifies exact deployable package	Hash recorded
Unzipped 22-file bundle	Provides runtime assets	Inventory complete
CLI verification transcript	Confirms exact runtime reproduction	PASS
Golden prediction report	Provides human-readable regression evidence	8 samples x 4 strains
requirements / environment	Reduces dependency drift	Fresh environment can run
Plant-DNABERT artifacts	Needed only for model-2 LIVE operation	15/15 loaded if used
Input manifest	Prevents SNP-order/allele mistakes	04_selected_2048_snps.csv available

4.16. Internal inference data flow

The dashboard and CLI present different interfaces, but both follow the same internal data flow. This flow is important because it identifies where scientific validation occurs and where the software is merely formatting output. Input validation and preprocessing are service responsibilities; card colors, charts, tables, and downloadable CSV files are presentation responsibilities.

Table 4.33 — Internal EGT inference data flow shared by CLI and dashboard.

Stage	Input	Operation	Output / invariant
1. Bundle initialization	Bundle directory	Read manifest, hashes, model spec, thresholds, prior embeddings	Verified immutable runtime assets
2. Input parsing	CSV/NPZ/golden/synthetic	Identify sample IDs and dosage matrix	N x 2,048 raw A1-dosage matrix
3. Input validation	Raw matrix	Check dimensional contract and numeric convertibility	Reject malformed input before model call

Stage	Input	Operation	Output / invariant
4. Preprocessing	Raw dosage	NaN -> 0; divide by 2; reshape to N x 1 x 2,048	Training-compatible tensor
5. Prior preparation	11 x 1,280 frozen matrix	Broadcast same prior tokens across batch	N x 11 x 1,280 tensor
6. Neural inference	SNP tensor + prior tensor	EnhancedGenomicTransformer forward pass	N x 4 logits
7. Probability conversion	Logits	Sigmoid	P(Susceptible) for C4/C5/V/P9a
8. Decision	Probabilities + thresholds	Per-task comparison	Four binary labels
9. Explanation	Input + selected task output	Gradient x Input and aggregation	SNP/region evidence
10. Presentation and export	Prediction records	Cards, heatmaps, tables, CSV	User-facing result without changing model values

The separation in Table 4.33 is deliberately strict. For example, the dashboard does not normalize dosage independently from the service; it passes the raw matrix to the inference layer. Likewise, the dashboard does not invent thresholds. It displays the thresholds returned by the model service, which originate from the final bundle. This prevents interface code from becoming an untracked source of scientific preprocessing.

4.16.1. Batch processing and memory behavior

At deployment time the model no longer handles the original 4.8-million-SNP matrix. A batch contains only 2,048 dosage values per sample, so CPU inference is practical for small demonstration and analysis batches. The frozen gene-prior matrix is also small and is broadcast across the current batch. The most expensive interpretability operation is Gradient x Input, which requires gradient computation for selected outputs, but the dashboard uses only a modest number of samples in interactive sessions.

The application stores completed results and the current input frame in Streamlit session state. This allows Prediction, Batch overview, EGT, and download views to reuse the same

model run. A tab switch therefore does not retrain the model or repeat inference unless the user explicitly presses Run prediction with a new input.

4.16.2. Probability, threshold, and class semantics

The four outputs are independent sigmoid probabilities rather than a four-class probability distribution. A sample can therefore be susceptible to several strains simultaneously, resistant to all four, or display any other combination. The final class for each strain depends on its own threshold. This design matches the biological problem: C4, C5, V, and P9a are separate inoculation outcomes rather than mutually exclusive categories.

Table 4.34 — Operational interpretation rules for dashboard outputs.

Concept	Correct interpretation	Incorrect interpretation to avoid
P(Susceptible)	Independent probability for one strain	Probability that the sample belongs to one of four mutually exclusive strains
Threshold	Validation-selected cutoff specific to the strain	Universal 0.5 cutoff
Four heads	Four related binary tasks	Single four-class softmax
Heatmap color	Visual encoding of susceptibility probability	Biological severity scale in centimeters
Attribution sign	Direction of model contribution for current prediction	Causal effect allele

4.17. Error handling and failure-mode design

A reproducible research application must fail clearly when it cannot guarantee the model contract. Silent recovery is dangerous in genomics because a file with the correct number of columns can still contain the wrong markers, allele orientation, or feature order. The prototype therefore favors explicit error messages for structural problems and leaves biological harmonization to a documented preprocessing step.

Table 4.35 — Main software failure modes and explicit handling strategy.

Failure mode	Detection point	Current behavior	Why this is safer
Bundle directory missing	Service initialization	Run fails and dashboard reports pipeline error	Prevents fallback to unintended model

Failure mode	Detection point	Current behavior	Why this is safer
Bundle asset/hash mismatch	Bundle verification	Verification fails	Detects corrupted or modified model assets
NPZ missing required key	Input loader	Error lists required/present keys	Avoids interpreting unrelated arrays as genotype
Wrong EGT feature count	Input validation	Reject input	Prevents shifted SNP mapping
Plant-DNABERT feature count != 3843	Model-2 upload loader	Reject input	Preserves trained feature order
NaN/Inf in model-2 feature CSV	Model-2 validation	Reject input	Avoids undefined scaler/model behavior
Plant-DNABERT artifacts incomplete	Artifact inspection	LIVE status not granted	Avoids benchmark values being mistaken for live inference
No uploaded file in upload mode	Run preparation	No-input status/error	Avoids stale-session predictions
Explainability unavailable	Attribution rendering	Informational message, prediction retained	Separates core inference from optional visualization

4.17.1. Scientific failure modes not fully solved by the prototype

Some errors cannot be detected from a 2,048-column CSV alone. The application cannot know whether an external user counted A2 instead of A1, reversed strand orientation, or populated the columns with the correct SNPs in the wrong order if no manifest-based harmonization record accompanies the file. These are scientific data-integration failures rather than syntactic file errors.

For this reason, the next production step should be a dedicated harmonizer that accepts VCF or PLINK input, matches chromosome/position/alleles against 04_selected_2048_snps.csv, reports ambiguous or missing markers, and writes the exact dosage matrix consumed by the service. The harmonizer should produce an audit table rather than only the matrix, so the operator can see which source variant was mapped to each model feature.

4.18. Dashboard design rationale for research communication

The dashboard is not designed as a generic business interface. Its layout is optimized for communicating a genomic prediction experiment to a researcher or thesis reviewer. The

interface therefore exposes probability, threshold, model comparison, and attribution instead of presenting only a single 'resistant/susceptible' badge.

4.18.1. Why the sidebar contains model paths and input source

Displaying the EGT bundle path and Plant-DNABERT artifact path makes the active model version visible during a demonstration. This is preferable to hard-coding one hidden directory because the operator can immediately detect that the application is pointing to the wrong export. The model-2 LIVE 15/15 indicator further distinguishes complete live inference from a benchmark-only state.

The input-source radio buttons separate three purposes. Golden samples are for reproducibility and defense demonstration. Synthetic data are for interface testing without disclosing or preparing a real genotype. Upload mode is for operational batch use. Keeping these modes explicit reduces the risk that synthetic data are later described as real biological predictions.

4.18.2. Why the Prediction tab shows both cards and explanatory tables

The colored cards provide an immediate answer, but the tables directly below them define P(Susceptible), threshold semantics, and the meaning of strain names. This combination is intentional: a visual summary is useful for presentation, while explicit definitions reduce misinterpretation of the probability and class polarity.

The resistance-gene evidence panel and top-SNP table are placed below the prediction rather than above it. The prediction remains the primary model output; attribution is supporting evidence. This ordering discourages the user from treating a highlighted gene or SNP as a causal diagnostic before seeing the actual classification.

4.18.3. Why the dashboard includes model comparison and Plant-DNABERT

A thesis demonstration should show not only the proposed model but also the experimental context in which it was selected. The benchmark tab therefore keeps EGT Ver4.1 next to the original NT500M and AgroNT runs and separately displays the stricter Plant-DNABERT GroupKFold results. The interface does not merge these values into a single ranking because the evaluation protocols differ.

The Plant-DNABERT tab also demonstrates that the software architecture can host a second exported model family without retraining. Because the feature contract and output

strains differ from EGT, it is presented as a separate tab rather than as a selectable backend behind the same input widget.

4.19. Final Chapter 4 summary

Chapter 4 converted the experimental models into a reproducible software workflow with explicit interfaces, input contracts, failure handling, and regression tests. The CLI provides verification and batch operation, while the bilingual dashboard exposes per-sample predictions, batch summaries, benchmark context, and attribution without altering the underlying scientific preprocessing. The internal data-flow and failure-mode analysis shows that the application treats the model bundle, SNP manifest, thresholds, and frozen priors as immutable deployment assets.

Testing is anchored by eight real golden samples and is linked directly to the four user use cases through a traceability matrix. The resulting system is suitable for thesis demonstration and reproducible local inference, while its documented limitations make clear what remains for a production breeding workflow: external genotype harmonization, broader population-aware evaluation, persistent project management, and versioned deployment of future model revisions.

CHAPTER 5. CONTRIBUTIONS, LIMITATIONS, AND FUTURE WORK

5.1. Summary of findings

This thesis investigated rice bacterial-blight resistance as both a genomic prediction problem and a reproducible software-engineering problem. Using 3K-RG genomic resources and the 340-accession phenotype panel evaluated against C4, C5, V, and P9a, the study constructed three complementary pipelines rather than relying on a single representation or evaluation scheme. Pipeline A established a broad benchmark across genome-wide SNPs, Plant-DNABERT gene-locus embeddings, and raw DNA; Pipeline B introduced a matched negative-control design for candidate resistance/QTL regions; and Pipeline C proposed the Enhanced Genomic Transformer (EGT), which integrates a compact selected-SNP representation with frozen resistance-gene sequence priors.

The experiments show that evaluation protocol is a first-order methodological choice. Random splits can exploit rice population structure and therefore yield optimistic estimates, whereas subgroup-aware GroupKFold provides a stricter assessment for Pipelines A and B. Pipeline B strengthened the biological interpretation by processing candidate and matched random control regions through the same extraction and embedding pipeline; candidate embeddings achieved a positive candidate-minus-control GroupKFold MCC for every strain. Pipeline C demonstrated that 2,048 training-selected SNPs can be fused with eleven Nucleotide Transformer resistance-gene priors in a multi-task architecture with strain-specific thresholds and model-level attribution. The selected Ver4.1 configuration reached a validation mean MCC of approximately 0.548 under the inherited 80/20 protocol, but this value is deliberately reported as validation performance rather than an independent test estimate.

5.2. Main contributions

5.2.1. Reproducible multi-representation BB340 benchmark

The study provides a common cohort and a documented path from public 3K-RG resources to SNP, consensus-sequence, gene-locus, candidate-region, and selected-SNP representations. This makes representation choice an explicit experimental variable instead of an undocumented preprocessing decision.

5.2.2. Homology-based extraction with sequence-quality control

The sequence pipelines avoid fixed-coordinate extraction from accession consensus genomes. Homology anchoring, chromosome-length checks, and %N quality control reduce

errors caused by coordinate drift, missing sequence, and ambiguous mapping. The resulting quality artifacts make exclusions auditable.

5.2.3. Candidate-versus-control experimental design

Pipeline B evaluates biologically nominated regions against matched random control regions processed by the same sequence model. This design provides stronger evidence that a predictive advantage is associated with the candidate regions rather than merely with population structure or sequence missingness.

5.2.4. Enhanced Genomic Transformer

Pipeline C introduces a multimodal architecture that encodes ordered SNP dosages with multi-scale convolution, Squeeze-and-Excitation, and self-attention, then uses cross-attention to read eleven frozen resistance-gene prior tokens. Four task-specific heads jointly predict susceptibility to C4, C5, V, and P9a.

5.2.5. Model interpretation at SNP, region, and prior-token levels

Signed Gradient x Input attribution is exported for selected SNPs and aggregated over resistance/QTL windows, while the prior branch can be inspected by gene token. The interpretation is explicitly framed as model attribution and candidate evidence, not as proof of causal variation.

5.2.6. Deployable scientific software and model bundle

The project delivers a CLI, bilingual Streamlit dashboard, golden-sample regression tests, and a 22-file model bundle containing the checkpoint, exact 2,048-SNP order, chromosome/position/allele metadata, preprocessing contract, fixed prior sequences and embeddings, thresholds, split information, validation outputs, environment metadata, and SHA256 integrity records.

5.3. Limitations

5.3.1. Limited phenotype sample size and high dimensionality

Only a few hundred accessions have the required BB phenotypes, while the original genotype and sequence spaces contain millions of potential variables. This small-n/high-dimensional setting limits model complexity and increases uncertainty in fine-grained comparisons.

5.3.2. Different evaluation protocols across pipelines

Pipelines A and B use population-aware cross-validation, whereas EGT follows the

historical teacher-style 80/20 train/validation protocol. The EGT validation set is reused for checkpoint selection, threshold optimization, and task-weight selection; consequently, its selected validation score cannot be treated as an independent generalization estimate or compared directly with GroupKFold values as if the protocols were equivalent.

5.3.3. Severe class imbalance for strain V

The V phenotype contains very few resistant accessions. Although focal loss, positive-class weighting, MCC, ROC-AUC, and PR-AUC reduce the risk of misleading accuracy, performance for the minority class remains less stable than for the more balanced tasks.

5.3.4. Run-specific supervised SNP panel

The 2,048-SNP EGT panel is study-derived and depends on the training split, label conversion, ANOVA scores, biological seed windows, and selection rules. It is not an independently published 3K-RG marker panel and should not be regenerated at inference time.

5.3.5. Missing-genotype encoding

The current EGT preprocessing replaces missing genotype dosage with zero before normalization. This is reproducible with the trained model but conflates missingness with zero A1 dosage. A dedicated missingness indicator or statistically justified imputation strategy may improve future versions.

5.3.6. Fixed resistance-gene priors

The eleven EGT prior sequences are fixed reference snippets shared by all samples, not full accession-specific resistance-gene reconstructions. They provide stable biological context but do not represent individual allelic sequence differences at those genes.

5.3.7. Consensus-sequence and mapping uncertainty

Sequence-based pipelines depend on BAM coverage, reference alignment, consensus calling, and homology anchoring. High %N and repeat-rich resistance loci reduce the number of usable accession-gene pairs and can correlate with population structure.

5.3.8. Attribution is not causal genetics

Gradient-based importance indicates which selected inputs influence the trained predictor. It does not demonstrate molecular causality, linkage independence, or a genome-wide association p-value. Functional validation and independent genetic evidence remain necessary.

5.3.9. External genotype harmonization is not yet automated

The deployed application expects 2,048 A1-allele dosages in the exact model order. The current prototype does not yet convert arbitrary external VCF/PLINK input into this contract with allele/strand checks, so external users must prepare harmonized input carefully.

5.4. Future work

5.4.1. Unified population-aware evaluation

A high-priority next step is to freeze one model specification and evaluate all pipelines under the same repeated subgroup-aware split design or an independent locked test set. Repeated seeds and repeated group splits would allow reporting mean and variability rather than selecting among small validation differences.

5.4.2. Larger and more diverse phenotype panels

Future studies should incorporate additional rice accessions, Xoo strains, environments, and replicated lesion-length phenotypes. More diverse labels would improve the reliability of rare-strain prediction and permit stronger external validation.

5.4.3. Stronger biological priors

The fixed prior experiment can be extended with sequence priors derived from verified rice-reference loci or accession-specific local windows, under an explicitly pinned DNA foundation-model revision. Plant-specialized DNA encoders can be evaluated as separate controlled ablations rather than replacing the current prior retrospectively.

5.4.4. Missingness- and allele-aware genotype preprocessing

Future EGT versions should distinguish true zero dosage from missing genotype and quantify the effect of different imputation/masking strategies. An allele-aware harmonizer should map external VCF/PLINK variants to the exact 2,048-SNP manifest, report ambiguous strand/allele cases, and preserve an auditable conversion table.

5.4.5. Robust biological follow-up

High-attribution SNPs and regions should be compared with independent GWAS, functional annotations, haplotypes, LD structure, and experimentally characterized Xa loci. Such analyses should be reported as separate biological validation rather than being inferred from neural attribution alone.

5.4.6. Production-grade software extension

The CLI/dashboard can be extended with project histories, authenticated multi-user access,

model-version selection, server-side job queues, automatic report generation, and persistent provenance records. These features should call the same validated inference service so that software expansion does not duplicate or change scientific preprocessing.

5.5. Final conclusion

The central conclusion of this thesis is that trustworthy genomic resistance prediction requires more than maximizing a single performance metric. A defensible system must preserve data provenance, control population-structure leakage, distinguish biological signal from sequence-quality artifacts, document supervised feature selection, and make the trained preprocessing contract deployable. Within that framework, the three pipelines provide complementary evidence: genome-wide SNPs remain a strong benchmark, candidate-region embeddings show predictive signal beyond matched controls, and the Enhanced Genomic Transformer offers a compact multimodal model that combines accession-specific SNP variation with explicit resistance-gene context. The resulting CLI, dashboard, and model bundle convert these experiments into a reproducible software artifact while retaining the scientific limitations needed for responsible interpretation.

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APPENDICES

APPENDIX A. ELECTRONIC SUPPLEMENTARY DATA FILES

The complete supplementary data files supplied with this thesis are preserved as electronic appendix files. To keep the printed report readable, large spreadsheets are indexed here and representative excerpts or compact summaries are reproduced below; the original files remain unchanged in the accompanying electronic appendix package. The two matched-accession CSV files are byte-identical and are retained in the package exactly as supplied.

Source folder for the supplementary EGT result figures: https://drive.google.com/drive/folders/1lhJo32gDv8L76stCMPm5W_YVaRe6ZBva?usp=drive_link

Table A.1 - Electronic data-file inventory.

File	Type / role	Dimensions	SHA256 (first 16)
340 assession SNP Bacterial Blight(1).xlsx	Excel supplementary/study data	Sheet1: 1630x10; Sheet2: 0x0	e986e6bf7fb533bb
12284_2021_462_MOESM1_ESM.xlsx	Excel supplementary/study data	Sheet1: 9x4	373309d9953af4d7
12284_2021_462_MOESM2_ESM(1).xlsx	Excel supplementary/study data	Sheet1: 349x17	4da709109bbd1a02
12284_2021_462_MOESM4_ESM.xlsx	Excel supplementary/study data	Sheet1: 5x3	29293cbafedc6964
9_gene_BB_thong_tin_khoa_hoc_va_trinh_tu(1).xlsx	Excel supplementary/study data	9 gene BB: 14x36	5e8ce0a8d03a7fce
dataset_and_model_summary(1).csv	CSV study artifact	18 rows x 2 columns	25001407337d0e5f
matched(1).csv	CSV study artifact	327 rows x 22 columns	d37e9204f9766c3f
matched(2).csv	CSV study artifact	327 rows x 22 columns	d37e9204f9766c3f

File	Type / role	Dimensions	SHA256 (first 16)
results_ALL_3branch(1).csv	CSV study artifact	171 rows x 12 columns	63d047623f242058
gene_quality(1).csv	CSV study artifact	7 rows x 6 columns	b593d4d81455a3d5

A.1. Dataset and model summary

Table A.2 - Dataset and model summary exported by the study pipeline.

Item	Value
Total matched accessions	340
Original SNPs	4817964
Selected SNPs	2048
Train samples	272
Validation samples	68
Evaluation protocol	Teacher 80/20 validation
DNA foundation model	InstaDeepAI/nucleotide-transformer-500m-human-ref
DNA embedding dimension	1280
R-gene prior tokens	11
Focal gamma	1.5
Learning rate	0.0003
Weight decay	0.02
Max epochs	60
Early stopping	Disabled (full 60 epochs)
Winning task-weight experiment	NT500M_C4_1.150
Winning weights [C4,C5,V,P9a]	[1.15, 1.0, 1.0, 1.0]
Best validation MCC	0.5479874455427887
Best validation epoch	36

A.2. Gene-level sequence quality summary

Table A.3 - Full gene_quality.csv content.

Gene	Mean %N	n_ok	Clean
Xa1	20.35	254	False

Gene	Mean %N	n_ok	Clean
Xa21	7.62	310	True
Xa27	51.05	187	False
Xa3_Xa26	45.98	173	False
xa13_SWEET11	0.53	327	True
xa25_SWEET13	3.76	327	True
xa5	0.37	327	True

A.3. Lu et al. supplementary primers

Table A.4 - Lu et al. Table S1 primers used in the bacterial-blight study.

Name	Forward Primers(5'->3')	Reverse Primers (5'->3')	Purpose
LOC_Os11g46250-ge	CCCACTGGGATCGAGGACTA	GTCTGCCACGTCAGCTTGCCG	Primers for amplification of genomic sequence of LOC_Os11g46250
LOC_Os11g46250-cp	CGGTACCCGGGGATCCCCCACTGGGATCGAGGACTA	GGCCAGTGCCAAGCTTGTCTGCCACGTCAGCTTGCCG	Primers for complementary construct
Hyg	GATGTTGGCGACCTCGTATT	GATGTAGGAGGGCGTGGATA	Primers for hygromycin
LOC_Os11g46250-test	ATTCTCGGTGACGGATTTAC	TGTAAAACGACGGCCAGT	Primers for testing the complementary construct
UBQ-qRT	GCTCCGTGGCGGTATCAT	CGGCAGTTGACAGCCCTAG	Primers for analysis of the interval control gene expressinon level
LOC_Os11g46250-qRT1	TACTACGTATTGTATTGCCT	TGTGGTATTTGATGCTGGAA	Primers for analysis of LOC_Os11g46250Hap1expression level
LOC_Os11g46250-qRT2	CACTACGTATTGTATTGCCTCCT	CGTGGTATTTGATGCTGGAG	Primers for analysis of LOC_Os11g462

Name	Forward Primers(5'->3')	Reverse Primers (5'->3')	Purpose
			50Hap2 expression level

A.4. BB340 phenotype-panel excerpt

Table A.5 - Representative excerpt from Lu et al. Table S2 (complete 340-accession file supplied electronically).

3K_ID	Variety	Origin	Region	Subgroup	C4 level	C4 LL	C5 level	C5 LL	V level	V LL	P9a level	P9a LL
B015	Kahamu	Romania	EUR	XI-1A	S	22.2222222222222222	S	22.362499999999997	S	25.333333333333332	S	18.747058823529414
B061	Guangluai 4	China	EAS	XI-1A	MR	9.944444444444445	MS	10.282352941176471	S	21.611111111111111	MS	13.618749999999999
B081	Heidu 4	China	EAS	XI-1A	S	20.777777777777778	S	15.972222222222222	S	32.555555555555556	S	17.516666666666667
B094	Zinuo	China	EAS	XI-1A	S	17	S	19.9	S	22	S	20.7
B149	Haoxiang	China	EAS	XI-1A	S	24.888888888888889	S	23.8	S	34.777777777777778	S	25.9
B157	88B	China	EAS	XI-1A	R	1	MS	12.055555555555555	S	31.444444444444443	MS	13
B197	Taizhongxianxuan 2	Taiwan	EAR	XI-1A	R	2.9444444444444446	S	15.525	S	30	R	3.159999999999999997
B208	Lucaihao	China	EAS	XI-1A	MS	12.444444444444444	MS	13.05	S	35.888888888888886	S	18.11875
B210	Nanxiongzaoyouzhan	China	EAS	XI-1A	MR	7.222222222222222	S	24.831250000000004	S	26.666666666666668	S	26.958823529411767
B216	Hanmadao4	China	EAS	XI-1A	MS	13.611111111111111	S	20.461111111111111	S	31.555555555555557	S	17.5
B221	Fanhaopi	China	EAS	XI-1A	MS	12.611111111111111	S	20.9	S	34.444444444444444	S	26.4
B261	Meihuanuo	China	EAS	XI-1A	S	19.444444444444443	S	18.433333333333333	S	32.111111111111114	S	23.046666666666667
B264	Sanbaili	China	EAS	XI-1A	MS	14.777777777777779	S	24.930769230769230	S	32.111111111111114	S	29.393333333333333

3K_ID	Variety	Origin	Region	Subgroup	C4 level	C4 LL	C5 level	C5 LL	V level	V LL	P9a level	P9a LL
CX3	BR 24	Bangladesh	SAC	XI-1A	S	19.5	MS	14.6555555555553	S	32.77777777777778	S	22.174999999999997
CX10	Gang 46B	China	EAS	XI-1A	S	24	MS	14.1	S	36.111111111111114	S	15.2
CX101	Hei Mi Chan	China	EAS	XI-1A	S	27.88888888888889	S	18.93125	S	34.88888888888886	S	18.73125
CX102	Tek Si Chut	Taiwan	EAR	XI-1A	MR	6.222222222222222	S	15.180000000000001	S	32.77777777777778	S	16.94
CX162	TN1	Taiwan	EAR	XI-1A	S	18.555555555555557	S	22.575	S	28.888888888888889	S	23.66875
CX270	TAICHUNG NATIVE1	Taiwan	EAR	XI-1A	MR	7.416666666666667	S	16.950000000000003	MS	14.333333333333334	S	19.511764705882353
IRIS_313-8454	LAI YIP ZIM::IRGC 4955-1	Taiwan	EAR	XI-1A	S	22.666666666666668	S	26.461538461538461	S	33	S	31.888888888888889

A.5. Significant-SNP annotation excerpt

Table A.6 - First 25 rows of the significant-SNP annotation file (complete file supplied electronically).

QTL Name	Significant SNPa	Chromosome	Position(bp)	P-value	Positive allele	Negative allele	Variation region	Corresponding gene
qC4-11	rs11_27482608	11.0	27482608.0	5.51e-08	A	G	upstream_gene_variant	LOC_Os11g45400
	rs11_27489727	11.0	27489727.0	4.33e-08	G	A	synonymous_variant	LOC_Os11g45400
	rs11_27490677	11.0	27490677.0	3.7e-08	C	T	downstream_gene_variant	LOC_Os11g45400
	rs11_27491623	11.0	27491623.0	6.24e-08	G	T	downstream_gene_variant	LOC_Os11g45400
qC5-4	rs4_31060503	4.0	31060503.0	2.36e-08	T	C	upstream_gene_variant	LOC_Os04g52260
	rs4_31071832	4.0	31071832.0	1.6e-08	G	A	upstream_gene_variant	LOC_Os04g52270
qC5-5	rs5_331582	5.0	331582.0	3.7e-08	A	G	upstream_gene_variant	LOC_Os05g01560
	rs5_335278	5.0	335278.0	3.7e-08	G	A	upstream_gene_variant	LOC_Os05g01570
	rs5_335846	5.0	335846.0	3.7e-08	C	T	upstream_gene_variant	LOC_Os05g01570

QTL Name	Significant SNPa	Chromosome	Position(bp)	P-value	Positive allele	Negative allele	Variation region	Corresponding gene
	rs5_336339	5.0	336339.0	2.68e-08	G	A	upstream_gene_variant	LOC_Os05g01570
	rs5_339828	5.0	339828.0	3.7e-08	T	C	intron_variant	LOC_Os05g01570
	rs5_340176	5.0	340176.0	3.7e-08	G	C	missense_variant	LOC_Os05g01570
	rs5_341755	5.0	341755.0	3.7e-08	A	G	upstream_gene_variant	LOC_Os05g01580
	rs5_341937	5.0	341937.0	5.92e-08	T	C	upstream_gene_variant	LOC_Os05g01580
	rs5_342109	5.0	342109.0	3.7e-08	A	C	upstream_gene_variant	LOC_Os05g01580
	rs5_343467	5.0	343467.0	3.74e-09	T	C	upstream_gene_variant	LOC_Os05g01580
	rs5_343510	5.0	343510.0	3.74e-09	C	T	upstream_gene_variant	LOC_Os05g01580
	rs5_343923	5.0	343923.0	4.55e-08	A	T	upstream_gene_variant	LOC_Os05g01580
	rs5_345033	5.0	345033.0	4.29e-08	G	A	upstream_gene_variant	LOC_Os05g01580
	rs5_346882	5.0	346882.0	2.27e-08	G	A	upstream_gene_variant	LOC_Os05g01590
	rs5_347288	5.0	347288.0	4.17e-08	G	C	missense_variant	LOC_Os05g01580
	rs5_347591	5.0	347591.0	3.88e-10	T	A	upstream_gene_variant	LOC_Os05g01590
	rs5_348262	5.0	348262.0	2.81e-10	G	A	upstream_gene_variant	LOC_Os05g01590
	rs5_350571	5.0	350571.0	1.16e-08	C	G	upstream_gene_variant	LOC_Os05g01590
	rs5_351345	5.0	351345.0	1.06e-09	C	A	upstream_gene_variant	LOC_Os05g01590

A.6. Resistance-gene reference manifest excerpt

Table A.7 - Resistance-gene scientific metadata/reference-sequence manifest excerpt.

Gene / locus	Chromosome	IRGSP Start	IRGSP End	Strand	Reference build
Xa1	4	31637907	31644734	+	IRGSP-1.0
Xa21	11	21273533	21277443	-	IRGSP-1.0
Xa3/Xa26	11	28399360	28402773	+	IRGSP-1.0
xa5	5	437043	443269	+	IRGSP-1.0
Xa10	11			-	Donor-specific allele
xa13/SWEET11	8	26725954	26728807	-	IRGSP-1.0

Gene / locus	Chromosome	IRGSP Start	IRGSP End	Strand	Reference build
Xa23	11			-	Donor-specific allele
xa25/SWEET13	12	17302127	17305326	-	IRGSP-1.0
Xa27	6	23653851	23654303	-	IRGSP-1.0

A.7. Matched phenotype-consensus cohort excerpt

Table A.8 - First 20 matched accessions (complete 327-row file supplied electronically).

id	name	origin	region	subgroup	C4_lvl	C4_LL	C5_lvl	C5_LL	V_lvl	V_LL	P9a_lvl	P9a_LL
B015	Kahamu	Romania	EUR	XI-1A	S	22.22222222222222	S	22.3625	S	25.333333333333333	S	18.747058823529 41
B061	Guangluai 4	China	EAS	XI-1A	MR	9.944444444444445	MS	10.282352941176471	S	21.611111111111111	MS	13.61875
B081	Heidu 4	China	EAS	XI-1A	S	20.777777777777778	S	15.972222222222222	S	32.555555555555556	S	17.516666666666667
B094	Zinuo	China	EAS	XI-1A	S	17.0	S	19.9	S	22.0	S	20.7
B149	Haoxiang	China	EAS	XI-1A	S	24.888888888888889	S	23.8	S	34.777777777777778	S	25.9
B157	88B	China	EAS	XI-1A	R	1.0	MS	12.055555555555555	S	31.444444444444443	MS	13.0
B197	Taizhongxianxuan 2	Taiwan	EAR	XI-1A	R	2.944444444444446	S	15.525	S	30.0	R	3.16
B208	Lucaihao	China	EAS	XI-1A	MS	12.4444444444444	MS	13.05	S	35.888888888888886	S	18.11875
B210	Nanxiongzaoyouzhan	China	EAS	XI-1A	MR	7.222222222222222	S	24.831250000000004	S	26.666666666666668	S	26.958823529411767
B216	Hanmadao4	China	EAS	XI-1A	MS	13.611111111111111	S	20.461111111111111	S	31.555555555555556	S	17.5
B221	Fanhaopi	China	EAS	XI-1A	MS	12.611111111111111	S	20.9	S	34.444444444444444	S	26.4
B261	Meihuanuo	China	EAS	XI-1A	S	19.444444444444443	S	18.433333333333333	S	32.111111111111114	S	23.046666666666667
B264	Sanbaili	China	EAS	XI-1A	MS	14.777777777777778	S	24.930769230769 23	S	32.111111111111114	S	29.393333333333333
CX3	BR 24	Bangladesh	SAC	XI-1A	S	19.5	MS	14.655555555555551	S	32.777777777777778	S	22.175
CX10	Gang 46B	China	EAS	XI-1A	S	24.0	MS	14.1	S	36.111111111111112	S	15.2
CX101	Hei Mi Chan	China	EAS	XI-1A	S	27.888888888888889	S	18.93125	S	34.888888888888886	S	18.73125
CX102	Tek Si Chut	Taiwan	EAR	XI-1A	MR	6.222222222222222	S	15.18	S	32.777777777777778	S	16.94
CX162	TN1	Taiwan	EAR	XI-1A	S	18.555555555555556	S	22.575	S	28.888888888888889	S	23.66875

id	name	origin	region	subgroup	C4_lvl	C4_LL	C5_lvl	C5_LL	V_lvl	V_LL	P9a_lvl	P9a_LL
CX270	TAICHUNG NATIVE1	Taiwan	EAR	XI-1A	MR	7.416666666666667	S	16.950000000000003	MS	14.333333333333334	S	19.511764705882356
IRIS_313-8454	LAI YIP ZIM::IRGC 4955-1	Taiwan	EAR	XI-1A	S	22.666666666666668	S	26.46153846153846	S	33.0	S	31.888888888888889

A.8. Three-branch model-result summary

Table A.9 - Aggregated summary of results_ALL_3branch.csv (complete 171-row result file supplied electronically).

branch	family	eval	n	Mean_MCC	Mean_AUC	Mean_Accuracy
DNA_tho	deep	CV_group	9	0.182	0.620	0.624
DNA_tho	deep	CV_random	9	0.272	0.690	0.649
DNA_tho	deep	independent_test	9	0.218	0.689	0.626
SNP	classical	CV_group	12	0.200	0.664	0.690
SNP	classical	CV_random	12	0.357	0.777	0.729
SNP	classical	independent_test	12	0.467	0.828	0.764
SNP	deep	CV_group	12	0.144	0.618	0.645
SNP	deep	CV_random	12	0.283	0.713	0.671
SNP	deep	independent_test	12	0.349	0.744	0.689
embedding	classical	CV_group	12	0.183	0.663	0.680
embedding	classical	CV_random	12	0.300	0.731	0.712
embedding	classical	independent_test	12	0.318	0.757	0.720
embedding	deep	CV_group	12	0.149	0.608	0.633
embedding	deep	CV_random	12	0.273	0.692	0.677
embedding	deep	independent_test	12	0.258	0.686	0.654

A.9. Lu et al. haplotype supplementary file

Table A.10 - Compact view of Lu et al. Table S4; the complete haplotype pattern is preserved in the electronic spreadsheet.

Haplotype	Variant-pattern preview	Accession No.
Hap1	----------...	208
Hap2	acagatgccgcaaagaaacatgcaacgcactcacccatgattggaacacggctacaccctatgctaactgggaaactgtaacaccgtccg...	30
Hap3	acGgatgccgcaaagaaacatgcaacgcactcTcccatgattggaacacggctacaccctatgctaactgggaaactgtaacaccgtctg...	18

APPENDIX B. ENHANCED GENOMIC TRANSFORMER SUPPLEMENTARY FIGURES

The following figures are embedded from the supplied Google Drive results folder. PNG renderings are included in the report for direct inspection; the source folder also contains corresponding PDF versions for vector-quality reuse.

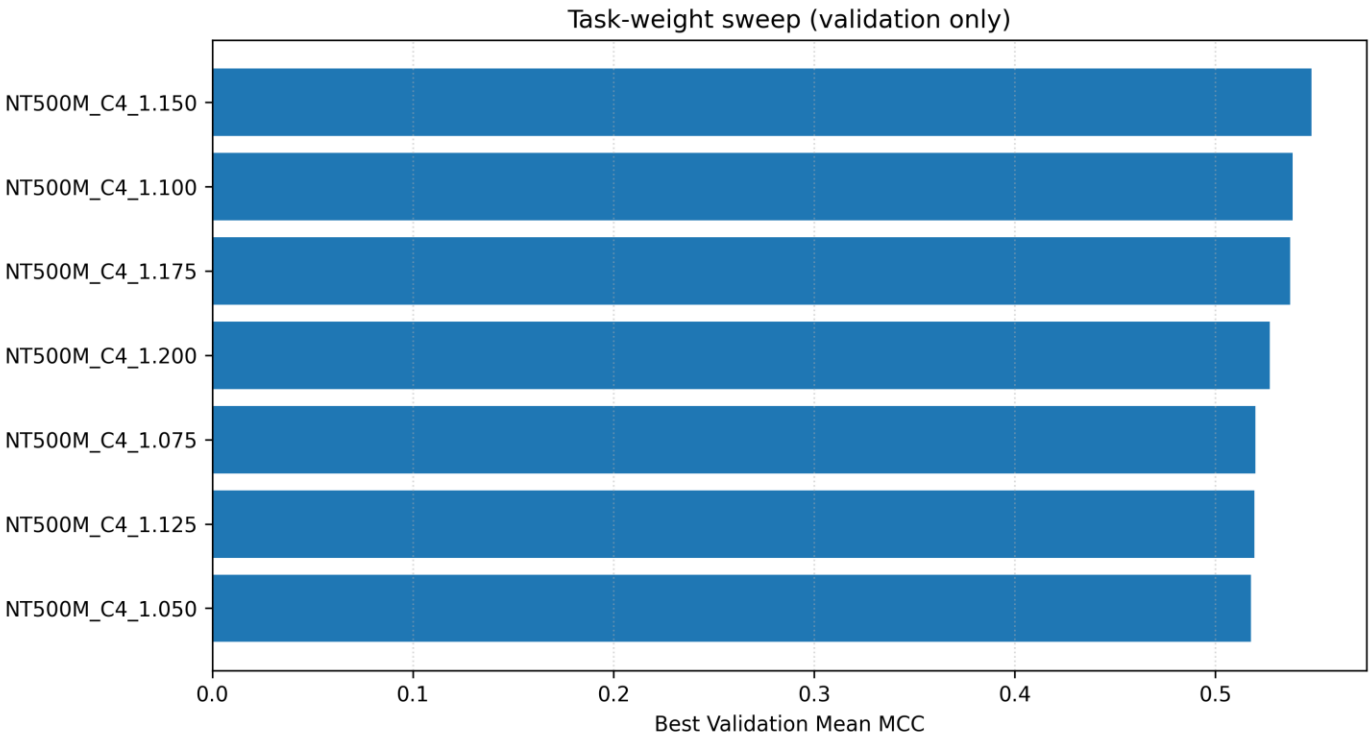


Figure B.1 - Task-weight sweep validation Mean MCC.

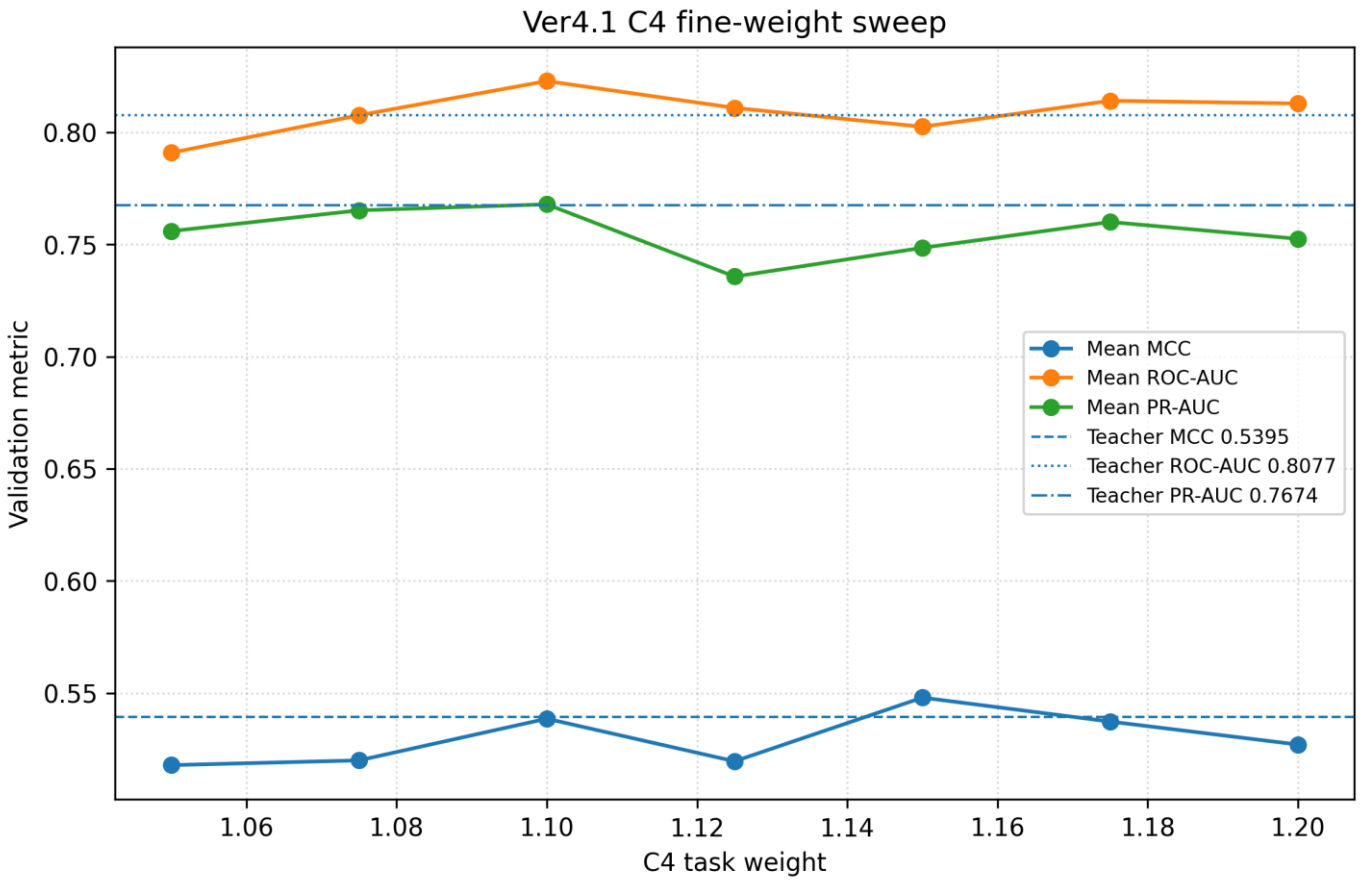


Figure B.2 - C4 fine-sweep metrics.

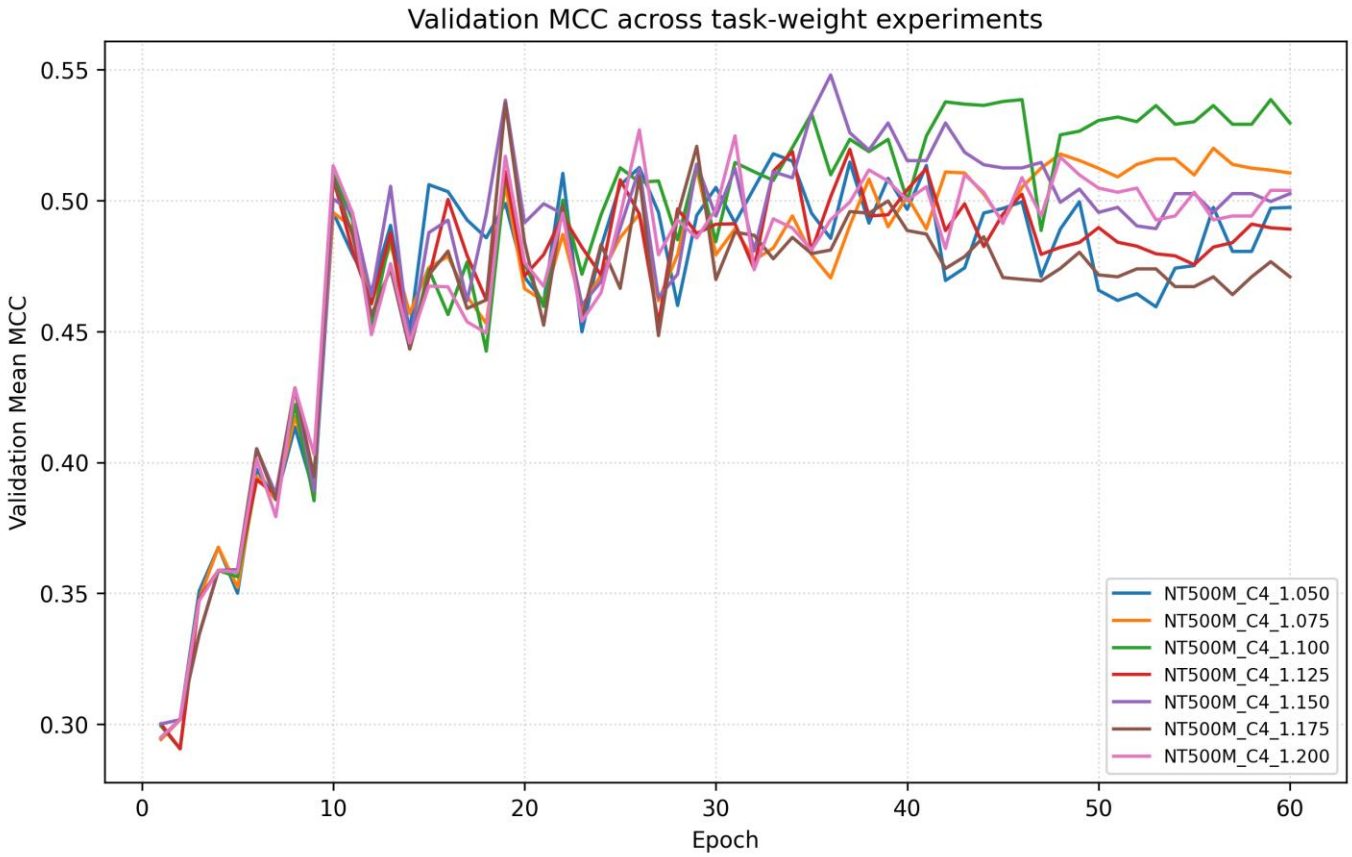


Figure B.3 - Validation Mean MCC by epoch.

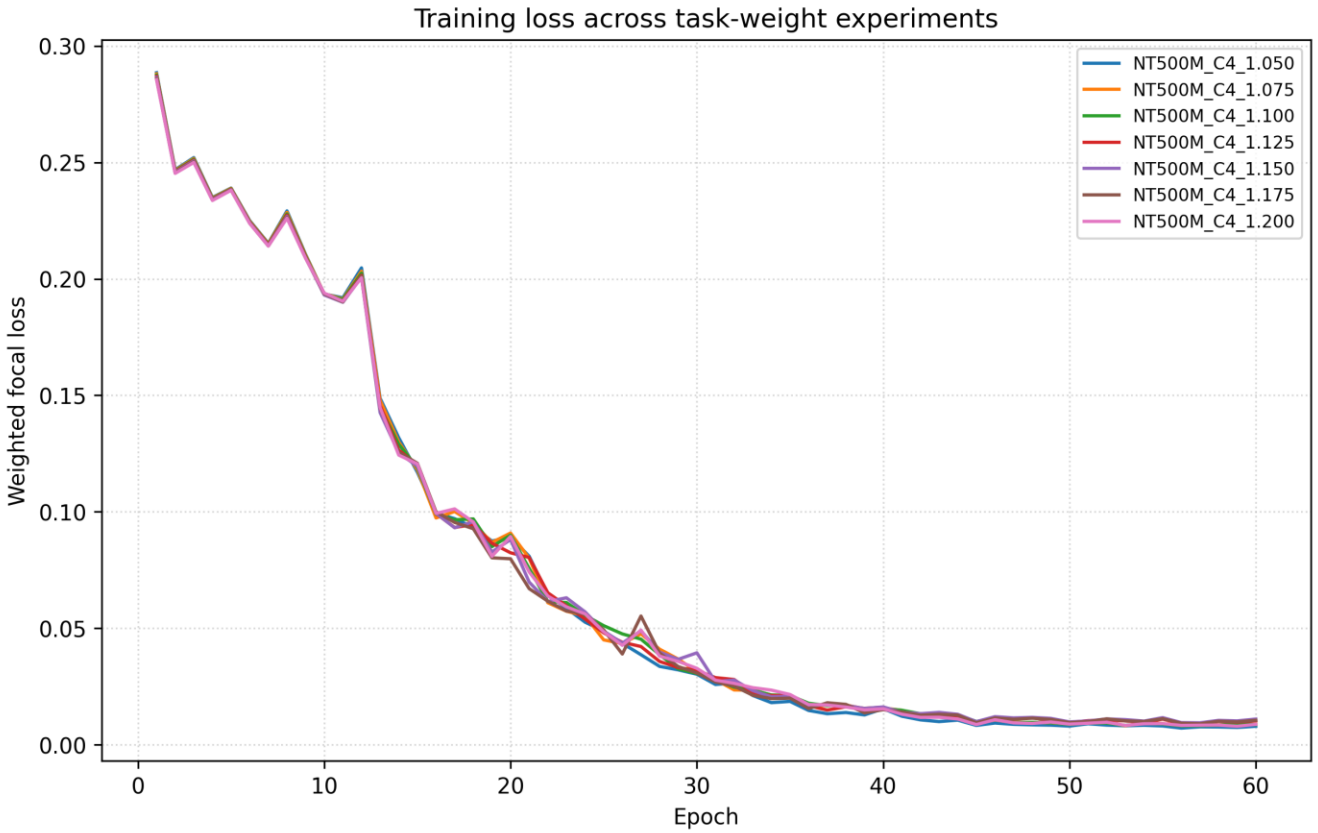


Figure B.4 - Training loss by epoch.

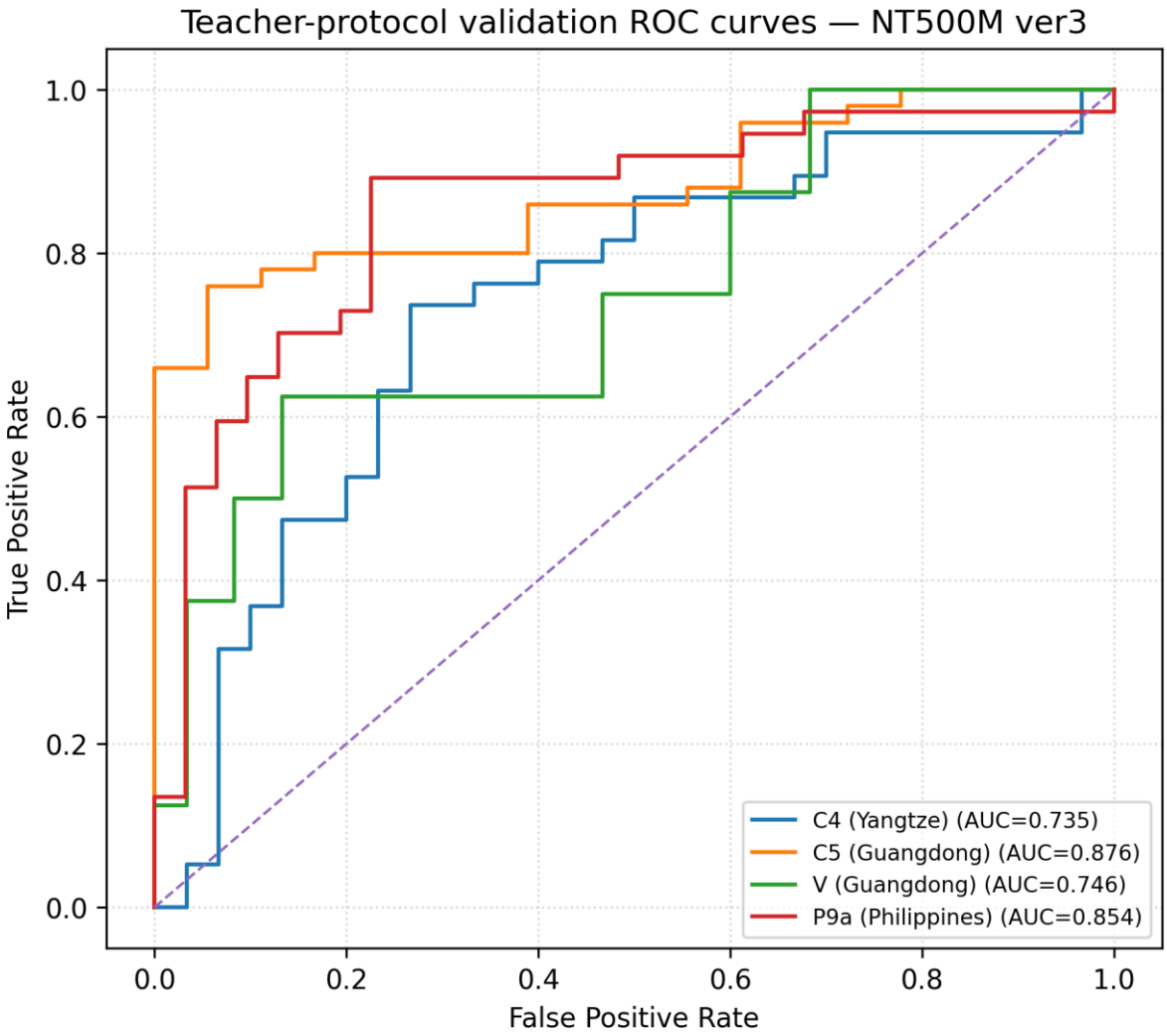


Figure B.5 - Teacher-protocol validation ROC curves.

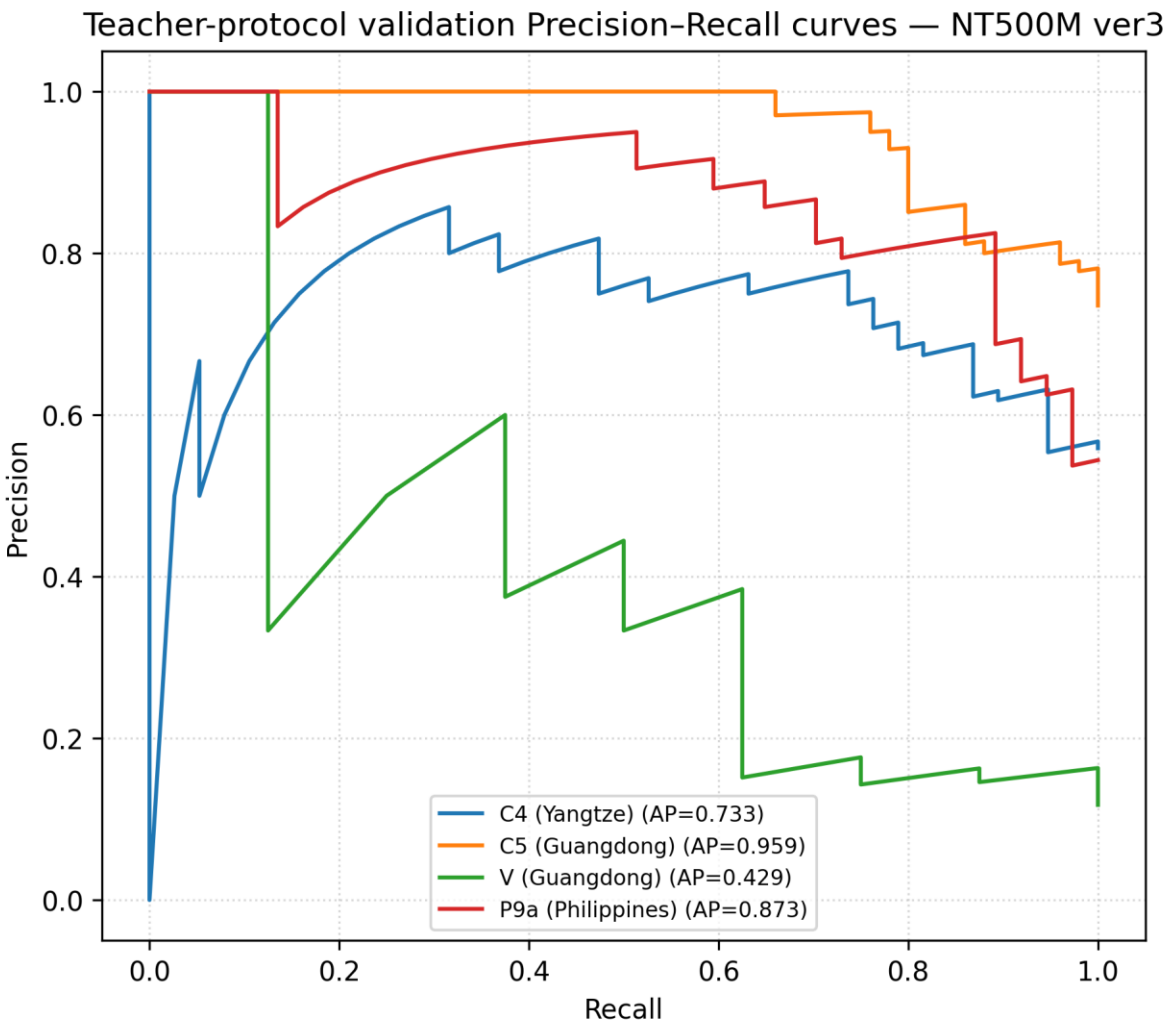


Figure B.6 - Teacher-protocol validation precision-recall curves.

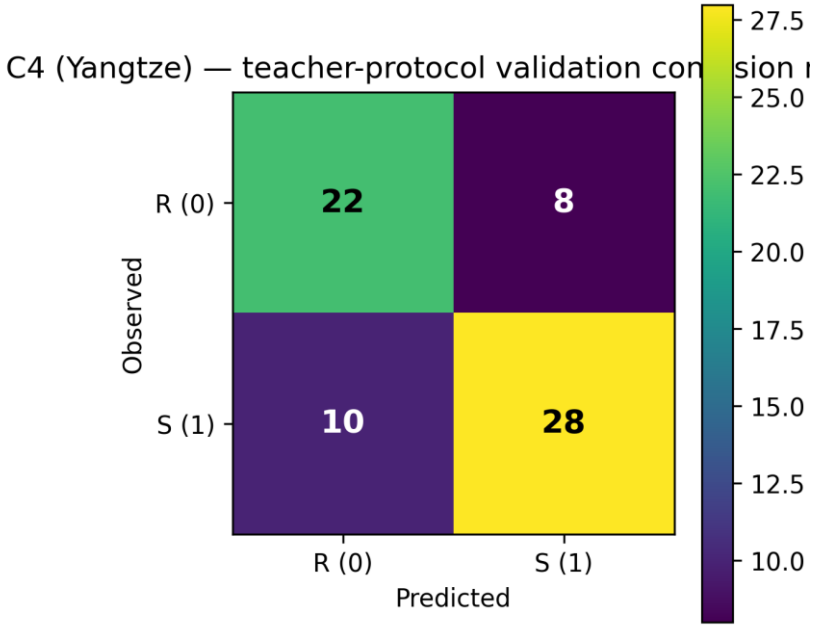


Figure B.7 - C4 confusion matrix.

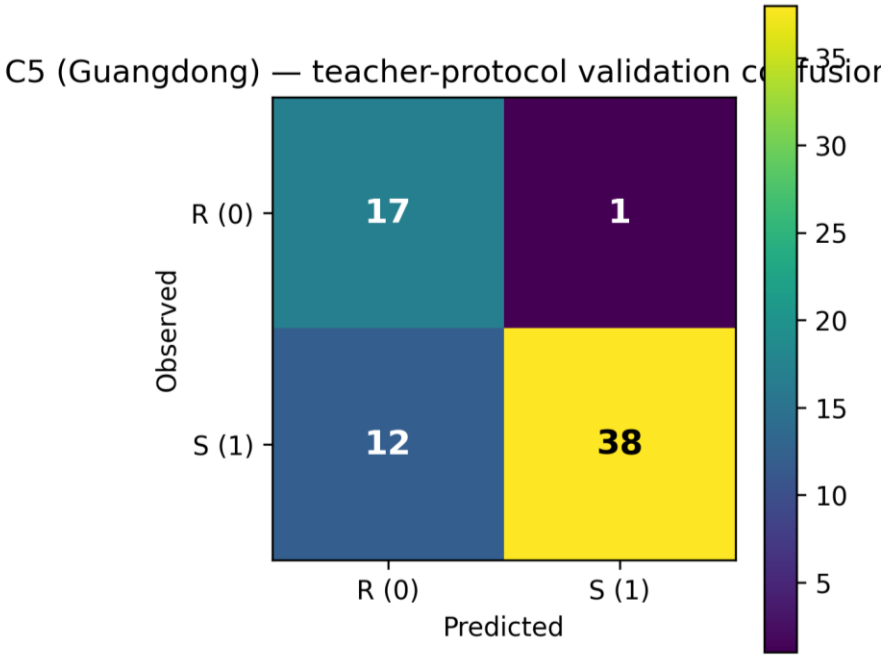


Figure B.8 - C5 confusion matrix.

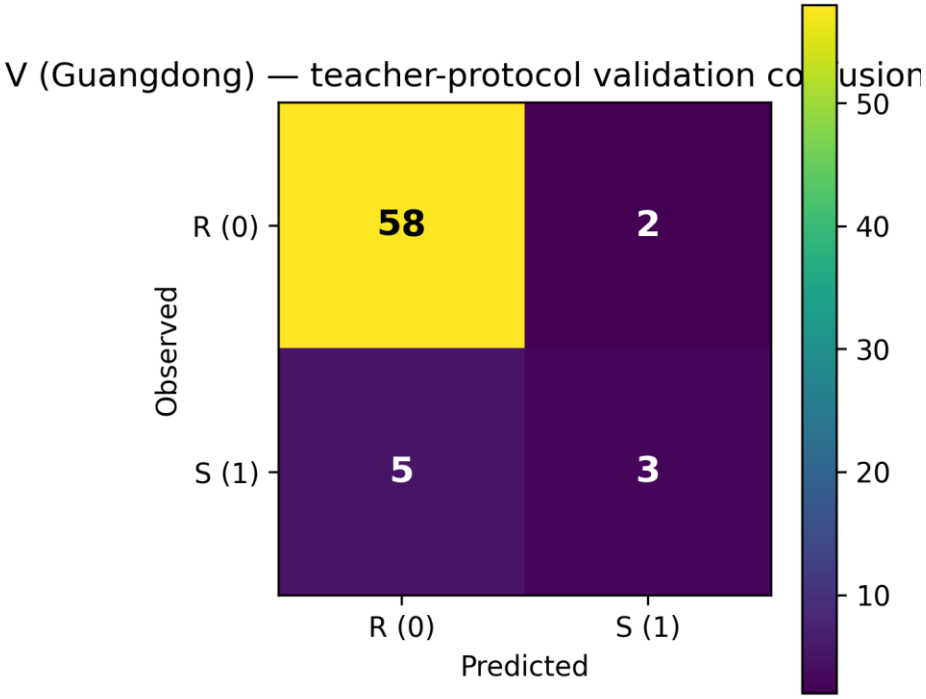


Figure B.9 - V confusion matrix.

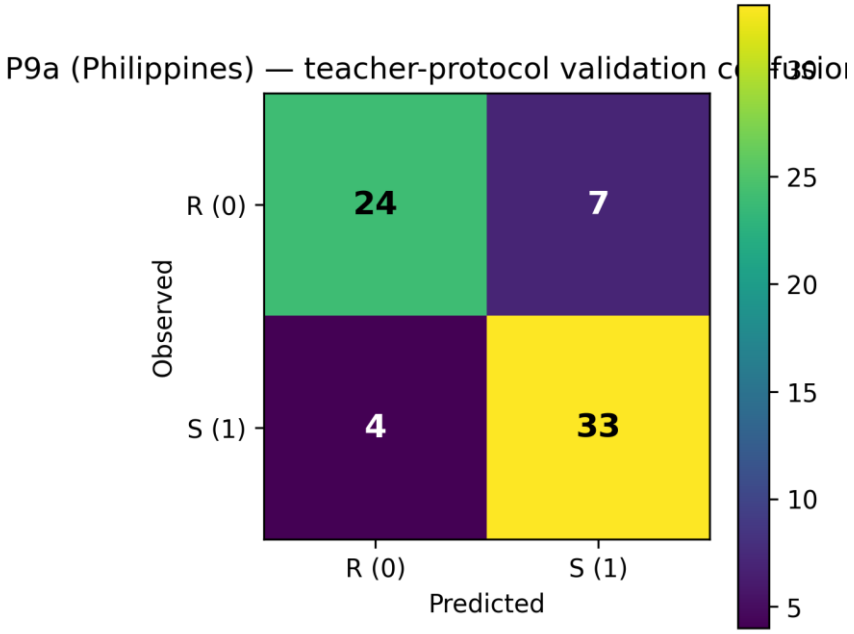


Figure B.10 - P9a confusion matrix.

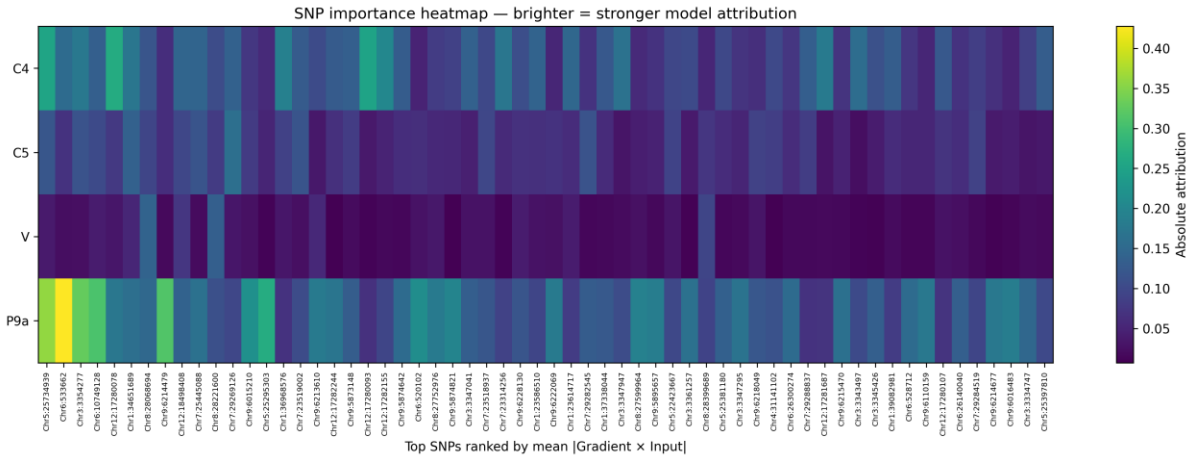


Figure B.11 - Top-60 selected-SNP importance heatmap.

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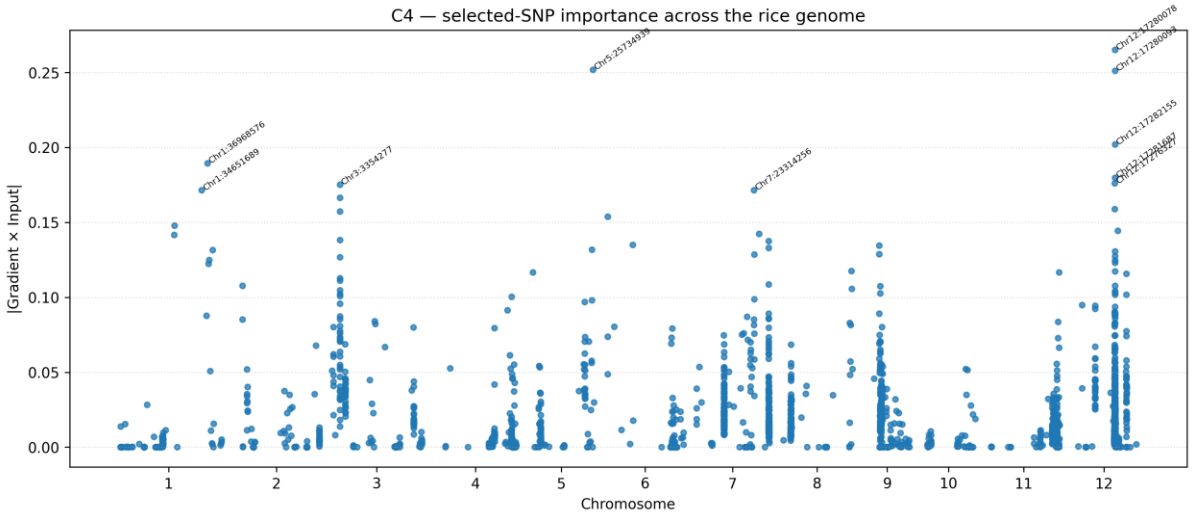


Figure B.12 - C4 Manhattan-like selected-SNP importance plot.

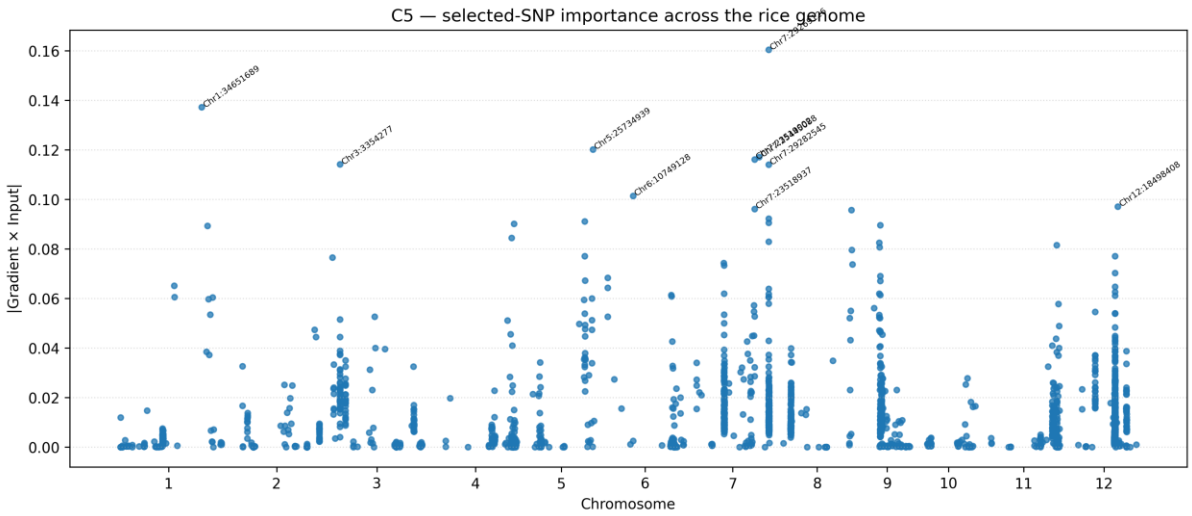


Figure B.13 - C5 Manhattan-like selected-SNP importance plot.

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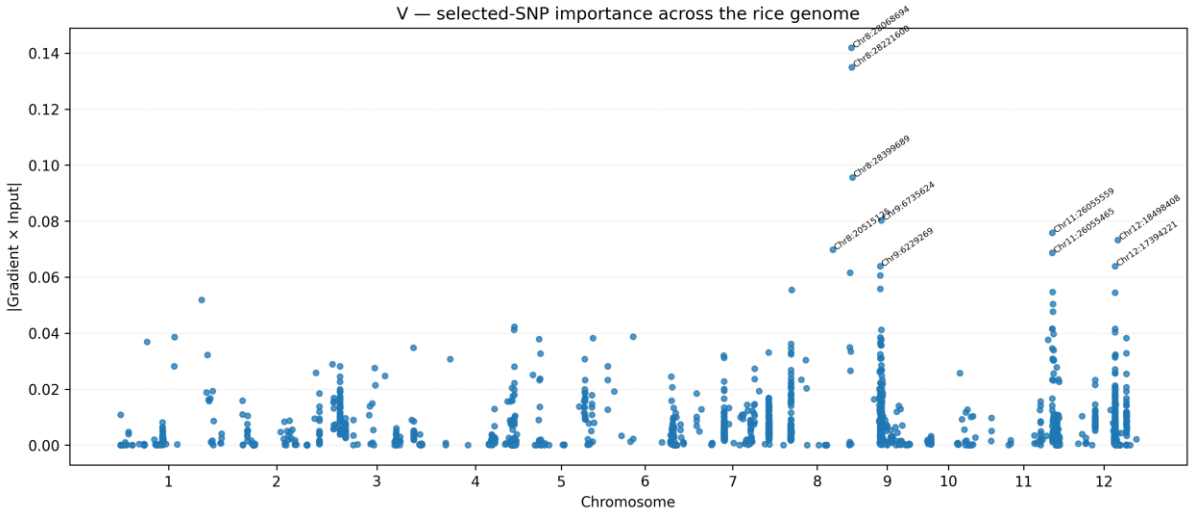


Figure B.14 - V Manhattan-like selected-SNP importance plot.

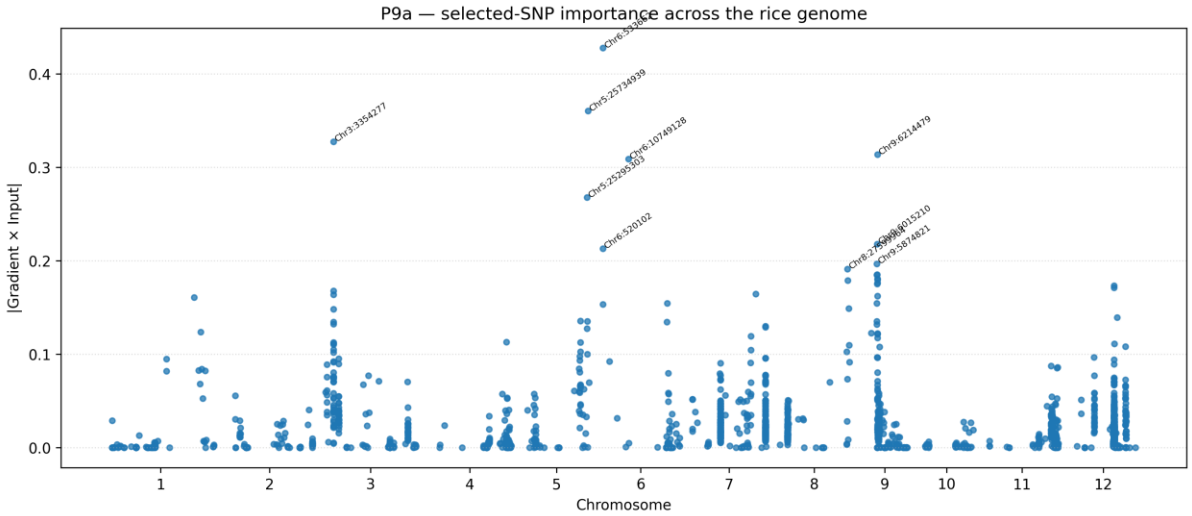


Figure B.15 - P9a Manhattan-like selected-SNP importance plot.

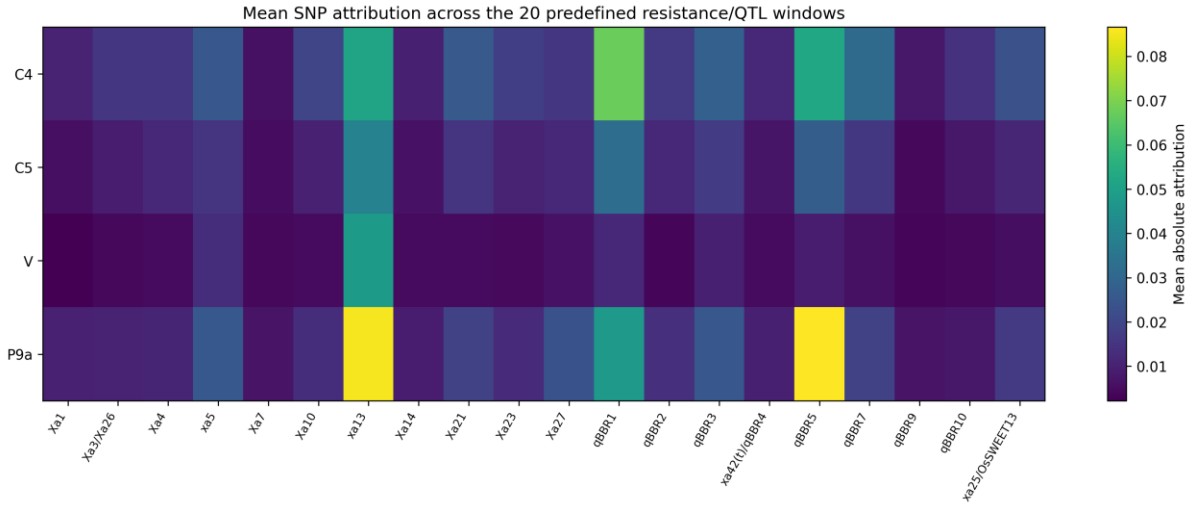


Figure B.16 - Resistance/QTL-region importance heatmap.

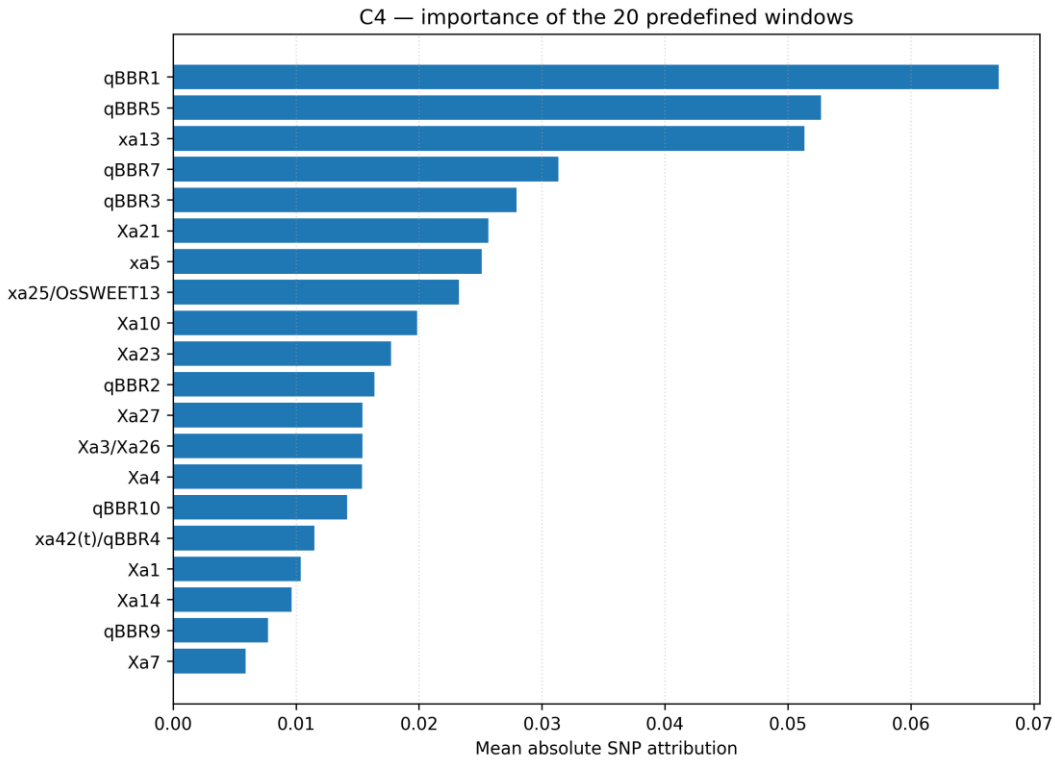


Figure B.17 - C4 resistance/QTL-region attribution bar chart.

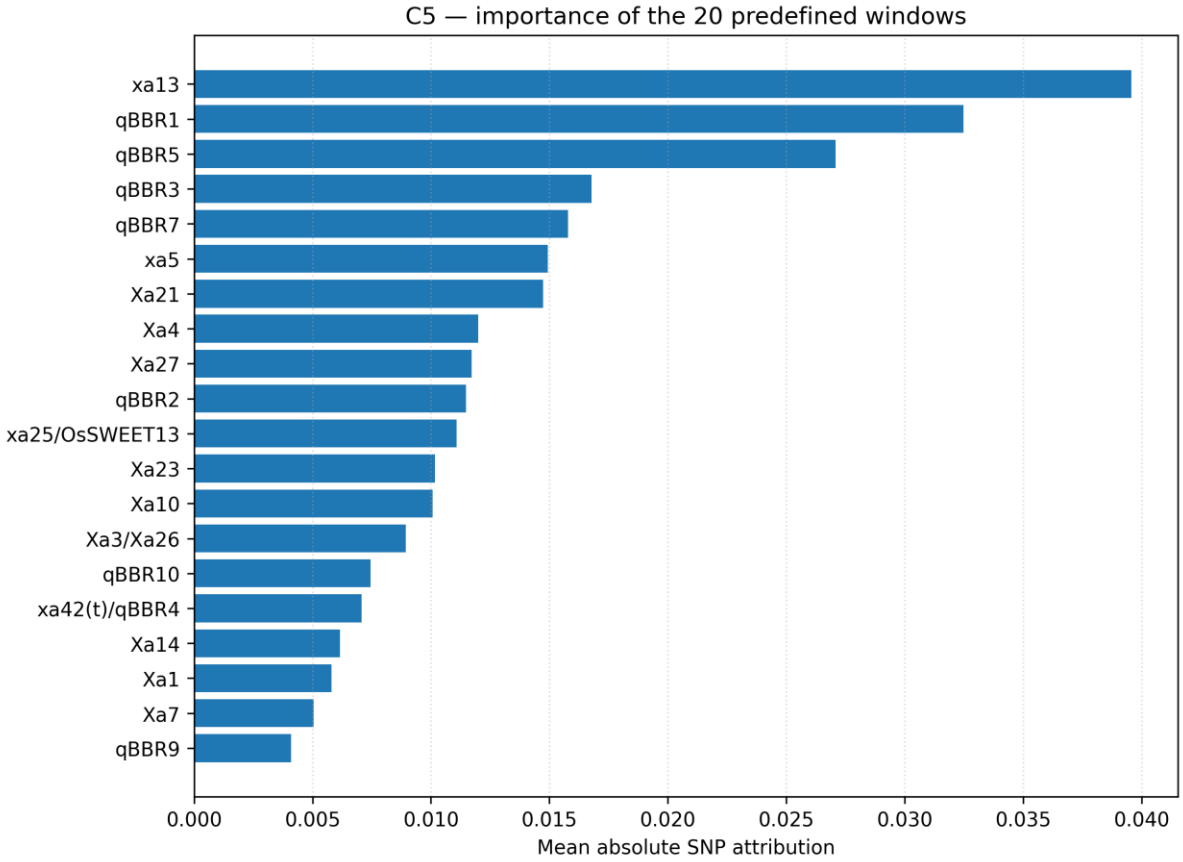


Figure B.18 - C5 resistance/QTL-region attribution bar chart.

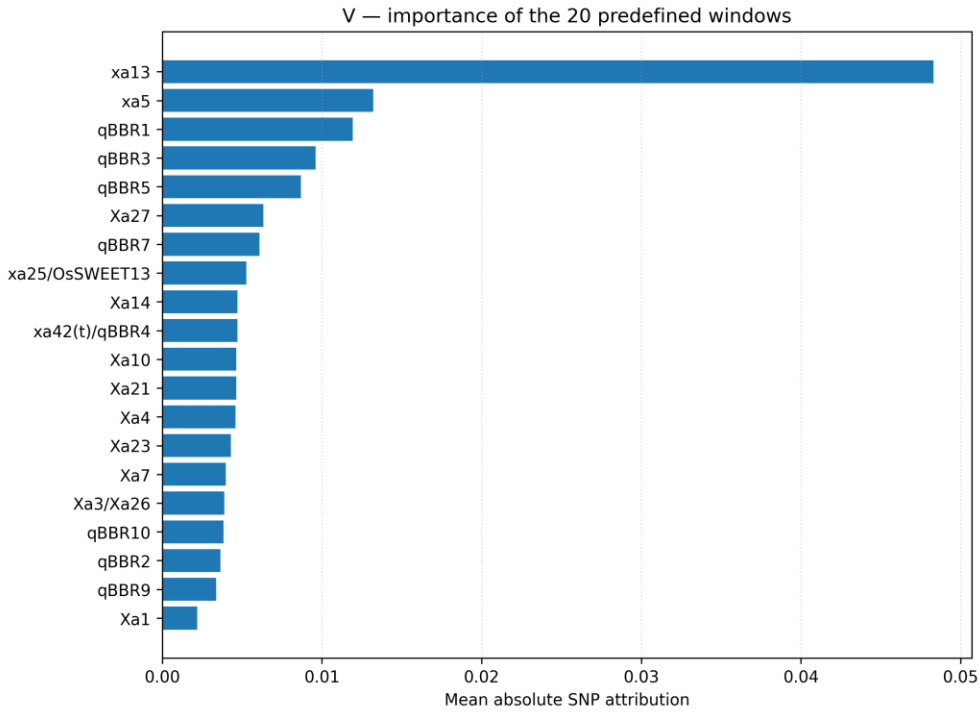


Figure B.19 - V resistance/QTL-region attribution bar chart.

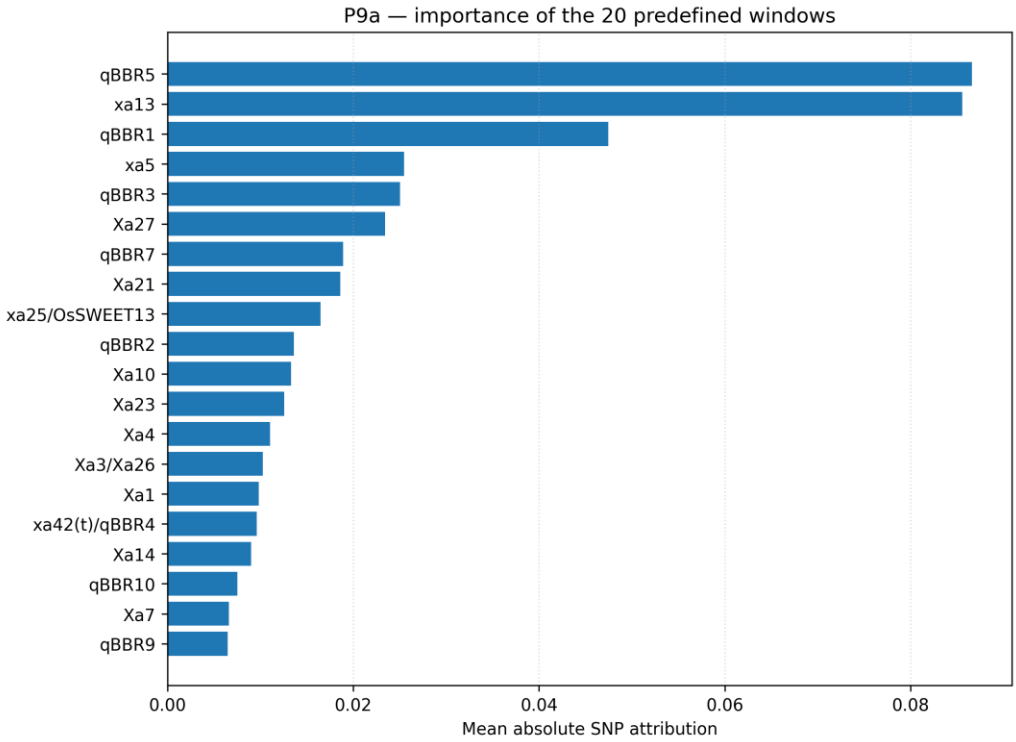


Figure B.20 - P9a resistance/QTL-region attribution bar chart.

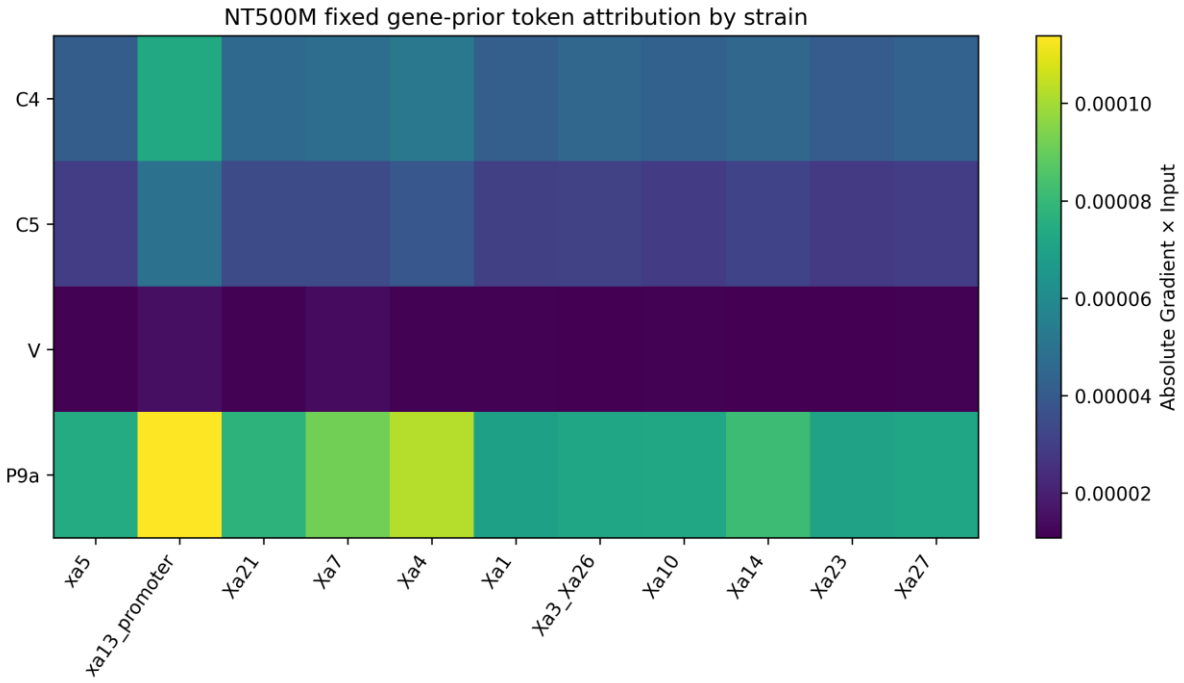


Figure B.21 - NT500M resistance-gene prior importance heatmap.

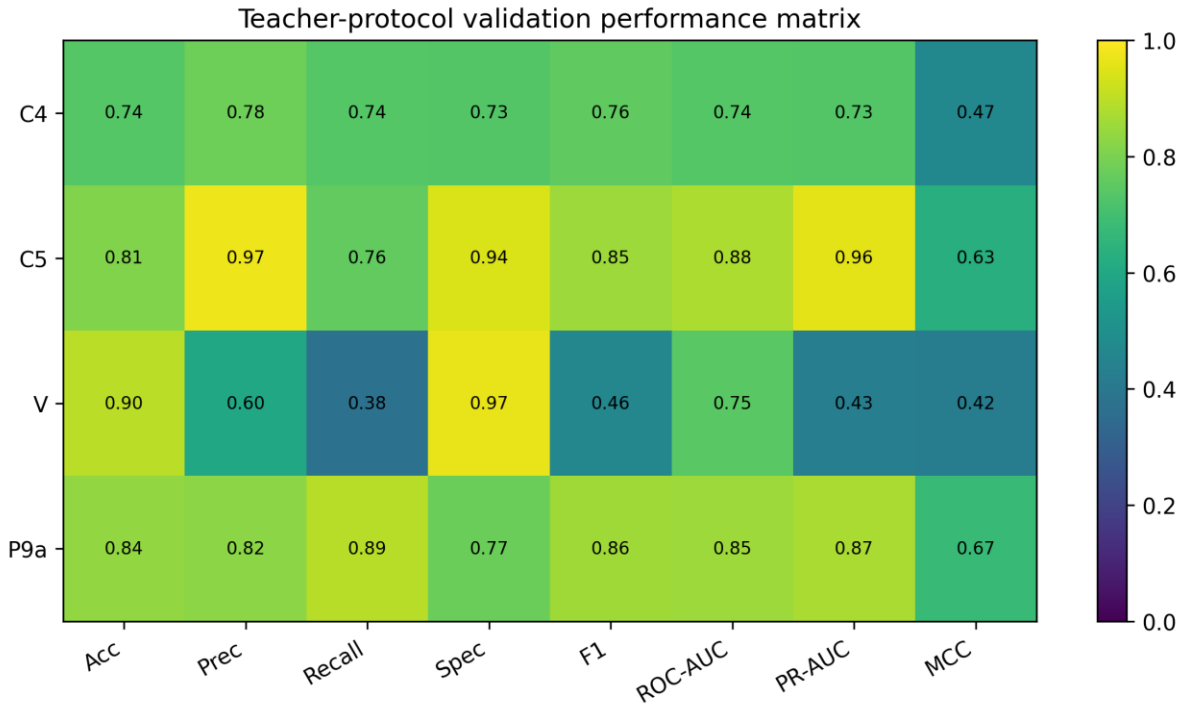


Figure B.22 - Teacher-protocol validation performance heatmap.